

E S S A Y S,

O R

DISCOURSES,

SELECTED FROM

THE WORKS OF FEYJOO,

A N D

TRANSLATED FROM THE SPANISH,

B Y

JOHN BRETT, ESQ.

VOLUME THE FOURTH.

L O N D O N,

Printed for the TRANSLATOR :

Sold by H. PAYNE, Pall-Mall ; C. DILLY, in the
Poultry ; and T. EVANS, in the Strand.

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PHYSICAL PARADOXES.

I. The Greek word Paradox, or Paradoxologia, does not properly signify a false or implicative proposition, but an improbable or incredible one; and therefore, this word is commonly applied to those propositions, which in the eye of the generality of mankind, at first sight seem false and contradictory, and difficult to be assented to; but upon examining them strictly and with attention, appear to be either true or probable. In this discourse, we shall treat of some of the things of this species, which are to be found in the line of natural philosophy.

PARADOX I.

Elemental Fire is not hot in the highest Degree.

SECT. I.

II. Vulgar physics, distribute the four qualities which are called first, among the four elements, assigning to each element one of them that is intense in the highest degree, and another which approaches, or comes near to the highest. Thus to fire they attribute heat in the highest degree, and dryness which is nearly approaching to the highest. To the air, humidity in the highest degree, and heat near to it. To the water, coldness in the highest degree, and moisture near to it. To the earth, dryness in the highest degree, and coldness near to it. This distribution, which if regulated, not by a nice examination of the nature of things, but only by an imaginary proportion, is exposed to three serious and weighty objections which are started against it by the modern philosophers; and it is only to the proposition that fire is hot in the highest degree, that they have hitherto made no difficulty of assenting; but that is the point I at present mean to controvert.

III. That elemental fire is not hot in the highest degree, may be evinced, by shewing there is another heat which is much greater, and that is the heat

of the sun, when its rays are concentrated in a burning-glass. It is certain, that the most vigorous effects and operations of elemental fire, does not come up in any degree of comparison, to those of that most ardent star. This has been proved evidently, by the experiments made with the burning-glass invented and fabricated by Mons. Villeté, an eminent artist of Lions in France; a description of which, was printed and published at Liege in 1715, and was afterwards inserted in the Memoirs of Trevoux of the year 1716. The instant the focus of this burning-glass was applied to any sort of wood, be it ever so green or moist, it set it as suddenly in a flame, as elemental fire would do dry tow. In less than a minute, it melted metals which were the most difficult of liquefaction, such as copper, iron, gold, and all minerals in general. The highest operation the chymists have found elemental fire capable of performing, is that of vitrification, which is so called from its reducing matter to a species of glass; but the most intense fire, besides its being tardy in this operation, can perform it only on particular subjects; whereas the burning-glass vitrifies in a short time, every kind of matter, tiles, bricks, cement, bones, stones of all sorts, even marble, and porphyry; and what is most extraordinary, it vitrifies those sort of stones also, with which they make the bottoms of furnaces for melting iron ore in, which although they will endure that intense heat for many years together, will begin to dissolve almost immediately, upon applying the focus of the burning-glass to them.

IV. Although this is very wonderful, what we are about to relate is still more so. The analytic resolution of gold, or what is the same thing, the separation or division of its principal component parts, had till then been deemed impossible; for the misers are not more tenacious of hoarding or preserving their gold, than the gold is of preserving its intrinsic texture; for in spite of all the tortures by fire, which the chymists could invent to break this texture, they could never make it lose its original form. Yet notwithstanding the valour of this generous metal, it submitted obediently to the power of the sun exerted through the burning-glass; as if that power to whom it is said to owe its existence, was the only one to which it could bear to be subject.

V. Mons. Homberg, of the Academy Royal of Sciences, was the first who experienced this rare phenomenon, by resolving into smook, with the focus of the burning-glass, at the Palace Royal at Paris, a great portion of its mass, which this celebrated chymist judged to be the mercurial part of the gold, and when this was gone, the residue which remained, appeared to be a terrestrial matter, mixed with somewhat of sulphur, that afterwards vitrified. So that in the opinion of Mons. Homberg, sulphur, and mercury, together with a portion of earth, is what gold is composed of, and although the two first are by their nature volatile, and capable of being dissipated and divided by the power of fire in all other metals, and in all other mixtures, they in gold are so intimately united, that no other force or influence can separate them, but that of the sun; it follows then, that the heat of the sun is much greater than that of elemental fire, and that this last cannot be hot in the highest degree, which is what we undertook to prove.

PARADOX II.

The Air ought rather to be esteemed cold, than hot.

SECT. II.

VI. The quality which Aristotle attributes to the air, is cold somewhat under the highest degree, or nearly approaching to the highest degree. Other philosophers, with more foundation of reason, esteem it indifferent to heat and cold; and I, without setting about to combat this second opinion, say it is more reasonable to suppose it cold than hot. Which I shall attempt to demonstrate in the following manner: in order to make a judgement of the qualities appertaining to a particular thing, we should consider it in a state, where it is divested of the influence of any extrinsic agent, by the operations of which it may be affected; the air then so circumstanced, will always be found to be cold; and from thence it should be inferred, that it naturally is cold. The minor of this proposition is proved, by the air being only warm while it is affected by the rays of the sun, and by its being found, that whenever that influence is withdrawn, it becomes cold again, and that it is by so much the more cold, in proportion to the absence of that influence. From whence it is observable, that in the temperate zones, the air is coldest

when the nights are long; and that in the sub-polar, and circum-polar ones, it is extremely cold, on account of the influence of the sun being withdrawn from them for the space of six months together; and that when it returns to enlighten them, because of the obliquity of its rays, it still continues cold, far beyond the degree of mediocrity.

VII. Nor is saying that in the absence of the sun, the earth is the agent that cools the air any reply to this; for if this was so, the lower region of the air would be colder than the middle one, as being nearer to the infrigidating cause; but this is contradicted by our experience; for we very commonly see, that the water does not freeze in the lower region, when in the middle one it is frozen into hail in the clouds; and very often also, that which was frozen above, thaws suddenly upon its falling down.

VIII. If it should be objected to this, that Aristotle and the Peripatetics, when they say the air is warm, speak of the elemental air, and not of the air of the atmosphere which is mixed with an infinite number of heterogenous corpuscles, by some of which it may be made cold, and especially by the many nitrous ones with which it is impregnated. To this I answer first, that in the country where I am now writing, there is not the least appearance of the air being nitrous, for that there is not a grain of nitre to be found in the whole country, notwithstanding which, the cold in winter is sometimes pretty sharp. I answer secondly, that we can only talk of the pure elemental air by conjecture, for no mortal has yet ever breathed any of it, nor is it possible that any one ever should, on account of this element being an open expanse, exposed to the inroads of the effluvias of all other bodies; and we should reason upon sensible qualities, by pursuing the thread of palpable experiments; and not by pursuing that of ideal proportions, as Aristotle did in making the division of his elemental qualities; for the Author of nature is not confined to pursuing only such proportions as fall within the compass of our comprehension. This is the false principle, upon which all the Pythagorean and Aristotelic philosophy has proceeded, and with respect to the doctrine of the four elements, that seems to be tainted with the same vice, as I shall shew more fully in another place. All that I shall say at present is, that Aristotle dealt out, and distributed the four qualities among

the four elements, as if he had been the absolute master of them all, and could dispose of and arrange them agreeable to his will and pleasure.

PARADOX III.

*Water considered according to its Nature, is rather disposed to be solid,
than fluid.*

SECT. III.

IX. This may be demonstrated, upon the same principle the antecedent paradox was; remove for any length of time, the interposition of any extrinsic agent that may contribute to warm the water, and it will always be found to be solid, that is, in a frozen state. This then being so, we may conclude, that is the state it is disposed to continue in agreeable to its nature. This is evinced by the sea under the Poles, and in the adjacent parts, being, during the whole six months the sun is absent from them, frozen to such a degree, that after the sun has revisited them, and continued its influence for the other six months, it is never totally thawed; and it is for this reason, that it has always been found impracticable to sail to China by that rout.

X. The water being under this influence, is a proof *ad hominem*, that it cannot, agreeable to the doctrine of Aristotle, be cold in the highest degree, for if that was the case it must ever continue in a frozen state; so that we can only say, that by its nature it is most disposed to solidity.

XI. Frozen water being easily made liquid by a moderate heat, is no proof that it is not naturally inclined to solidity. Metals may be liquified by an intense heat, but that does not shew that they are not of a solid nature; and a thing being more easily, or with more difficulty dissolved, makes no difference with respect to this mode of reasoning; and therefore the water being made liquid with a less intense heat, than is found necessary to liquify metals, does not prove, that it is not disposed in its nature to be solid as well as them.

PARADOX IV.

Either all Qualities are occult, or none of them are so.

SECT. IV.

XII. The school philosophers call all those qualities occult, which are not reckoned among the four elemental ones, and which do not result from the various combinations of these elemental ones, because they suppose their operations move in a line superior to those which may be supposed to proceed from humidity, dryness, cold, heat, hardness, softness, colour, favour, &c. And although it is true, that some by pursuing the system of assigning second qualities, which result from the various combinations of the first; or third qualities, resulting from the various combinations of the second; and among these third, have placed the wonderful virtues of the load-stone, the operation of purges, and others of those which they call occult, and which by these means they have attempted to reduce to manifest ones; this mode of proceeding, has been abandoned by the generality of philosophers, and with reason; for it is very clear, that combine, and recombine, the qualities of humidity, dryness, heat, and cold, in what way you will, you will not be able to find out, that they are capable of directing the load-stone to the Pole, or of causing it to attract iron.

XIII. It is not my intention, to examine the nature and origin of all, or either of these qualities; but shall only say, that they are all equally occult, or are all equally manifest. In order to demonstrate this, we will compare the calefactive virtue of fire which is looked upon as the most manifest, with the attractive virtue of the load-stone, which is reputed the most occult. All that is known, or that is taught by the peripatetic doctrine, of the calefactory virtues of fire, may be reduced to this, that it is a property appertaining to that substance, or a quality that springs from, or arises out of its form, that produces the effect which we call heat, and that the action with which it causes it, is called calefaction; but so it is, that it is just in the same manner that we know the attractive virtue of the load-stone; viz. that it is a property or quality, springing from, or arising out of the form of this entity, that produces the sensible effect of drawing the iron to it, and that the

action with which it causes it to approach towards it, is called attraction; it follows then, that we know just as much of the attractive virtue of the loadstone, as we know of the calefactory virtues of fire; and therefore, both of them are equally occult, and equally manifest.

XIV. And if we were to speak the truth, how could we deny, that the quality we call heat is occult, when it is even occult, whether it is or is not a quality? Not only the corpuscular philosophers deny it to be possessed of all quality or form, but many of those who admit those properties to be contained in it, constitute the heat to proceed from a vertical, or vibratory motion of the insensible particles of the body. And so long as we shall remain without an argument to convince us of the truth of these opinions, we shall not know which of them is right.

PARADOX V.

It is false, in a general and comprehensive sense, that Virtue by being united becomes the stronger.

SECT. V.

XV. The axiom *Vis unita fortior*, I apprehend, applies more justly to civil and political things, than to natural ones. If we attend, we shall find that two agents, each of which is strong in proportion to four, by being joined together, will not be stronger than in proportion to eight. If two men separately can support but four half hundreds each, by joining them together, you would find them not able to support above eight. It is true, that a man who breaks arrows one by one, cannot break a bundle of arrows tied together, which is the comparison that Scilurus availed himself of (Plut. in Apophth.) to persuade his sons to continue in fraternal union; but we should not understand by this, that by binding them together, each arrow separately acquired any additional strength, for this example was only intended to shew, that he who could with ease break each arrow separately, was not sufficiently strong to break the whole bundle united together. Suppose, for instance, the bundle was to consist of twenty arrows, if there was any thing less than a twentieth part of the force exerted to break a

single arrow, than it was necessary to exert to break the whole bundle, it is as certain that that single arrow would not be broke, as that the whole bundle would not. Thus this example does not prove, that there is any additional virtue or strength added to the wood of which the arrows are made, by uniting them together, but only, that twenty joined together can resist a greater force than one singly, which is a thing that is self-evident.

XVI. Nothing can be plainer than this; but abstracted from the force of this reasoning, experience has shewn us, that in some agents, in contradiction to the common opinion, their union has diminished their strength. It is the general sentiment, that two threads twisted together and united in one cord, will support more weight than they will separately; and that a rope made of many fine threads, is stronger than all these threads divided. Mons. Reamur, of the Academy Royal of Sciences, demonstrated in 1711, that the fact was quite otherwise, and that the threads separate would support a greater weight than they would united. He made this experiment with two single threads, and found that each by itself would support nine pounds and a half, which put together make nineteen pounds; after this he caused them to be twisted together, and they were broke with sixteen pounds. He made another experiment with three threads, one of which supported six pounds and a half; another eight, and another eight and a half; the sum of all which put together amounted to twenty-three pounds; and after causing them to be made into a cord, they would not sustain more than seventeen.

XVII. It may perhaps be answered to this, that the threads in twisting them together might be stretched and made finer, and their strength weakened; and that besides this, they might have been strained by supporting the weight in the first instance; but this objection, although specious, is insufficient; for by attending to the relation of this matter in the History of the Academy Royal, we shall find, that by order of the Academy, another experiment was made in a quite opposite way, for a small silk cord which supported a little more than five pounds, was afterwards untwisted, and they found that the single threads among them all sustained six pounds and a half.

XVIII. The true cause of this phenomenon in my judgment, is, that the threads in the twisted cord do not bear equally alike, because it seems to me morally impossible in the twisting, to preserve the exact length or bearing of all the fibres, but that some of them must be more tense or tight, and some of them more slack than others; in consequence of which, the bearing of some must vary more from the perpendicular or central line of gravity, than that of the others, and that some part of them are upon the stretch, and others rather slack at one and the same time. From hence it follows, that the weight at first is not supported by all the fibres, but only by those which are the most tense, and the least distant from the central line of gravity, which not being of themselves sufficient to resist the strain, give way, and that the weight afterwards falling upon the others, they do the same; and that this is the case may be evidently seen, by observing when more weight is hung to a rope than it is able to sustain, it does not break instantly, but successively, and although the total separation is performed in a very little time, there is space enough before it is compleated, for us to perceive, that some threads break first, and others afterwards.

XIX. But although my opinion should not exactly coincide with the experiments made in the academy, I conclude, that the judgment we should make of this matter is, that both in this, and all other physical agents, virtue united is the same that it is separate; for the cord breaking with a less weight than the threads sustained separate, did not result from these last being stronger separate than they were united, but from the fibres of the threads acting more together in their separate state, than they did in their united one, as in the last case, the resistance of the fibres to the force, was made more in succession than it was in the first; or to speak more properly, although the threads were united in the cord, their power or virtue of resistance to the force, was not united or exerted together.

PARADOX VI.

*The Sun in consequence of its own proper Nature, and intrinsic Disposition,
heats and enlightens unequally at different Times.*

SECT. VI.

XX. The common causes of our experiencing more or less heat, or more or less light from the sun, are the serene or loaded state of the atmosphere; the oblique direction with which the sun's rays are reflected on us; the position, or situation of places; the length or shortness of the days; the tranquillity or agitation of the winds, and our vicinity to cold or warm places, such as snowy mountains, or warm subterraneous effluvia. But, abstracted from any of these sublunary or inferior causes, I say, that in the sun itself, is contained a cause, that occasions it to reflect more or less light or heat, and that it actually does reflect more or less light and heat, at some times than it does at others, and that this is in virtue of its own proper nature and disposition.

XXI. The reason of which phenomenon is, the transitory spots, which the astronomers have for some time past observed in the sun. These are a sort of darkish places, unequal in size and duration, which seem as if they were on the superficies of the star, and which at different times are more or less numerous, although it has happened, that for years together none of them have appeared. Some believe that the antient Chaldeans had some knowledge of them, for in the Book of Job, we read the following expression of Job's friend Eliphaz to him, that the Heavens themselves are not exempt from their spots: *Cæli non sunt mundi in conspectu ejus*. On the other hand, the want of telescopes among the ancients, did not render their making observations on these things impossible; for some of these spots are so large, as to be visible without a telescope; as for example that was, which was seen in the year 1706, whose superficies according to the computation of the astronomers, was thirty-six times larger than the whole earth; and when they arrive at this magnitude, or if they are even much less, they may be discerned, by looking at the sun with a glass tinged with any dark colour.

XXII. But the first person we have any account of, who observed these spots, was Father Christopher Scheinerus a jesuit, who applied himself to it with such diligence, that between the years 1611 and 1627 he had made fourteen hundred observations of them, which he gives an account of in his *Rosa Ursina*. The celebrated Galileus Galilei began also to observe them about the same time that Scheinerus did; and since them, the most laborious astronomers of the last and the present century, have applied themselves to

making the same observations; so that this is a matter, which at present there is not the least doubt about; and although some have entertained suspicions whether these spots were in the body of the sun, or at a little distance from it; others have removed those difficulties, by demonstrating, that they are inherent in, and exist on, the superficies of the sun; for they not only revolve in the same proportion the sun does, but they most probably remain upon it during the whole time it takes in making an intire revolution, which it does in twenty-seven days, and they are to be seen compleat for half the time of that period; which could not happen, if the spots were inferior to the star.

XXIII. And whether these spots are sooty or smoaky vapours, which arise from the great furnace of the sun, as some think, or are any other different thing; it is clear, that during their continuance, the light and the heat of it reflected on the lower elemental regions, must be diminished in proportion to the size and number of the spots; and to this cause may be attributed, some of the notable diminutions of the light and heat of the sun, which we find recorded in history, provided they were not occasioned by some obstructions in the atmosphere. Mayolus tells us, that in the time of the emperor Justinian, the light of the sun for the greatest part of a year was so dim and faint, that it scarce exceeded that of the moon; and Plutarch tells us, that at the time of the death of Julius Cæsar, the light of the sun was equally feeble for a whole year together; which Virgil also takes notice of in the following verses of the Second Book of his Georgics:

*Ille etiam extincto miseratus Cæsare Romam
Tunc caput obscura nitidum ferrugine textit:
Impiaque æternam timuerunt sæcula noctem.*

PARADOX VII.

The Rays of the Sun reflected on a Concave Body, produce a greater Heat in Winter than in Summer, which Heat is greater still in Proportion to the coldness of the weather.

SECT. VII.

XXIV. The experiments to prove this, were to the admiration of all the by-standers, repeatedly made with the burning-glass of Mons. Villete, which we have spoken of before; and it was observed at the time of making them, that by so much the more cold the burning-glass was, by so much the quicker and stronger was the operation of the focus; and that by so much the warmer it was, by so much the more tardy and remiss were its effects. Among those who read these accounts, some looked upon them as wonderful, and others as incredible.

XXV. But with all this, the reason of the phenomenon is not very hidden. It is certain, that cold condenses bodies, and that heat dilates them. It is also certain, that the more dense a body is, the more apt and fit it is to cause reflection, and that it is least so, the more lax it is. From these two premises it may be clearly inferred, that the effects we have mentioned were things of course; but for a further explanation of this matter, I will say they were so from two causes; for the burning-glass by being in a warm state, must consequently be less compact and hard, and the operations of its focus more remiss and languid. In the first place, because a great portion of the rays were absorbed in the pores of the metal, which had been dilated by the heat, and made no reflection at all. Secondly, by their being dilated, and as we may say, become more spongy, the concave superficies of the metal were less smooth and equal, from whence it would follow, that many rays by being obstructed in their descent by some insensible prominences, would not make their reflection in a right line on the point of the focus. This may be better understood, by attending to the reflections of a ball, that is struck against a wall with an unequal surface; and it is a position agreed to by all the mathematicians who have treated of catoptrics, that light and heat in their reflections, follow exactly the same rules, that heavy bodies do.

XXVI. Nor is there any difficulty in supposing, that a body as hard as metal, may undergo some insensible rarefaction when it grows warm; for in the first place, if a very intense heat can dilate metal, so as to make it break all its ligatures and become fluid, a less heat would have the same effect, in proportion to the degree of it, and might dilate and rarefy it a little. Secondly, experience teaches us, that every sort of metal is more sonorous when it is cold, than when it is warm; from whence it may be evidently

inferred, that heat and cold do somewhat alter its texture; it being certain, that upon the texture, depends its being more or less sonorous.

PARADOX VIII.

A flame extending itself upwards in a pyramidical or conic form, is occasioned by a violence done to the flame itself.

SECT. VIII.

XXVII. The endeavour of the flame to ascend, is the proof which is vulgarly exhibited, of there being an element or sphere of fire above us, by those who maintain that position; but we shall shew, that this is a very feeble proof, even admitting their own state of the case, because every thing that is lighter than the fluid that surrounds it, will, if it is not by violence restrained from doing so, rise above it, without being invited to ascend by a sphere of its own species that is aloft; and it is from hence, that flame, smoke, vapour, an infinity of elemental effluvia, and divers other species of things ascend, without the assistance or intervention of any other cause, except that of their being lighter than the inferior gross air.

XXVIII. To this we shall add, that there is not in the flame, the propensity to ascend which they suppose, and which seems to be indicated by its rising upwards in a conic form; for this ascent is violent, and not natural to the flame. We shall illustrate this paradox, by an experiment which my Lord Bacon mentions in the first Book of his Centuries. A small wax candle, fitted into an iron tube, and placed perpendicular in a vessel filled with spirit of wine, in which it must be immersed to such a depth, that when they are both set fire to, the flame of the candle shall not rise higher than that of the spirit; and you will see, the flame of the candle and that of the spirit being distinguishable by their different colour, that the flame of the candle will appear in the middle of that of the spirit, extending itself in breadth, and not in a pyramidical, but a round form, that occupies four or five times as much space, as that which it used to occupy when burning in the open air. This experiment proves, that the pyramidical form which the flame regularly assumes, is caused by the pressure of the air that surrounds

it; and this is the sentiment of the modern philosophers, who without advertent to the experiment we have mentioned, would make no scruple of pronouncing, that the form it assumes is a violence done the flame.

XXIX. By this example it may be seen, that experiments, provided they are made with judgment, and attended to with subtil reflection, are the only means by which we can attain any certain knowledge of natural things; but if they are made in a lumping way, and without due attention, and exactness, they are the occasion of innumerable errors. Many have arisen, from attributing to a native inclination, or intrinsic virtue of some body, effects, which are only caused by the impulse of some other neighbouring body. Before the gravity and elasticity of the air were discovered, it was looked upon as a thing demonstrated by experience, that the water's rising in the pump, was from its inclination to obstruct or oppose a vacuum, and now it is a thing evinced by experience, that the air is what impels it to that ascent.

PARADOX IX.

In the composition of all vegetables, there is contained a portion of metal.

SECT. IX.

XXX. This is a novelty in physics, which has not been discovered till of late years. Mons. Gofredy of the Academy Royal, having examined the ashes of many different plants, found in them all, some fine particles that would adhere to the load-stone; from whence he concluded, that these particles were either particles of iron, or else were particles of the load-stone itself. But, as it was a doubt which had never yet been cleared up, whether the load-stone would not attract other metallic particles, which were not part of itself, nor of iron; the Lemeriers father and son, made some fresh enquiries, which removed all doubts respecting the matter. They with a burning-glass melted the particles which the load-stone had attracted from the ashes of the plants; which after much sparkling, liquefied into the very form and substance of the load-stone, and of iron also; and when grown cold, formed a lump of the consistence and hardness of metal. Even in

honey, after it has been distilled, have been found these particles which were attractable by the load-stone; from whence it may be inferred, that this metallic composition, exists in, and is diffused through the most subtil juice of flowers and blossoms.

XXXI. But, after all this had been done, it still remained to be ascertained, whether these particles pre-existed in the plant; or were the production of the fire, and the result of their calcination; which second opinion appeared the most probable, because there seems no difficulty in supposing, that the fire might transmute into metal some of the particles of vegetables; but there appeared a very great one, in conjecturing that so heavy a metal as iron, could rise to the summit of trees, and pervade the most subtil fibres of their leaves.

XXXII. Mons. Lemeris the son cleared up this doubt, by various subtil and curious experiments, which not only evinced the volatility of iron, but also excited a belief, that this metal contributed greatly to promote the vegetation of all kinds of plants. The most remarkable experiment he made was the following one. Having poured spirit of nitre upon the filings of iron, there succeeded a violent effervescence, which after some time ceased, and left the metal dissolved into a red liquid; by afterwards pouring oil of tartar per deliquium to this liquid, another fermentation was excited, which went on increasing, till, at last, it formed on the sides of the vessel, various subtil branches, which after all the sensible fermentation had ceased, continued to grow till they reached the top of the vessel.

XXXIII. And although the first time he made this experiment, he obtained only the rude lineaments of a tree, by varying afterwards the quantity of the oil of tartar, till he hit upon the just proportion of it, he obtained a perfect metallic vegetation, and produced a tree completely formed, with its roots, trunks, branches, leaves, and flowers. This able chymist concluded from his various observations on this matter, that both the volatility, and the vegetation, were owing to the filings of the iron; for without them, he could only produce some crystalline lumps at the bottom of the vessel, which he concluded proceeded from the nitre that had been dissolved. Whoever is desirous of knowing more of the manner and effects

of these operations, may read the accounts of them in the Journals of the Academy Royal, of November 1706.

XXXIV. But we are not to conclude from hence, that metallic vegetation is only to be produced by iron. The Abbé Vallemont in his first volume, *on the natural Curiosities and Art of Agriculture*, says, that there were exhibited at Paris, such like artificial metallic vegetations, produced from gold, silver, iron, and copper. But that which was the most common, and the most in vogue there, was the vegetation procured from silver, which the chymists called the *Tree of Diana*; and which was produced in the following manner. Dissolve an ounce of silver, in two or three ounces of spirit of nitre. Evaporate this dissolution in a sand-heat; till nearly half of it is consumed. Mix the remainder in a vessel, properly proportioned and suited for the business, with twenty ounces of clear water, and two ounces of quick-silver. By leaving this mixture afterwards in a state of rest for forty days, a silver tree will grow, and in that space of time will be compleatly formed, very nearly resembling a natural one, in shape and figure. Mons. Homberg, a celebrated chymist of the Academy Royal, by making use of the same materials, contrived to form a metallic tree in less than a quarter of an hour; the receipt for doing which, together with the physical explanation of this phenomenon, given by Mons. Homberg, may be seen in the Memoirs of the Academy Royal, of the thirteenth of November 1692.

XXXV. These metallic vegetations, joined to the discovery beforementioned, of our having found iron in the ashes of all sorts of plants, not only prove that metals may, by virtue of certain fermentations, be made sufficiently volatile to enable them to rise upwards, and pervade all the tubes through which the alimentary juices of plants circulate, but also make it probable, that to this metallic mixture, they in great measure owe their vegetation.

XXXVI. This is the substance of what, in support of this paradox, I have found in the Works of the before-quoted philosophers; to which I shall add a conjecture of my own, which appears to me very efficacious to render credible, the formal existence of particles of iron, or load-stone, in all vegetables whatever; whether they are of load-stone, or iron, makes very

little difference, as all experimental philosophers are agreed, that load-stone is nothing else but a fat or rich vein of iron.

XXXVII. My conjecture is founded on a theory, which is embraced at present by all the mathematicians, and confirmed by conclusive reasonings, which are founded on the earth being endued with a magnetic virtue. The truth of this theory, is proved by innumerable observations. It has been found, that the magnetic needle poised in equilibrio, accommodates itself to the meridian of the earth, in the same manner the load-stone does, that is, it does not point to the poles of the heavens, but to those of the earth; for it has been observed in the northern regions, that it does not appear to elevate or point towards the pole of the heavens, but rather seems to depress, or point down towards that of the earth; and generally, in all, and every particular, the magnetic needle has been found to be affected in the same proportions with respect to the terraqueous poles, that it is with respect to the load-stone. The various declinations that it is liable to, from the pole of the earth, in different places, cannot be attributed to any other cause, than that of the unequal magnetism, of the terraqueous globe in different regions; and the different declinations of the poles of the load-stone are attributed to the unequal magnetism or perfection of the parts of the stone. It has been found, that the earth itself, can communicate the magnetic virtue to iron; for if you take a bar of iron red-hot from the forge, and place it perpendicular in the earth, and there let it remain till it becomes cold, it will manifestly acquire the magnetic virtue; and if it is afterwards poised in equilibrio, it will point to the poles of the earth, in the same manner as if it had been touched by a load-stone. It will also acquire the same virtue, if it has remained for many years in a perpendicular position, without its having been placed so in the earth red-hot; and the same has been experienced with respect to bars, that have been set before windows in an upright position. The same effect will also be produced, by laying a hot bar upon the ground, exactly in a north and south direction, till it gets cold; but if it is so laid without being previously heated, and suffered to continue in that position for several years together, it will be found to be impregnated with the magnetic virtue. Whoever is desirous of seeing an account of these observations more at large, and to satisfy himself how they evince the magnetical quality of the earth, should read the mathematical authors who

treat of the load-stone, and he will find, that all the modern ones make some remarks upon that particular.

XXXVIII. This magnetic property of the terraqueous globe being admitted, it may be inferred from it, that excepting the exterior crust of the earth, which is composed of such heterogeneous parts, as are necessary for the growth and increase of the various productions and mixtures with which it abounds, all the rest, is nothing else, but a solid quarry of load-stone; and this is the opinion that is strenuously maintained by some; although there are others, who think that the magnetic virtue, is distributed through all the parts of the terraqueous globe.

XXXIX. Both these opinions may be right, for they do not seem incompatible one with another; but in favour of the second, which is that which conduces best to support my sentiment, we may urge another celebrated piece of experience, which evinces efficaciously, that this very exterior earth which we touch and stand on, is impregnated with many insensible particles of load-stone or iron; and this is manifest, from the earth itself being endued with this magnetic virtue, or inclination of pointing to the pole; for bricks that are made from it, and are well burned and purged or freed from all foreign humour or moisture, and especially if they are made long and narrow, by being touched with a load-stone will acquire the virtue of pointing to the poles; and they will even acquire it without being touched, if they are laid in a north and south position, and suffered to continue so for many years together. (See Father Dechales, lib. 1 & 2, de Magnete.) It being then certain, that this quality of pointing to the pole, is peculiar to the load-stone, or the iron, and incommunicable to all other substances, it must evidently be inferred from this property in the bricks, that the very earth we touch and stand upon, is impregnated with particles of load-stone or iron. All vegetables then receiving their nourishment from the earth, it is not wonderful that there should be found in them all, some of these particles.

XL. I would recommend it to those who take delight in philosophy, to endeavour to inform themselves, whether it is not probable, that all other mixtures are impregnated with these particles; for if that should be found to

be the case, we should discover the cause of the descent of heavy bodies; for there being in the earth the magnetic virtue, and in all other mixtures particles of iron, in spite of all our endeavours to separate them, they always would be brought together again by attraction. But as he who is a lover of the truth, should conceal no objection that may be made to any supposition that he advances, I will not dissemble, that I find a terrible one against this idea of mine; which is, that according to this system, iron should be heavier than gold; for although we should admit, that some iron, or magnetic particles are mixed with the gold, it is not credible, that the quantity of them should be equal to those in the iron itself; for if that was the case, the load-stone would attract the gold, as it does the iron. But whether there is in the terraqueous globe, another attractive quality distinct in kind from that of the load-stone, in virtue of which, all bodies may be disposed to approach it which we call heavy, by having in themselves a propensity to do it correspondent to that there is between the load-stone and the iron, is very difficult to be proved.

ON SCEPTICAL PHILOSOPHY.

SECT. I.

I. There is so great a latitude in scepticism, and its gradations are so different, that according to the various extension that is given to the meaning of the word, it is used to describe the most extravagant error, and the most prudent mode of philosophizing. Rigid scepticism is a wild delirium, moderate is prudent caution. But those who in this age, have undertaken to combat the moderate sceptics, have either from ignorance or malice, confounded them with the rigid. When I reflect how gross a thing ignorance must be in such a case, it makes me think they acted from malice, and when I consider how detestable a thing malice is, it makes me conclude what they did was through ignorance.

II. Although the Greek word *scepsis*, from whence sceptic and scepticism are derived, signifies inquisition, investigation, and speculation, custom has in a great measure altered the meaning of the term; on which account, sceptical at this day, implies doubt, or doubting; and scepticism, that particular avowal which sceptics make of doubting, or withholding their assent to all matters, that are liable to be controverted or disputed.

III. This doubt or suspension of assent, may be more or less rational, in proportion to the greater or less extent that is given to it, or according to the matters to which it relates. Thus, although doubting of many things would be prudence, doubting of all would be madness.

SECT. II.

IV. And although authors commonly represent to us, some subtil philosophers of antiquity, as obstinately persisting in suspending their assent to every thing that was proposed to them, and which was evident both to their reason, and their senses, and likewise represent them, as earned

defenders of universal scepticism, to which opinion they refuse to admit of the least exception; yet for all this, it is very doubtful to me, whether these were their real sentiments; and I am rather inclined to think, they did it, to shew their ingenuity, and to display their talent for disputing; or that from some other motive, they spoke differently from what they thought. The most famous among this tribe, were Archesilaus, Carneades, and Pyrrho. But the first, if we believe Sextus Empericus, was in reality a Platonist, and only a sceptic in appearance; whose method it was, always to dispute problematically upon every thing in public, but was ever careful to suggest in private, to such of his disciples as he found were the most capable, the Platonic doctrine. Cicero says, that his zeal for contesting every thing that was advanced by his school-fellow and rival Zeno, carried him to the dangerous extreme, of endeavouring, although it was against the conviction of his own mind, to refute whatever was advanced by Zeno. We may add to this, that according to the testimony of Diogenes Laertius, Archesilaus never went to the extreme of doubting the evidence of his senses, but rather treated with contempt and derision, those who carried scepticism to that length.

V. Carneades was a most subtil philosopher, and so eminent as an orator, that Cicero in various parts of his works speaks of him with admiration and envy; and assures us, that by the keenness of his ingenuity, and the fertile torrent of his elocution, he could persuade his hearers to believe whatever he thought fit, and Quintilian, and Numenius, declare the same; and it is probable, that his propensity for disputation, and his ambition of shewing his ingenuity, excited him to contest the most established axioms; which together with his affecting to distrust the evidence of his senses, made him pass for a rigorous sceptic. But we may venture to assert, that if an historical anecdote which is related of him by Numenius is to be credited, Carneades believed his eyes as much as any man living; for having surprized a concubine of his in the embraces of his beloved disciple Mentor, he was so offended at his treachery, that he broke with him for ever, and excluded him from being the successor to his academy. How happened it then, that like a good sceptic, he did not doubt whether the representation of that obscenity, might not be a *deceptio visus*? I for my own part am inclined to think, that there never yet was a sceptic in the world, who after the

exhibition of such evidence, could maintain his heart and his mind in a state of suspence.

VI. It is commonly said of Pyrrho, who was by so much the most famous of all the sceptics, that he in a manner obscured the others, by giving his name to the system of universal doubt, which at this day is called Pyrrhonism, and the sectaries of it Pyrrhonists; and he was so positive in maintaining, that we should suspend our opinion, with respect to all we saw, or felt, that he would not turn the least aside, to avoid being run over by a mad horse, or bit by a mad dog; nor would he slacken his pace, although he was warned that he was advancing near the edge of a precipice; and that he would have perished a thousand times by these dangers, if his friends, who were attentive to his safety, had not prevented his running headlong into them. But although this account of him is much vulgarized, I do not know whether any other antient author, except Antigonus Carystius, a Greek historian, who was contemporary with, or lived very nearly in the days of Pyrrho, has vouched the truth of it; at least the learned La Mothe de la Vayer, quotes him as the only author, who attests the relation; and it is even doubtful whether Antigonus Carystius asserts the thing positively, because in the *Præparat. Evang. lib. 14. cap. 18.* of Eusebius, we find this author quoted for a fact, which contradicts that relation; which is, that Pyrrho, when a dog was once about to run at him, climbed up a tree to get out of his way, and avoid the danger. Upon which occasion, those who were present, raillied him upon the difference there was between his doctrine and his practice.

VII. But let Antigonus Carystius, who is an author I have not seen, or any other whatever, say what they will, to give sanction to this relation, we may without the apprehension of being accused of injustice, condemn as incredible, the account of Pyrrho's having carried his extravagance to such a length. This philosopher lived to the age of ninety, and through the whole course of so long a life, it is not probable, he should always have friends at his elbow, ready to save him from the many dangers, to which a man of so rash a conduct must unavoidably have been very frequently exposed, and especially, in the long voyage he made to India to consult the Gymnosophists. Diogenes Laertius, who is the person that gives us the

account of the great age to which Pyrrho lived, and of his voyage to India, assures us, that Pyrrho led a very solitary and retired life, which is not very compatible with his having always been attended by his friends, nor is it reasonable to conclude, that a man who was so very ridiculous as he is described to have been, could have very many, or very sincere ones. Finally, the citizens of Elis, his countrymen, elected him the high priest of their religion. Now how is it credible, that they should have confided such an employment to a person, who if his scepticism had been carried to the length we have mentioned, they must certainly have looked upon as a mad man? From this circumstance, it may also be inferred, that the charge of impiety, which was generally imputed to him, was an unjust one; because his countrymen would hardly have elected him the supreme minister of their religion, if they had known that he professed to be of no religion at all, and that he doubted the existence of a deity. What devotion or zeal could they expect for the service of the temple, from a man, who either was ignorant of, or doubted whether the object of their worship existed?

SECT. III.

VIII. I not only think, that these philosophers did not in their hearts, adopt the system of universal doubt; but I also believe, there never was any one who sincerely did it; because there are objects, with respect to which, there can be no doubt but by implication. No one, for example, can doubt of his own existence. The very act of doubting, is affirmative of a certain knowledge, because he who doubts, must certainly be convinced that he does doubt; and if the sceptics are not certain that they doubt, how can they affirm they do with such obstinacy and stiffness? Therefore we should conclude, that it is not from sentiment, but from their fondness for disputation, that some of them defend the system of universal scepticism; and if there ever was any one who truly assented to it, we should not consider him as a philosopher, but as a lunatic; for it would be improper to call maintaining such a system, a particular mode of philosophizing, as we ought with more justice, to term it a particular mode of raving.

IX. It is most likely then, that the most rigid sceptics, and those who were truly and *bona fide* such, admitted of some exceptions to the

universality of their system, or understood it in some determined sense, which limited or restrained it. Socrates, whom some look upon as the primitive father of the sceptics, said of himself, that he knew nothing with certainty, except it was his ignorance of all things. This however, was placing some sort of limitation to the system, although it was but a very small one. But I am of opinion, that Socrates, who was a very modest man, only meant to say, that what he knew amounted to but a small matter, and that therefore we ought to understand his declaring himself totally ignorant, as an hyperbolical expression. Saint Justin the martyr, and other fathers who highly applauded that philosopher, would not have done it, if they had taken him for a rigid sceptic, which would have amounted to the same, as his having been an impious man; because it is evident, that he who doubts of every thing, cannot be a person of any religious principles; but he was so far from such a kind of man, that it is most probable, the Athenians condemned him to death, because he affirmed the existence of one only God. It is at least certain, that he treated the multifarious number of the heathen gods with great derision; from whence we should conclude, that he knew this important truth, that the Godhead admits of no multiplication.

X. Other sceptics, who said that they doubted of every thing, and that we ought to doubt of every thing, did not perhaps mean to exclude all certainty, but only scientific and demonstrative certainty, which except in mathematical cases, we must confess is very rare to be found. Even many mathematical demonstrations, and especially those which are very complicated, are not exempt from the suspicion, of there being some occult fallacy contained in them, on which account they ceased to be true demonstrations. How many have presumed to demonstrate they could square the circle, whose modes of doing it, and their reasonings upon the subject, when they came to be submitted to rigorous examination, have been found to have some sophism involved in them, or to have been built upon some supposition which was taken for certain, when in reality it was not so. The geometrical demonstrations, with which they prove the infinite divisibility of the continued quantity, are sufficiently simple; notwithstanding which, there are authors, who because the infinite divisibility of the quantity is imperceptible, suspect, that there is some occult sophistry involved in those demonstrations.

XI. Others have refused to credit the evidence of their senses; but not so grossly, but that they have condescended to make use of them to direct their actions in the common affairs of civil and human life. They govern themselves by them for the purpose of living, but not for that of philosophizing. The information of their senses serves them to chuse the useful, and avoid the pernicious, but not to determine the theory of an object.

XII. The arguments they assign for this distrust of their senses in the last instance, may be reduced to three. The first is, the distinction we should admit between the impression which objects make on our senses, and the absolute essence or quality contained in the objects themselves. For example, we say that hemlock is bitter. If by this expression, we mean to declare, that this herb makes on our palate a determined impression or sensation, which we call bitterness, we speak properly; but if we mean to say that the herb in itself contains an absolute quality, which we describe by the same name, we speak improperly; for if this was the case, all the animals who relish hemlock would find it bitter; but it happens otherwise, for the goats are fond of it, and eat it greedily. In the same manner do all those reason, who adopt this mode of arguing with respect to all other sensible species. They say, the fire produces in us that kind of impression, which we call heat; but that we ought not from thence to conclude, that it is hot in itself. Thus although by approaching very near to it, it produces pain in us, still it is not sensible of any itself. Therefore we ought not for this reason, to say the fire contains pain, it being at most only capable of occasioning pain; neither ought we to call it hot, but only capable of producing heat; nor can we call it hot but by equivocation, as we say, Medicine is healthy, because it produces health in the animal.

XIII. This distinction is the fundamental maxim, by which the modern philosophers maintain their rejection of all the sensible qualities, which are assigned by Aristotle and his followers to objects; and they carry the matter so far, that they will positively assert to you, that neither snow is white, nor charcoal black, nor a bell sonorous, nor a flower fragrant; and if you mean to express by these denominations, intrinsic qualities, or such as proceed from some accidental intrinsic form contained in objects; they will not

allow such descriptions of things to be good and valid, but only so far as they signify certain determined impressions, which by means of the physical and corporeal impulse of the insensible particles of matter, are made on our organs, which are of the same use to us, to enable us to chuse the good, and avoid the pernicious, as our knowledge of the real intrinsic forms would be. They proceed to assert, that men would equally abstain from eating arsenic, if they believed the moderns, who say this mineral kills by dissolving the blood, by the rapid motion of its particles; as they would by believing Aristotle, who imputes all the mischief it occasions to be the effect of a venomous quality existing in the arsenic, and that they would be equally anxious to acquire gold, by believing the moderns, who say its brilliant yellow hue is the effect of a determined impression made by the light on the retina, where it appears reflected in that colour, from the particular texture of the insensible particles of the gold, as they would, by believing Aristotle, who says, the colour proceeds from an accidental form, inherent in the gold itself. I know very well, that a learned philosopher said, a little while ago, that the ladies had great reason to complain of Descartes, for having taken from their faces that delicate smooth whiteness which makes them look so tempting, and for having placed it in their admirer's eyes. But although this may do very well as a joke, it is certain, that the charm has an equal effect to make them appear desirable, whether it is stamped on the eyes, by the particular reflection, which, by the determined texture of the insensible particles of the skin, is made by the light, or whether it is produced by the intrinsic quality, which the Aristotelicans assign as the cause of colours; and I much doubt, whether to this day, the Cartesian philosophy has served any man as a preservative, against the venom of that sweet poison, which we call beauty.

SECT. IV.

XIV. The second reason assigned for distrusting the evidence of our senses, is the experience we have of the alterations which are produced on sensible species, either by the interposition of a medium, or the different dispositions of our organs. The species, which by passing through a uniform, or homogeneous medium, appears straight, by being passed through a different one, will seem crooked, as may be evinced by passing a straight wand from the air through water, which, by virtue of the refraction

of rays produced in its passage from one element to the other, will make it appear crooked. To him who has a jaundice, every thing he looks at seems yellow; and although it is true, that this is the effect of a preternatural accident, we are not certain, whether abstracted from all morbid dispositions, there are not in various individuals, different temperaments and configuration of parts, which are capable of producing different sensations with respect to the same object; and it seems most probable, from all that is evident and manifest to our observation and experience, that the thing is so, for we don't see any one individual, who is perfectly and exactly like another; and we have known men, whose right eyes have represented objects, both with respect to colour and magnitude, different from their left.

SECT. V.

XV. The third ground for the said distrust, is derived from the erroneous representation of things made by the imaginary faculty, which fancies external sensations of objects that do not exist. Such, for example, as the imagination of him who has had a leg cut off, representing to him a pain in the leg or foot that has been amputated; or that of an hypocondriac person, who fancies himself to be glass, earthen ware, a wolf, or a dog, whose imagination represents those strange forms, as evidently and manifestly palpable to his senses; so that he who fancies himself glass, will swear with invincible firmness, that he perceives in his own person the transparency, and feels on his own skin the smooth surface, which is proper and peculiar to that artificial composition.

XVI. This error is common to all men in their wanderings when they are asleep; because he who dreams, believes he perceives those objects with his senses, which he only perceives with his imagination; and from hence, the rigid sceptics deduce a most pernicious argument, by way of proving, that we should doubt of every thing; because they say no man can be positively certain, whether he is asleep or awake; according to which mode of reasoning, no one can be certain, whether he sees, hears, or touches any thing whatever; for notwithstanding his having the greatest confidence that he is awake, it is possible that he may be sleeping; and that the things which

are represented to him as seen, heard, &c. are all imaginary. For example, I consider myself as now writing, and reading what I write. But what certainty have I that I am writing and reading? Have not I dreamed a thousand times, that I was reading and writing? And at those times, such occupations appeared to me, not as things that I was dreaming of, but as exercises that I was really and actually practising; and therefore, that may be the case with me at present.

XVII. I have said already, and with reason, that this argument is a most pernicious and dangerous one; for whatever answer you make to it, your adversary turns upon you, and urges the opposite opinion as forcibly, as you can support your reply. At least I have never seen any solution of this matter, which could more or less vanquish the force of the difficulty. It has been objected to this sort of reasoning, that the argument proves too much, and tends to involve in, and reduce to the same state of doubt, all the sacred dogmas of religion. This remark is a just one, for he who comes to doubt, whether all he hears or sees is not a mere imaginary representation, must necessarily doubt of the reality of all the instruction he has received, both with respect to religion, and every other matter. But what advantage shall we gain, by insisting upon this against a sceptic, whose intention perhaps is to destroy all religion? And although he should not argue with this depraved view, but should reply to these objections out of wantonness, or from motives of vanity and ostentation, and a fondness of shewing his parts, it would be wasting time to dispute with him, or to press the argument, because these instances, notwithstanding they are good ones and may be fairly insisted on, are not answers.

XVIII. It is certain there are some truths, that maugre the clearness and confidence with which the understanding assents to them, are not exempt from being exposed to difficult objections; or to speak more properly, there is no truth whatever, be it ever so firmly established, against which, some sophistical objection may not be raised. It would therefore not be right upon all occasions, to give up a maxim, whose truth we clearly perceive and are convinced of, only because we can't reply to every argument that may be urged against it. There are truths of such a nature, that although they would strike, and be clear to any man of ordinary understanding, still, the giving

an answer to every objection that could be raised against them, might possibly be found a difficult task to a person of very subtil ingenuity. And notwithstanding we should not be able to hit upon an argument, that would confront or silence those which are used by the sceptics, to persuade us to doubt whether we are sleeping or waking, we should not abandon ourselves to such a doubt, but support ourselves in a firm confidence, that we are not mistaken with respect to the state we are in. And in truth, the arguments used to make us doubt of it, are not of such a nature, as that a clear, solid, and unembarrassed answer may not be given to them.

XIX. And in order to do this, I shall begin with supposing, that evidence in this case may be of two sorts, mediate, and immediate. An evident proposition, supported by immediate evidence, is, when without the assistance of any kind of proof, it presents itself so clearly to our understandings, that the mind is constrained by invincible necessity to assent to it. An evident proposition supported by mediate evidence, is, when of itself, it is not represented to our understandings with all this clearness, but is necessarily inferred from some other self-evident proposition.

XX. I will suppose, secondly, that immediate evidence should be divided, into metaphysical and experimental. The first is governed by universal principles, which of themselves persuade or convince the understanding; such as these, the whole is greater than a part; two contradictory propositions can't be both true at one and the same time. Experimental evidence is derived from certain singular truths, which strike every individual with infallible conviction; such as these, that I at present have such or such a desire, or that I think of such or such a thing, that I suffer some pain, or that I am affected with some particular sensation; for example, joy, sadness, or anger.

XXI. That every individual feels this experimental evidence in some particular instances, no one can doubt, and even if any person, by giving to his scepticism all imaginable extension, should be desirous of doubting of every thing, he could never stifle the experimental evidence that he doubted. And here I can't help remarking, that experimental evidence is a thing of such moment to the Carthesians, that all their metaphysical

testimonies depend on, or are derived from it; because, from that first maxim or proposition, I think, is inferred immediately, the proof of their own existence; and mediately, all the other demonstrable truths, are established upon experimental evidence only.

XXII. It is also certain, that from the truths which are established upon experimental evidence, you can deduce no demonstrative reasoning; or at least of that sort, which the logicians call *à priori*. The reason is, because they are evident of themselves, or else are supported by immediate evidence, and are not dependant upon any other thing from whence they are inferred. Therefore, although for example, I at present am convinced that I am desirous of eating such or such a thing, I cannot persuade any other person that I fancy it, by any immediate demonstration; because this is evident to me, not from any principle that is notorious to all mankind, and from whence may be inferred the existence of such a desire; but because the appetite itself, is intimately connected with my spirit, and I feel it so forcibly, that I cannot doubt of its existence: the same thing happens in the cases of those truths, which are established upon immediate metaphysical evidence. If I should be asked, how I know that the whole is greater than one of its parts, I should answer, that I do not know it from any principle from whence it may be inferred, but because the truth, *that the whole is larger than a part*, is so clearly evident to my mind, that it is as incompatible with doubt, as the light of the sun with midnight darkness. If any one should deny to me, the truth of the maxim, that two contradictory propositions cannot be both true at the same time, it would be impossible for me to prove the truth of the maxim, either by arguments deduced *à priori*, or *à posteriori*. The reason is plain, because if I was to argue the point, the most I could do, would be insisting on the manifest contradiction, and asserting the impossibility of the thing, which is the last resource of logical dialect. But observe, when we are got thus far, my adversary, to act consistently with his first caprice, thinks fit to change his ground, and to admit both extremes of the contradiction, and to insist they are both true. Now how, in this case, am I to prove that they cannot be so? Why I can do it by no other means, than by having recourse to the axiom, that it is impossible two contradictory propositions can be both true at the same time. But this is begging the question from the beginning, and proving what is denied, by the same proposition which is the subject of the dispute.

XXIII. The suppositions we have just made, contain all the solutions we are capable of giving, to the above argument. I say then, that I myself, and I say the same of all others, who are in my present situation and circumstances, have experimental Evidence that I am now awake: because the state of wakefulness, which consists, in the intimate and ultimate disposition of the faculties to exercise themselves in proper operations, is an object, which presents itself to my mind, with such clearness, that if I was never so desirous of doing it, I cannot entertain a doubt of my being at present in that state; nor can any other reason be required of me, nor can I give any other, for assenting to this truth; neither can I for my assent to a first principle, give any other than the beforementioned, nor alledge any other, to explain the existence of any reflection, which my soul is at present intent upon.

XXIV. I will not however dissemble, that after this answer is given, there still remains a serious difficulty to surmount, which may be stated in the following manner. The experimental evidence, from whence we deduce this persuasion, is fallacious; because when we sleep and dream, we are under the same persuasion that we are awake, and our senses then represent to us, that we are engaged in actual exercises in such a manner, that if it was at that time to occur to us, to make reflection upon what we were doing, we should conceive, that we had experimental evidence of our speaking, seeing, hearing, &c. Therefore the idea that I now entertain upon reflection, that I have at present experimental evidence of my being at this instant awake, and engaged in study and writing, affords me no absolute certainty that I am not asleep, or that I am actually so employed.

XXV. This is the utmost extent to which the matter of doubting can be carried. But by way of answer to it, I shall reply, that the persuasion we have of being awake when we are dreaming, differs greatly from that we have of being awake, when we actually are so. This last is a clear, firm, determined, invincible persuasion, and such a one, as is required to constitute experimental evidence of the certainty of the fact; which it does in such a manner, that notwithstanding all the reflections we can make, and all the doubts and difficulties we can suggest, our assent to, and persuasion of the truth of it, remains firm and unshaken. On the contrary, when we are

dreaming, our reflection is obscure, and the persuasion produced in consequence of it, feeble and wavering; which will evidently appear to be the case, if in the course of our dreaming, it should occur to us to make a doubtful reflection, whether what our imagination represents to us is reality, or only a dream; and the result to a man who was to do this would be, a faint and doubtful determination of whether he actually dreams, or only believes he is dreaming, and he would remain in a state of doubt, whether what his fancy represents to him be real or not. I say in all these cases, his decision of the matter, would not be resolute and firm, but wavering and languid. This doubt of whether I am asleep or not, often occurs to me in dreams, which never fails to produce one of the two following effects, either to certify to me that I am dreaming, or to make me suspend my assent to the fact. And I will venture to assure any one, who shall persist for a few moments, in proposing these doubts to himself when he is dreaming, that he will find the same thing happen to him.

XXVI. We may make use of the same reasoning, if the argument is applied to the deliriums of maniacs. Every man, who after having been deprived of his reason, comes to be restored to his senses, finds a great difference with respect to the persuasions he feels, and the clearness of the opinions he forms in a state of sanity, and those he entertained when he was mad. Mad people seldom make any reflections, either on the state of their mind, or the subject of their madness; and when they do happen to make them, are generally more or less blinded by their apprehensions; of which I myself have known some instances; and it has fallen in my way, by the force of lively representations, first to prevail on some mad people to doubt of the truth of their apprehensions, and afterwards to reason them out of them; among whom there was a nun, who had been exceedingly mad for some years, and whose life was thought to be in danger, although in reality it was not so; and I being called to administer the sacraments to her, was so happy as to reduce her mind to a state of calmness, that rendered her capable of receiving them as a penitent. This I accomplished, by using various reasonings and arguments, tending to undeceive her, and at last was so fortunate as to hit upon one, adapted to the nature of her disease, and the state of her mind, which had its effect: but in doing this, great attention should be had to the tone of the voice, and the energy and vivacity of the looks, and care should be taken, that every action should be such, as is

likely to give efficacy and force to the arguments you use, in order that they should make an impression on their minds; great caution should also be used not to irritate them; and by these means, they are often brought to intervals of sense; and although it is true, that they seldom last long, still the gaining an hour of calm reason, is a matter of great importance, as it was in the case of the nun we have just mentioned.

XXVII. The delicacy and curiosity of this subject, has caused me to dwell on it longer than was necessary; for I am so far from apprehending, that the arguments which are used in support of universal scepticism, will prevail with, or have any weight on the world in general, that I do not believe there ever yet was a man, who in reality was convinced by them.

SECT. VI.

XXVIII. The limitations with which rigid scepticism may be mitigated, are innumerable; consequently, scepticism will appear more or less absurd, according to the various exceptions with which it is corrected; but this is a matter of such extent, that to reason upon it with any degree of precision would fill a large volume. I shall therefore proceed to treat of scepticism, as confined within the line of physics, which is the point, to which I proposed at first to restrain my arguments on this subject.

SECT. VII.

XXIX. I have always wondered, and do not yet cease to wonder, at seeing modern philosophers attack scepticism as a physical error, and am much astonished, to find them condemn it in that sense as an error, dangerous to the dogmas of faith. Nor can I comprehend, how such a charge can arise, otherwise, than from gross ignorance or malicious spite; unless it is applied to some sceptic, who from explaining himself loosely, may have given occasion for such an imputation.

XXX. The system of physical scepticism, asserts, that physical and natural things, do not admit of demonstration or scientific certainty, but that they are all matter of opinion; and consequently, that we should not call

natural philosophy a science, because strictly speaking it is not one, but rather a habit, or acquired facility, of reasoning with probability upon natural things. We shall understand science in this place, in the sense which Aristotle and the schoolmen understand it, who define it, *to be an evident knowledge of the effect by the cause*. But in our definitions, we shall not exclude experimental certainty, or certain knowledge, acquired by the experience we have had, and the observations we have made, of physical matters; but we shall rather maintain it, as the only method by which we can arrive at attaining truth; although I doubt whether we shall ever attain by it, a discovery of the internal, or intimate nature of things.

XXXI. Neither will we deny, that with respect to physical objects, many propositions may be advanced or supported, which are deducible with infallible certainty from metaphysical principles: as for example from this, *the whole is greater than its part*, is evidently inferred, that a man is bigger than his head; and from this, *being is the effect of operation*, is inferred, that my father existed when he begot me. But these, and other innumerable demonstrations of this sort, afford no physical knowledge whatever; because they do not manifest in any degree, either great or small, the nature of the very entities which are the objects of them. Why do I say manifest the nature of the entities? They do not even display to the understanding, a single truth, which may not be comprehended without their help, by the most rustic man upon earth. So that the syllogistical conclusions upon infallible truths, which the school philosophers so much boast of, do nothing more, than explain by circumlocution and in terms of art, the very things, which may be immediately comprehended, and naturally explained, by any rational man who never studied. Further, how can they call by the name of demonstrations, things which demonstrate nothing? that is, they manifest nothing, but what was manifest without their assistance. A logician will say, thinking he says something to the purpose, that he knows by means of the art of demonstration, that which he could not know artificially without this aid. But I answer, that this artificial knowledge is totally useless, because it neither manifests to me any one truth of which I was before ignorant, nor does it illustrate to me with greater clearness, the things I was before acquainted with; it being certain, that it is as evident and clear to the rustic, and that he assents with as much firmness, that the whole tree is bigger than one of its branches, without the help of artificial logic, as I

do, assisted with my whole armory of syllogisms. If a learned person, should undertake to instruct a man who walks well, and with a good grace, by explaining to him all the laws of motion, to step scientifically; and also, by explaining to him the number and use of the muscles, should teach him the application of those laws to every member of his body concerned in that exercise; should not we be apt to say, that besides his taking a very tedious one, he had also taken a very needless trouble, it being certain, that his pupil, after all this pains, would not walk a jot better, if so well as he did before; the cases are similar.

SECT. VIII.

XXXII. The thing then being understood, in the sense we have explained it, the conclusion I draw from the premises is, that science, or scientific certainty is not to be found in physics. Doctor Martinez, in the second volume of his *Medicina Sceptica*, (Convers. 27.) proves this conclusion abundantly, both by the authority of scripture, and by the sentences of many fathers; though as the works of this author are easily to be met with, I shall not here insert the authorities he quotes, but shall content myself with adding to them two very remarkable ones that he has omitted. The first is, that of my father St. Bernard (in Cant. Cantic. Sermon. XXXIII.), who when he is speaking of philosophers, says, *Vagi sunt, nulla stabiles certitudine veritatis, semper discentes, et nunquam ad scientiam veritatis pervenientes*. And it is somewhat remarkable, that the Saint says further, philosophers never arrive at attaining a certain knowledge of the very truths they themselves are in search of, and are desirous of teaching to other people: *semper discentes*; and lest some should conclude he spoke of supernatural truths, I thought it necessary to hint here, that these are not the objects of philosophical inquiries. Neither should we suppose, that he spoke of the moral philosophers, because many of these, even including the Gentiles, investigated many truths, that came within the compass of their plan, with intire certainty; and it is evident, that if Aristotle, had written with as much precision upon physics, as he did upon ethics, we needed not to have wished for any thing better.

XXXIII. The second authority, is that of Lactantius Firmianus, an illustrious person, and a venerable member of the church; this great man (Lib. III. Divin. Instit. Cap. iv. v. and vi.) treats largely of the scepticism of Archesilaus, of whom we have spoken before, and after combating and confronting this philosopher effectually on the subject of universal doubt, admits without reserve, that he would have been right, if he had limited his scepticism to matters of physics, because there is not, nor ever can be any science, which will teach mankind, the causes and reasons of natural things: *Quanto faceret sapientius, ac verius, si exceptione facta diceret causas, rationesque duntaxat rerum cælestium, seu naturalium, quia sunt abditæ, nec sciri posse, quia nullus doceat, nec quæri oportere, quia inveniri quærendo non possunt.*

XXXIV. Some of the sceptics themselves prove our conclusion, for they say, that the things appertaining to physics are singular, and that from singular things you cannot derive a science. But this reason does not satisfy me. In the first place, notwithstanding physical things are for the most part singular, they in some instances may cease to be singular. Thus, although every real entity is singular, it may, contemplated metaphysically, depart from its singularity. In fact, the schoolmen together with St. Thomas say, that physics may depart or be separated from their singular, but not from their sensible nature; and that mathematical things, may depart or be separated from their singular and sensible, but not from their intelligible nature; also, that metaphysical ones, may depart or be separated from their singular, sensible, and intelligible. In the next place, the axiom that from singular things you can derive no science, should be understood with some grains of allowance; that is, of such singular things as are particularly necessary to the individual, and are accidental to the species; but from those which appertain or belong to the species, science may be deduced, even when they are connected with the individual. For example. If I know scientifically; that man according to the general received opinion respecting him, is risible, I also know scientifically that Peter is risible, for in the following syllogism; *every man is risible; Peter is a man; and therefore Peter is risible*, is implied the truth of the premises, and the consequence deducible from it is, that it is scientifically evident. Further, if ever a philosopher should arise, who should know with certainty, the specific nature of all material entities, and from that knowledge, should be able to

demonstrate all their properties, and the respective operations appertaining to each species, and in virtue of this extraordinary penetration, should be capable of giving reasons *à priori* for all the phenomena of nature, we should not be able to deny, that such a philosopher possessed physical science, notwithstanding the object of his science, related only to the species, unconnected with individuals. What then remains to be proved, is, that in physics there is to be found no sort of science, or evident knowledge of the matters appertaining merely to physics, abstracted from all other considerations; and truly the dogmatic naturalists, would be very happy, could they be permitted to possess a knowledge that amounts to as much as this; nor would it disturb them, to have it dinned in their ears, that the knowledge of common conceptions is metaphysical, and not physical; for they would tell you, that both physics, and metaphysics, may depart or be separated from their singularity, and that the principal distinction between them is, that the last regards its object with a greater degree of abstraction; that is, as exempt from all matter, and considers only those reasons, which may subsist abstracted from matter, such for example as spiritual entities; on the contrary, physics only contemplate material and corporeal entities; the most sublime of which conceptions, is that which regards the reason of the body, and the lowest, that which is attached to the corporeal part of it. But call this knowledge by what sort of name you will, either physical or metaphysical, the way to bring the matter to a short issue is, to shew that there is not in reality, any such knowledge.

XXXV. And what is more easy to be proved than this? I reason thus; physics contemplate the nature of an entity that has motion, or circulation; this may be considered, either according to the specific, or generative nature of the thing; and I assert, that nothing is known with certainty of the nature of any entity that has motion or circulation, either in one or the other of the before named senses.

XXXVI. And to begin with the specific, who can deny that in this sense, we know nothing of any one entity? I defy all the philosophers to tell me what is the constituent or component physical substance, of any one of the material entities that is to be found in the universe, and will give them leave to chuse that which they have most examined. St. Basil has some words

(Epist. 168. *ad Eunomium*) that are admirably applicable to this purpose, and which I shall here insert: *Itaque qui se existentium scientiam assequutum esse gloriatur, exponat nobis quomodo, quod minimum esse eorum, quæ in lucem prodierunt, natura habeat.* Let the presumptuous philosopher, who boasts of his physical knowledge, explain to us the nature of the least entity which God has created. Let him, adds the same Father, who is ostentatiously vain of having penetrated into the secrets of nature, tell us what is the nature of an ant: *Dicat formicarum nobis naturam, qui eorum, quæ in natura sunt scientiam cum fastu se prædicat assequutum.* But why should we waste time? There is not, nor ever was to this day, any one, who by means of acquired science, could penetrate the constituent or component physical substance of any living or inanimate entity; the reach of our capacities not permitting us to go further, than to distinguish by some very extrinsic accidents, one thing from another; and this mode of distinguishing, is confined principally to those we call naturalists, and rarely extends to those in the schools, who acquire the appellation of philosophers, and who generally content themselves with distinguishing some few, which they commonly do in the unhappy manner we shall proceed to point out; but when school philosophers descend to treat of specific conceptions, their philosophy is so miserable and confined, that they only attempt to give the colour of a definition to a few species of brutes, whose voice they describe by some particular name, and then explain their conception of them, by some term derived from that name; thus they call the lion a roaring animal; the dog a barking one, and the horse a neighing one; but the fish, because they are very numerous, and are for the most part mute, cannot well be comprehended within this rule, and therefore are obliged to go without a definition.

XXXVII. The school philosophers cannot be ignorant, that these are not definitions, but should rather be termed mere shadows of definitions, which they make use of for want of true ones, to express logically, what definition is, what species, what gender, what difference, and other things appertaining to logical dialect; but is it not clear, that defining a horse by calling him a neighing animal, can give us no better conception of the creature, than may be conveyed to us by the most stupid clown, who would explain the thing more simply, and without school jargon, by telling us that

a horse *is an animal that neighs, or is capable of neighing*? Oh! what a penetrating philosophical description is this of the nature of a horse!

XXXVIII. If any one, notwithstanding what has been said, should chuse to reply, that nature, as the root of all operations, should be explained according to the order, or habit of those operations; and thus, that a horse is well and physically defined, by the radical order of the act of neighing; if any one, I say, should reply to me in this manner; I apprise him in the first place, that all substantial nature has its absolute being, which is understood to be antecedent to the order of operations, for the first, is supposed to be the cause or reason of the other; that is to say, because such a thing has such a being in nature, from thence is derived the habit or aptness for such operations. I apprise him secondly, that if we should permit nature to be well defined by the precise or characteristic order of operation, this should not be extended to every sort of operation, but should be confined to the primary operation, which marks the foundation of the species; but alas, we are ignorant of what that is. For example, if a man, as is generally thought, is well defined by his rational faculty, or radical power of reasoning; because the being able to reason or reflect, is the principal, or primary operation of man; a horse then should be defined by his radical habit to that act of perception, instinct, or knowledge, which is proper to his species, and distinct from that of all other animals. But who has penetrated what this is? Or who has ever known the innate difference, that there is between the instinct of a horse and a dog? And thus, as it would be ridiculous and absurd to define a man by the radical order of elocution, by saying, that he is an animal who can talk and discourse; because this absurdity would be incurred, on account of the act of elocution being posterior, or secondary to that of reason or reflexion; and it would be much more ridiculous, to define him by the order of his voice, which you describe by some particular name, and to define him in the manner you define a horse, which you call a neighing animal; nor is it less absurd to define a horse by the radical order of his neighing. I apprise him thirdly, that if such definitions are admitted as good and legitimate, it is the most easy thing in the world, to define every substantial entity whatever, because in order to do it, you have only to observe its operations, give the most remarkable one a name, and define it by that name. By the help of this instruction only, you may make every

peasant a consummate philosopher, and enable him to define the nature of all the entities in the universe.

SECT. IX.

XXXIX. These reflexions answer no other purpose, than that of confronting here and there a superficial and bastard schoolman, for all capable people know and confess, that we are unable to give a definition of any one substantial entity, except man. To what straight limits then is our philosophy confined!

XL. But the misfortune is, that we have no certainty, that the general received definition of a man, to wit, that he is a *rational animal*, is a good and unexceptionable one; for we are clear it would not be a good one, if this faculty appertains to other animals as well as him, and it is matter of doubt whether it does or not; but I will not, nor cannot, in support of this doubt, avail myself of the authority of Porphyry, who in his Treatise on Predicables, supposes God to be a rational animal; and in order to distinguish between God and man, defines man to be a mortal rational animal, because he thought that without the addition of the word mortal, the definition would be applicable to God as well as man. Neither will I avail myself of the authority of Aristotle, from whose second book, De Sect. Pythagora, Jamblicus cites these words: *Animalis rationalis aliud quidem est Deus, aliud autem homo*. But I may for this purpose, avail myself of the authority of some Fathers, among whom is St. Austin, who all affirm, that the angels are corporeal, or at least have doubted of their incorporeality; to which doubt, appertains that, of whether angels are rational animals, for in order to suppose them so, there wants nothing but their being corporeal, and consequently it is doubtful, whether the definition of rational animal, appertains solely to man.

XLI. I may be told, that the corporeality of Angels has been condemned, and their incorporeality defined, in the second Council of Nice, and the fourth Lateran one. But in the first place, is it certain, that these Councils declared the incorporeality of angels to be undeniable, and that they declared affirming the contrary is erroneous; for it is somewhat doubtful,

whether the incorporeality was defined in them or not; because, although mention was made in these Councils of this matter, it was not done with an express design to discuss the point, but was only touched upon, as we may say by *incidence*; which is an exception that all eminent theologians admit, to our being bound to observe things that are canvassed in that way even in Councils; as we are not obliged to suppose any matters defined in them, that are not debated and settled with an express intention. For which reason, the most learned Cano (*lib. 5. de Considerat.*) took the liberty of saying, that the opinion that angels were corporeal, although it might be false, was not heretical; and long before him, St. Thomas in his book *de Malo*, (*quæst. 16. art. 1.*) had said, that this question was no part of the Catholic Dogmas. My father Saint Bernard (*lib. 5. de Considerat.*) goes further; for he seems to think, the opinion which denies the corporeality of angels, to be void of probability; and it will not be improper to remark here, that he was much posterior to the second Council of Nice, and Saint Thomas, not only to the second Council of Nice, but to the fourth Lateran one also. This may likewise be alledged as an answer, to the objections that are made in consequence of some texts of Scripture, which give the epithet of spirits to the angels; for it is certain, that the fathers who considered it as defensible to maintain the corporeality of angels, were not ignorant of those texts; the exposition of which, in truth is not difficult, for we may fairly conclude, the Scripture calls them by this name, on account of their being airy or light bodies, and we may also suppose, that it is for this reason, that in various places it gives the name of spirit to the air. *Spiritus procellarum. Advenientis spiritus vehementis, &c.*

XLII. The second reply I shall make, is, that supposing it to be determined that the angels are incorporeal, this truth is not established by philosophy, but by faith; and as upon the knowledge of this truth, depends the certainty, that the definition of a rational animal does not belong to an angel; it follows, that by the help of philosophy alone, we should never have hit upon giving a definition of a man; and with respect to defining other things besides a man, its insufficiency is admitted. What sort of philosophy is this? It ought rather to be called a total want of philosophy.

XLIII. Not only with regard to the angels, but on the part of brutes also, we find motives for doubting, whether the definition of *rational animal*, may not be applied to them as well as to man. If *rational animal* signifies an animal capable of reflexion, the brutes are rational animals in the opinion of all those, who suppose them to be endowed with reason and reflexion: and as this sentiment may be supported by strong arguments, it remains somewhat doubtful, whether rationality is not diffused in different proportions to other animals as well as men, or whether it is confined to them only. It is true, that if this opinion should be admitted, we should grant, that the reason of man is distinct from, and of a superior nature to that of brutes; but if in the definition, we do not point out the character which distinguishes them, we are reduced to assign as the difference, a generical idea.

SECT. X.

XLIV. As we ascend the predicamental tree, and advance from the species, to the genuses or kinds; we perceive, that philosophy does not discern these more clearly, than it did the others; for we find in both cases, equal ignorance, and equal uncertainty. If we ought to have scientific certainty of any one genus in preference to another, it should be of that under which we ourselves are comprehended, which is the animal kind, for that is the thing which most immediately relates to us, and because also, we employ more consideration and attention upon that, than we do upon the others. We call the order animal, which comprehends man, and every other species of terrestrial and aquatic brutes, as likewise every kind of fish or fowl. And what do we know of animals according to this general description? Why that they are *sensitive living beings*; for that is the definition we give of them. But do we know this with any degree of certainty? By no means; for it has been doubted whether every animal is sensitive; and it has also been doubted, whether the sense of feeling does not belong to other entities as well as to animals.

XV. The first doubt is built upon the opposition and arguments of the Cartesians, who pretend, that all brutes are inanimate machines, and that there is no other sensible animal but man; for which reason, in their

opinion, a sensible being is not a generical entity, but a specific one, and proper to the human species in a quarter proportion. I am thoroughly persuaded, that this opinion of the Cartesians is a false one; but I have not yet seen any evident argument or demonstration to confront it; nor has any person hitherto, hit upon one that evinces the falsity of it; for their principal foundation is not so weak, but that it has puzzled the most able Aristotelites to give an answer to it. But I do not find, that this is any obstacle to our giving our firm assent to the sensibility of brutes; although we have no reason to glory in our proofs of this matter, when the contrary opinion, besides the arguments that are produced to support it, finds so many partizans; and among them some of excellent ingenuity. Nor have we any reason to suppose, what I have known many people maintain, that all the Cartesians, think differently from what they speak in this matter; for some of them, are as full of the caprice that brutes are insensible, as we are of the persuasion that the contrary is the fact. A few years ago, certain ladies who were present at a bull-feast, seemed to express great concern for the sufferings of one of the bulls, which the people employed for that purpose, worried excessively. A French lady, who was a Cartesian philosopher, that sat near them, begged they would not grieve, for says the good Cartesian with great gravity, *I believe as firmly as I do in God and this cross, that the bull feels no more pain than the bench I sit on.* I do not know whether the other ladies believed her, but I am persuaded, that many are stedfastly of the same opinion with the French lady.

XLVI. The second doubt, whether other entities besides animals are not sensible, is patronized by Campanella, who labours by a variety of arguments, to prove in divers parts of his Works, that all elemental things are sensible. This doubt is also supported, and with more colour of reason, by those philosophers, who allow feeling to plants. And that this opinion may not appear extravagant; for the information of those who adopt the common sentiment, it will not be amiss to let them know, that Aristotle did not look upon this thing in that light, for he rather inclines to patronize the doubt; for in his first book *de Plantis*, he says, there is no certainty whether plants are, or are not, endued with feeling, appetite, and knowledge: *Nec enim constat, habeant ne plantæ animam, appetendique facultatem, doloris item, & voluptatis, & rerum discretionis.* In the third place, the naturalists, who build upon experimental observations, attribute feeling to some

determined species of plants, which on that account, they call sensitive plants.

SECT. XI.

XLVII. If of our own proper genus, we know nothing with certainty, what must our knowledge be of others? The genus most immediately next to our own, is that of plants, and in this, notwithstanding our proximity, we can discern nothing but our ignorance; for we cannot venture even by conjectures, to point out the constitutive difference between them. This is not only invisible to the eyes of evidence, but impalpable to the essays of opinion. We commonly define the genus of a plant, by the term insensible living entity. But the word insensible, which we use to describe the difference, only signifies want of sensibility; and a positive entity, such as a plant is, cannot be ascertained by a negative. Besides this, as we observed before, it is somewhat doubtful, whether plants are, or are not sensible. We call them also *living vegetables*. But by this idea, we do not assign to the plant a different nature from the animal, for that this also is a living vegetable. If I am told, that the difference consists, in that the life of the animal is vegetative and sensitive, and that that of the plant is purely vegetative, I reply, that the adverb *purely* in this place, signifies nothing but the want of sensitive life, which is meant to be expressed in the extreme, and a want, cannot be made use of as a descriptive term, to point out the constitutive difference between positive entities. Nor will it be of any use to answer, that the term want, means a want or defect in the mode of explanation, and does not apply to the thing signified; for till it can be ascertained what the thing signified is, we must remain totally in the dark; and it is also false, that this term *want*, is not applicable to the thing signified; for negative expressions are positive with respect to the thing signified, when they are used to deny any imperfection in the object they relate to; for the want of an imperfection, is the want of a want, it being certain, that all imperfection consists in the want of positive perfection; for which reasons these words, infinity, immensity, indivisibility, although negative with respect to the mode of expression, are positive with respect to the thing signified; but the term insensible, or insensibility, applied to a plant, signifies want of perfection, and thus is negative with respect to the thing signified.

XLVIII. Besides this, some express a doubt, whether plants are vegetable or not; and some are also doubtful, whether the property of vegetation does not likewise belong to stones and metals. If we consult the Cartesians upon this point, they will tell us, that all we call vegetation or nutrition of plants, is pure mechanism, and that the attraction of the nutritious juice which we attribute to them, is a solemn chimera. If we forsake the Cartesians, and apply to the experimental philosophers, we shall find many of them, who will tell us, that metals and stones grow by means of vegetation; which is an opinion, that not long ago, was illustrated by Joseph Pitton de Tournfort, a celebrated naturalist of the Academy Royal of Sciences at Paris, and more especially, by the celebrated observations he made upon marble in the wonderful cave of Antiparos; and with respect to metals, we, to the authority of others, shall add that of Aristotle, who in his book *de Mirabilibus Auscultationibus*, says, that in a territory of the island of Cyprus, they were used to sow iron, and that it grew and vegetated like other plants.

XLIX. And now we are speaking of Aristotle, we will not omit mentioning an authority of his, which is very applicable to our present subject, because it is totally opposite to a received opinion in the schools, to wit, that plants and animals are of two different genuses, and that the distinction between them is, that the animal is a living sensible entity, and the plant a living insensible one. Aristotle in his book *de Plantis* says, that oysters and other shell-fish, are both plants and animals: *scimus autem, quod conchyla animalia sunt cognitione carentia: quapropter plantæ sunt, & animalia*. I ask now, how one species can be classed under two diametrically opposite genuses? and how an oyster can be both sensible and insensible? for as an animal, it ought to be a living sensible entity, and as a plant, a living insensible one. Nor can it be said, that Aristotle, when he called an oyster a plant, spoke in a metaphorical sense, for this mode of speaking is not allowed to philosophers, but is only permitted to poets and orators. Besides, the cause he assigns, shews, that he spoke in a rigorous philosophical sense; although, to speak the truth, I do not comprehend who could reveal to Aristotle, that oysters and other shell-fish are destitute of that sort of knowledge, which is proper and natural to the most stupid brutes.

SECT. XII.

L. From the lowest species, we proceed to the next above, and enquire, what is the nature of a living entity; what we are to understand by the term living, and what is life? To this, we shall be answered from the schools, that life is motion *ab intrinsico*, and that living, means that which moves *ab intrinsico*; that is, its motion is produced by some intrinsic faculty, or virtue, which it has within itself.

LI. This definition is incumbered with greater difficulties than the antecedent ones. All the modern philosophers oppose it, although for distinct and contrary reasons. Gassendo, father Maignan, and the other atomists, attribute motion *ab intrinsico* to atoms, from which dogma it will follow, that motion *ab intrinsico*, is not distinctly peculiar to living animals. The Cartesians are very positive in maintaining, that nothing can give motion to itself, but that all motion in the universe, proceeds from that impulse, which God in the beginning gave to matter, which does, and ever will subsist, without being impaired or injured; and that by virtue of that impulse, motion is communicated from one part of matter to another; so that all which was before at rest, upon its beginning to move, received its impulse from some other body which was in motion before, and transferred its influence to that which was then at rest; and either partly, or in the whole, gave it the power of beginning to move; and they say further, that man, which is the only living corporeal entity they admit, when he moves, cannot be properly said to give motion to his members, but only directs their motion by his will, in virtue of the power antecedently impressed by the impulse of other bodies on the animal spirits.

LII. It cannot be denied, that this doctrine is terribly fortified, by the celebrated maxim of Aristotle, which says, *that every thing that moves, is set in motion by something else*: and although the followers of the common opinion, explain this maxim, so as to render it not incompatible with the definition they give of living entities, there results from this explanation, the inconvenience, that it weakens the principal force of the axiom which they avail themselves of to prove the existence of a first mover, to stimulate a body at rest; for if we suppose, that a living entity has the power of moving

itself, we cannot maintain the necessity of a divine influence to produce this motion, without admitting at the same time, the existence of a first mover in other respects, to be established. Thus it seems to me, that the Cartesians can with some colour of reason pretend, that religion is interested in understanding the axiom in all the rigour, which they profess to understand it themselves.

LIII. But be it as it may, with respect to this difficulty, and all others, which the moderns by virtue of their principles can oppose to the doctrine of Aristotle; it is certain, that there may be very serious ones urged against their definition of living entities. Heavy bodies move *ab intrinsico*, and are not living entities. Fire moves *ab intrinsico*, and is not a living entity. The fermentative motion also, according to common physics, is understood to be *ab intrinsico*. And I have remarked and proved in another part of this work, that what the Aristotelites say of heavy bodies being moved by communicated impulse, in the form this explanation is intelligible, is also applicable to the motion of living entities.

SECT. XIII.

LIV. There now only remains within the compass of the predicamental scale, one other thing to consider, which is the highest, and most sublime within the reach of physics, and that is, the nature and composition of the body; but where will this lead us to, the point being equally doubtful with all the rest? The composition of the body, is divided into elemental and mixed; but as this last is composed of the other, unless we know what the elemental is, it is impossible to know of what the mixture consists. Very well; but who knows the nature and number of the elements? To this question, methinks I hear four answers, from four sects of philosophers, which are all different, although each pretends to exclude the others from a competent knowledge of the matter. The Aristotelites say, that they are air, fire, earth, and water. The chymists, that they are salt, sulphur, mercury, earth, and water. The Cartesians insist, that they are composed of the subtil, the globulous, and another matter which is more gross, and which they call the third element; and the atomists, that they consist of the atoms. These are the opinions, which are at present the most prevailing, although there are

numbers of others which I shall omit to mention, because they have but few votaries. Now which of these opinions is the true one? Perhaps neither. At least there is but one sect out of the four that admits either of them to be true, and whichever that sect adheres to, the other three assert to be false; which amounts to the same, as saying that one witness deposes to the truth of the matter in question, and that three give testimony against it. Thus no judge to whom the decision is confided, can give sentence in favour of either of the parties; it being out of his power to affirm, that any one of their systems is right and true.

LV. As the audience before whom I propound this reflection, is composed chiefly of the followers of Aristotle, methinks I hear them exclaim, that if we were to collect the suffrages of the academic professors, we should find most votes in favour of his opinion. To this, I reply in the first place, that although a plurality of sectaries, gives greater extrinsic probability to an opinion, still it does not afford certainty, nor intrinsic probability; and the question here, is not whether Aristotle's opinion is the most probable, but whether it is certain. I reply secondly, that it is doubtful, if we were to include the professors who teach physics in all nations, whether we should in reality find a majority of votes in favour of Aristotle; and unless we could prove by some authentic instrument, that God has limited the knowledge of philosophy to our nation only, and has excluded all others from inheriting the science, I do not know upon what pretension we can ground a right, of admitting none but Spaniards to vote in the question. Some of our antient professors say, that we should not regard the opinions of strangers, because they are novelists; but at the same time, strangers reply, that we should take no account of what the Spaniards assert, because they are obstinate and wrong-headed, and that there is no proof, be it ever so clear, that will make them depart from their old opinions; to which they add, that in Spain, they do not follow Aristotle from choice, but from necessity; for that it requires an heroic spirit, to contradict any thing that has been advanced by Aristotle in that kingdom, where whoever dares venture to oppose him, is instantly assailed by tempests of injuries, which fall on his head like violent hail storms. Nor is even an heroic spirit, sufficient to enable a man to support such an opposition, because the duty of obedience, obliges people not to depart from the line of their school; and this is also frequently the case, in other countries; and from hence, some Anti-

Aristotelites have insisted, that the plurality of votes in favour of Aristotle should not prevail, because they cannot be supposed to be free.

LVI. But abstracted from the quantum of extrinsic probability, contained in the doctrine of Aristotle respecting the elements; I assert, that upon a strict examination, you will not find more truth contained in his sentiments, than in those of his adversaries. This opinion, I deduce from the position, that his first qualities consist of four, to wit, heat, cold, wet, and dry, from which in exact proportion, he attributes one to be impregnated in the highest degree with each element, and another near to the highest. This proof, goes lame in innumerable particulars. First, the giving to these qualities the title of first, is conferring a mere voluntary dignity on them, and especially, when we know the invincible difficulty, that attends establishing, that all the others result from them. Secondly, it is very doubtful, whether all the four beforementioned, can properly be called qualities; for many Aristotelites, and with great colour of reason, deny humidity and dryness to be such. That which is humid, is not so from any quality it possesses in itself, but because it has imbibed into its pores some foreign liquid substance, and let that be evaporated, and it will remain dry; from which we may conclude, that humidity is a substance, and that dryness consists in the want of that substance. Thirdly, the appropriating these as two of the four elements, is done without any foundation or reason. In the next place, how can they prove that water is cold in the highest degree? If this was true it would kill us; neither can they prove it so, in a more moderate, or secondary degree, for we are taught by experience, it is indifferent to both cold and heat, according to the agent that is applied to it. It is heated by the fire; and remove it from the fire, and it becomes cool again, not because it has any propensity or natural disposition to be cold, but because it is made so, by the cool atmosphere which surrounds it. There are many other very grave objections to be urged against this doctrine of the four qualities; and therefore the foundation upon which it is built, and the reasoning deduced from it to establish the quaternian system of the elements, are both very slight and futile.

LVII. The second argument, is built on the correspondence of the four humours of the body, with the four Aristotelic elements; that is to say, the

blood with the air, the choler with the fire, the melancholy with the earth, and the pituitous with the water. But this puts us in a worse situation than we were before. In the first place, it is a doubt among the medical people, whether the humours of our bodies consist precisely of four. Some say they are more, and others that they are less. Some add to the four, the lymph, the pancreatic juice, and the nervous fluid; and some again will not admit, that there is any other humour except the blood. Secondly, if the four elements correspond only with the four humours, we are left without an element to correspond with the solids, which on account of the hardness of their substance, might with more propriety be compared to the earth, than to the melancholy humour, which is beyond comparison less hard and solid than the other. Thirdly, in the same voluntary manner, with which they assign four elements to correspond with the four humours, they may assign one element to correspond with the flesh, another with the bones, another with the marrow, another with the fat, or adipose substance, another with the tendons, and so on. Fourthly, in order to reason justly upon the human body, or animal species, we should not seek for four substances that are analogous to the four humours, but for four that enter into all the mixtures; for the question, relates to elements that partake of the composition of all mixtures in general, and not precisely of the animal. But what traces can we discern, of four humours, or four substances, equivalent to these, either in minerals, or plants?

LVIII. The third argument in support of this quaternian system, is pretended to be derived from experience; for it is alledged, that when a piece of wood is burning, we see it resolve itself into the four Aristotelic elements. At first it emits a small portion of water; then it takes fire; the fire is followed by smoke, which we know to be of an aerial nature, by its ascending to the airy region; and last of all, there remains a portion of earth in the ashes.

LIX. Although as Etmuller says, in matters of physics and medicine, *præstat unum experimentum centum rationibus*, still, the experiment just alledged is so defective, that it is not of more value than the arguments before urged. In the first place, dry wood, may as properly be called a mixture as green wood, notwithstanding which, it emits no water upon

being laid on the fire. Secondly, as we are here treating of the elements in general which enter into, and make a part of the composition of every species of mixture, the fire should produce the same appearances, and have the same effect upon them, that it has upon the wood; but this does not happen, for minerals laid on the fire emit no water, unless it is when they have imbibed some foreign moisture. Thirdly, the chymists by means of fire differently applied, extract from wood and other mixtures, various substances, which differ from those four that are produced from the wood in the ordinary way of burning it; and it seems reasonable from hence, that we should augment the number of the elements. Fourthly, we do not know whether these substances pre-existed in the wood, or whether they were produced *de novo* by the fire. Fifthly, the ash is not earth, nor an elemental or simple body as is supposed, for there may be separated from it a large portion of salt, which is a distinct substance from either of the four; for it is neither earth, air, water, or fire. Sixthly, neither is the smoke air, as is manifest from the soot which condenses in the chimney. If I should be told, that in the smoke is contained a variety of particles, some of which compose the soot, which remains in the chimney, and others which mount higher, and are impregnated with the atmosphere, and become air; I reply, that in consequence of this, they should admit of a fifth element of soot; or to speak more properly, of five or six additional elements; for Mr. Boyle informs us, that the chymists can extract from soot, five or six different substances. Finally, all that becomes ashes, existed before in the form of fire; from whence it follows, that the form of ashes is a new production, for matter cannot exist under two substantial forms at one and the same time; and consequently, the elemental form of earth which the Aristotelites ascribe to the ashes, could not pre-exist in the mixture, but must be a new production. This objection militates principally against the Aristotelic principles; but others might be formed in different shapes, against every system whatever.

LX. I have combated only the Aristotelic opinion of the elements, not because the others are not incumbered with equal difficulties, and exposed to equal objections, but because in Spain, we suppose the others to be perplexing, and even improbable, and conclude that of the four elements, to be just and right; I therefore chose to attack this system, to let my

countrymen see, that we know nothing with certainty respecting the elements.

SECT. XIV.

LXI. I have remarked before, that if we are ignorant of what the elemental bodies consist, we cannot possibly know the nature of the mixtures. But even if we could ascertain what the elements are, we should still remain in profound philosophical ignorance with respect to the component parts, of both them and the mixtures; for admitting the four following, air, fire, water, and earth, to be elements of all the mixtures; who has yet ascertained the nature of those four bodies? Aristotle only reasoned upon their qualities, and this he did with so little precision, that all he said may be looked upon as doubtful, for he had no solid principle, from whence he inferred, that they possessed the properties which he attributed to them; but drew all his conclusions, from an ideal proportion which struck his own imagination; and respecting which, it has since been proved, that he was mistaken. He says that air is hot, and fire dry in the highest degree; but in our essay on physical paradoxes, we have proved that the air is not hot; and according to Aristotle's definition of humidity, we may infer, that a flame is humid, for it cannot be contained within its own bounds, but pervades other regions. We also proved in our essay on physical paradoxes, that elemental fire is not hot in the highest degree; and to what I said there, I will add in this place, that experience shews us, one fire is hotter than another, and has greater power to heat or melt substances, either in consequence of its greater bulk, or on account of the matter with which it is made, or the manner the place is disposed and contrived, in which it is lighted; from all which it may be inferred, that fire in its nature is not hot in the highest degree, for if it was, as in every fire the nature of fire is preserved, every fire would be hot in the highest degree, and of course, could not be exceeded in heat by another fire.

LXII. Aristotle however, did no more than assign false or uncertain qualities to his four elements, and left untouched, substantial nature which is the root of them; and those who succeeded him in all after-ages, if they aimed at more, attained no more. The sectaries of Aristotle content

themselves with saying of the elements, what they say of all other natural compositions; that is, that they are constituted of matter in physical forms, are real incomplete entities, and evidently distinct from each other. By all which, admitting this description to be right, they teach us nothing, till they explain to us, of what the physical form consists, and what is the specific nature of the physical form, in every natural composition. But their system explained in the general way, in which they themselves define it, is strongly combated by the modern philosophers, who find an insurmountable difficulty in the generation of the material forms, not being able to comprehend, how their production can be any thing else but a new creation; for the arguments the Aristotelites fly to, of deducing them from the power of the matter, contain nothing but words, void of all real signification; and truly Aristotle himself having said, that the form is one of the principles of a natural entity, and that the principles are those things, which are neither made from their own substance, nor from that of any other entity whatever, shew them to be words which have no fixed meaning: *Quæ nec ex se, nec ex aliis, sed ex quibus omnia fiunt*; how then can we reconcile this doctrine, with the form being made of the matter?

SECT. XV.

LXIII. But have the moderns who exclaim so much against Aristotle, hit upon the truth? By no means. They have reasoned more boldly, but not with better success. They tell us, that the texture, collocation, figure, and motion of the particles, produce all the operations of nature, without the necessity of having recourse for this purpose, to accidental or substantial forms; but by saying this, they expose themselves to the same fault which they reprehend in the Aristotelites, which is that of speaking too generally; for as these last, do not explain or define, of what the substantial form consists which distinguishes one entity from another, neither do the others determine, the texture, co-ordination, and figure of the particles which appertain, and are proper to each composition; in consequence of which, they all involve themselves in innumerable difficulties, which they reciprocally object against each other. The Cartesian system appears chimerical to the Gassendists, and the Maignanists; and these two last parties, although they agree in assigning the Atoms as the principles, and elements of all material things, oppose one another in various particulars, it

being a principle with the Maignanists, that the Atoms are different in species, and with the Gassendists, that they differ only in figure; all of which systems, are exposed to be combated by arguments that are terribly strong.

SECT. XVI.

LXIV. From all that has been said, it may be evidently concluded, that we know nothing of the nature of that principal object of physics, *an ens mobile*, either taken as confined to individuals, or considered with respect to the species, or contemplated in an abstracted sense, as relating to the gradations, of lowest, inferior, or supreme. What one sect affirms, another denies, and the worst is, that, attend to the reasoning of which of them you will, you will find the arguments against the system of each, stronger than the proofs in favour of it. On this account, Lactantius said wisely, that philosophers have swords, but no shields: *Gladium habent, scutum non habent*. They produce penetrating arguments wherewith to assail the opinions of their opponents, but not solid solutions, wherewith to defend their own. What can we do then in such a case? Why nothing but suspend our judgment, till some angel shall decide the contest.

LXV. Perhaps some one will remark to me, that the substantial nature of things is at a great distance from our view, and that therefore it is not wonderful, that we have not yet penetrated into the inmost recesses of philosophy; for that without advancing so far, we may find sufficient matter whereon to exercise our speculations, in contemplating the ordinary phenomena of nature, and by endeavouring to discover their proximate causes; which attempt, may possibly be crowned with success, by observing and reasoning upon every species of movement which is performed by an entity, that has motion or circulation, according to the nature of such an entity.

LXVI. I will acknowledge, that philosophy helps us to reason upon natural phenomena, and to enquire into their more immediate causes; but in doing this, we frequently wander in the dark, and are much exposed to be interrupted by impediments, produced by ignorance and doubts; except it is

in the case of a few truths, the discovery of which, we owe to the light of experience; and this fact is evinced, with respect to the very instance of motion which has been just alledged.

LXVII. With regard to the movements of generation, corruption, alteration, augmentation, and the rest, which are considered as distinct from local motion, you can find nothing that is not questionable, both in the schools of the Aristotelites, and in those of the modern philosophers. The very definition of motion in general, which was given by Aristotle, some reject as obscure, others as perplexed, and others as nugatory. The movements we are now treating of, in the opinion of Aristotle, are acquisitions of new forms, either substantial, or accidental; but the Moderns, who deny the existence of all material forms, object to this definition of those movements. Even among the Aristotelites themselves, it is not agreed, whether motion is not influenced by passion; nor whether the first, is not obedient as a passive agent to the last. And thus in every thing else, it is all question, and all dispute.

SECT. XVII.

LXVIII. And why should we be surprized that with respect to these movements, which as we may say, nature executes behind the curtain, human understanding has made so little, or next to no advances? what we ought more to wonder at, is, that it happens the same with respect to all local motion, which lies so plain and open to our observation.

LXIX. The motion with which heavy bodies descend, is the most frequent and common to our view. And what do we know of this? Of its properties very little; of its causes nothing. We know it acquires some degree of acceleration in its passage, because we see it; but what is the proportion of the increased acceleration, is matter of great debate, both among the philosophers, and mathematicians. We know that it is a motion of descent, but we do not know whether its course is directed to the centre, or the axis of the earth. The cause of this motion remains so hidden, that the philosophers to this day, have given no opinion concerning it, that I will not venture to pronounce an absurd one. The Aristotelites, by saying the cause

of this motion proceeds from an innate disposition to move in every thing, say nothing, unless they point out the particular virtue or faculty, which excites motion in heavy bodies, for what we have just mentioned, is the reason they generally assign for all species of movements. They should not be allowed to dispute or controvert what they themselves have advanced; and if they should attempt to give a more rigorous definition of their opinion, they would fall into a still greater absurdity; which made the learned Father Saguens say, *Quis non palpat crassitiem hujus chimericæ opinionis?* The Cartesians to account for this phenomenon, recur to the vertical motion of the subtil matter, which separating itself from the earth, and pursuing the angle or direction of tangents to the circle, impels heavy bodies to descend. But this reasoning has been confuted by most efficacious mathematical arguments. Gassendo invented an effluvia composed of terraqueous corpuscles, which mount into the air, and penetrate the pores of heavy bodies, which they generally encounter, and after turning their first course into a descending one, impel them downwards. Nothing has so much convinced me of the great difficulty of this question, as seeing a man of the subtil ingenuity of Gassendo, recur for the solution of it, to a fiction destitute of all probability; and which is exposed to invincible objections. Father Maignan, and his followers also, make use of the terraqueous effluvia for the solution of this difficulty; they do not allow that they act by impulse, but that by their sympathetic and magnetic virtue, when they come in contact with heavy bodies, they dispose them to descend.

LXX. It is very probable, and perhaps more than probable, that the ascent of light bodies, is caused by the descent of heavy ones; because the heavy body has power, in consequence of the impetus of its descent, to occupy the lower station, where meeting with the light body, it obliges him to leave that situation, and to mount upwards; for many argue with great colour of reason, that there is no such thing as absolute lightness to be conceived in any body whatever, nor is such a quality necessary, a respective or comparative lightness, being sufficient to answer every purpose. Thus we say a body is light, not because it is void of gravity, but because it is less heavy than the one with which we compare it. In the same manner we say the air is light, not because it is not ponderous, but because it is less so, than earth, water, and all the other bodies that surround us; and that no other levity but the respective, is necessary to cause bodies which

we call light to ascend, may be clearly seen in the case of oil, which notwithstanding that it is heavy, if you pour a quantity of water into the vessel where it is, the water on account of its superior gravity, will occupy the inferior station, and oblige the oil which was at the bottom to ascend. The same thing happens with regard to air. If you dig a ditch to any depth in dry ground, the air will descend and occupy the whole of it; and there is no other way to dispossess the air from the bottom, and to make it rise upwards, but conveying water into the ditch, or by throwing into it some body that is heavier than the air.

LXXI. It is not owing to the principles of physics, but to experience, that we understand the little we know of this matter; and with respect to this little, there still remains great difficulties for philosophers to contemplate; and the greatest of all, is ascertaining the cause of the ascent of vapours to the region of the air. It is certain, that vapours are nothing else but water, resolved into exceedingly minute particles. The water however being heavier than air, how can the water arise to the height occupied by the clouds? Every particle of water notwithstanding it weighs very little, is much heavier than a particle of air of equal size, and the greater or less gravity of liquids, for the purpose of their impelling one another, is computed in the gross, and not according to the proportion of globules of equal size; and we know that a pound of water, will cause a quarter of an hundred of oil to rise in a vessel.

LXXII. Some philosophers, who were aware of this serious difficulty, upon reflexion, conjectured that some portion of ethereal matter or pure air, might adhere to each particle of vapour; and in consequence of this conjunction of the two, the whole might become lighter than an equal quantity of the inferior gross air of our atmosphere; and on this account, might mount on the top of it; for although iron is much heavier than water, if we fasten a small portion of iron to a bit of deal board, it will swim, because the quantity of iron and board together, is lighter than an equal bulk of water. Francis Bayle, adopts this opinion of a portion of the ethereal matter adhering to the vapour; and on the other hand, Father Pardies a French jesuit supposes, that a particle of extended vapour, in the form of an air bubble, contains in its cavity the ethereal matter. All this is far from

certain; but I shall not detain myself in controverting, either the one or the other of these modes of reasoning. Others again imagine, that various igneous particles, which arise from the earth, after separating from the water or some other liquid those small particles which we call vapour, with their continual impulse and agitation, force them upwards. But neither does this system appear to me very defensible, any more, than that of the vulgar philosophers, who say, the sun by its activity attracts the vapours; for if this was so, the vapours would not stop till they reached the sun, or at least till they were obstructed in the heaven of the moon, by the moon itself, or some other solid body; for the attractive power is stronger, the nearer the thing attracted advances to that which attracts it; and the first would never cease to move towards the other, till it came in contact with it, if it was not interrupted in its course by some obstruction; and besides, the attractive virtue is a certain something, which nobody can comprehend, and is therefore in a manner wholly banished from philosophy.

LXXIII. Who would not be surprised, that physics should not have helped us to attain a knowledge of so common a phenomenon, as that of the ascent of vapours? but so far has it been from attaining it to any degree that comes near to truth or certainty, that hitherto, we have not been able to discover any thing upon the point, that is satisfactory to the understanding. And with respect to all other species of motion, we are in the same predicament.

SECT. XVIII.

LXXIV. Have we attained a knowledge of the cause of the elastic motion, which is the property that makes a wand or a sword-blade which is forcibly bent, recover of itself, the straight form it had before, or if it was naturally bent or crooked, and was opened and made straight by force, makes it return to its original crooked shape? Descartes, for the explanation of this phenomenon, recurs to his common asylum, the impulse of the subtil matter, which not being able to penetrate the pores of the wand or sword-blade on the hollow side, where the pores are closed by the inflexion, by its great efforts to open and penetrate them, causes the wand or sword-blade to recover its former figure. But who does not perceive, that for this purpose,

it will be necessary to suppose the subtil matter should be for ever moving in opposite directions, from west to east, and upwards and downwards, &c. for the wand or sword-blade, to whatever aspect you turn the hollow side, recovers its natural figure equally the same? Besides, Descartes supposes the subtil matter to be infinitely fluid, from whence it follows, that it will be impossible to shut the pores in such a manner, as that they will become impenetrable to it.

LXXV. Others say, that the same impulse which is impressed on a wand or bow by him that bends it, is the thing which opens it afterwards. But against this opinion, it may be urged in the first place, that he who bends it, does it gradually and by slow degrees; and the impetus with which it opens, is violent and quick. Secondly, the archer who bends a bow, is not possessed of a force equal to that, with which it reverts to its former figure, for that is so great when the arrow is drawn near the head, that it is capable of giving it an impetus, that will pierce a substantial thick body through and through; and how can any one communicate a force or impulse, which is greater than that he possesses?

LXXVI. The Aristotelites, who are great adepts at the easy invention of giving the name of quality, virtue, or faculty, to express the cause they are enquiring after, and by the addition of an adjective to that name, which is a denomination taken from the effect, say, that the cause of the elastic motion, is an elastic virtue which is inherent in the wand or sword-blade. This is in reality, having found out a master key, to unlock all the mysteries of nature; for there is nothing so obscure, that with this invention may not be made manifest. If you ask, what is the cause of the marvellous properties of the load-stone, the answer is, the magnetic virtue; if you ask, what are the causes that perform in us the operations, of the concoction of aliments, the expulsion of excrements, and that of nutrition, &c. the answer is, that they are performed by a concoctive virtue, an expulsive virtue, and a nutritive one; and according to the same mode of reasoning, the cause of the winds, is a ventilating virtue, that of lightning a fulminating one, and that of the flux and reflux of the sea, is owing to two opposite virtues, the one fluxive, and the other refluxive. By this cheap mode of philosophizing, every thing is demonstrated at a glance. But to speak seriously, what is this, but

answering in the very language of the question? Saying that the cause of the elastic motion, is the elastic virtue, is in effect the same as saying, the cause of the elastic motion, is the cause of the elastic motion; and saying the magnetic virtue, is that which causes the load-stone to attract the iron, is answering in the stile of pleasantry, which some children have studied, and are very ready at; who when a person asks one of them, *My lad, whose son are you? the boy answers my father's.*

SECT. XIX.

LXXVII. The cause of the progressive motion, is also very difficult to be accounted for. It is hard to comprehend, how the motion of a stone thrown from the hand subsists, after the action of throwing it ceases. Who moves the stone after the hand is still? What many of the Aristotelites say, is, that the action of the hand produces a quality in the stone, which they call impetus, and that this quality causes the stone to move after it is thrown from the hand. But this solution, is destitute of all appearance of truth. If every violent motion, as the Aristotelites maintain, proceeds from an extrinsic cause, how can the motion of the stone thrown in a mounting direction, which is a violent one, proceed from an intrinsic quality communicated to the said stone? If all generation, or accumulation, according to the doctrine of the same school, supposes corruption, what quality or accidental form in the stone was corrupted, to prepare it for engendering that new quality, which they call impetus? What dispositions preceded this generation? Or what time is there for their preceding it, when a globe of great bulk, with its motion impels a little one? It being certain, that the contact of the two, does not continue longer than an instant, what property then has that quality, to occasion so quick a corruption? Is it perchance, the gravity of the stone itself? This however, as it subsisted at the time of receiving the impulse, if it is a property opposite to that quality, would at the instant, have prevented its generation, as it is afterwards said to impede its conservation. We might make many other reflexions, to prove, that that quality is chimerical. Others recur to the air, for the means by which the motion is continued, which they say, being violently divided by the fore part of the stone, takes a sudden turn to the hinder part, and impels the stone forward. But omitting many other objections, which would render this mode of philosophizing totally improbable, I shall just remark, that it

would follow from hence, that a stone could not move through a void space, give it what impulse you will, which is what I imagine nobody can believe. Descartes, reconciles this difficulty, with his maxim respecting the general law of the communication of motion, established by the Author of nature at the creation, which we shall not combat minutely, in order not to waste time; but shall content ourselves with observing, that that maxim applied to the present matter, and thoroughly investigated and explained, would signify, that the stone thrown from the hand, moves because God has ordained that it should move; but to solve difficulties in this way, it is not necessary to study philosophy.

SECT. XX.

LXXVIII. Finally, there is no motion whatever, about the cause of which, the philosophers do not altercate. What contests have there been among them, to explain how the movements of rarefaction, and condensation are performed? Some fancy, that rarefaction consists, in the same quantity of matter occupying a larger space; which explanation, others looking upon as unintelligible, assert, it consists in the opening of the pores, and the extension in consequence of that opening, of various parts of the body, by the introduction of a subtil liquid substance, in the same manner that water enters a sponge, or as rarefied air enters water, or the ætherial matter insinuates itself into the air. This is the doctrine of the Cartesians; but in the opinion of the Maignanists, and Gassendists, such reasoning is nothing to the purpose; for these, as they admit in nature, not only of the possibility, but of the necessity of a vacuum, or many voids distributed in small interstices, find no inconvenience in supposing, that in bodies, there are small empty spaces, unpossessed by any matter whatever.

LXXIX. Fermentation, this solemn instrument of nature for the performance of an infinite number of her works; consists in nothing, but an intestine motion of the insensible particles of the mixtures, by which is solicited, a new combination of their elements. But from whence does this motion proceed? The Moderns, since the discovery of the acid and the alkali by Otho Takenius, attribute all the fermentations, to the meeting of these two substances. But this is only pointing out the matter, on which this

motion has its effect, and we do not enquire here for the material cause, but for the efficient one. Who, or what, impels the acid and the alkali to this conflict? The juice newly expressed from the grape, will quiet this combustion for some little time; but after that, the tumult begins again. What new agent can we discover here, to excite the second fermentation? This is a secret, which only the Cartesians have presumed to unravel, by recurring to their invisible fairy, *the subtil matter*; whom they make the author of all this domestic sedition. It is with some propriety that I call it a fairy; for as the vulgar attribute to the fairies, all the nocturnal noises and commotions, whose causes they are ignorant of; so the Cartesians impute all those motions, which on account of the darkness that obscures their causes, may be truly termed nocturnal, to the impulse of the subtil matter.

LXXX. I am so far from believing, that the subtil matter is the *primum mobile*, or first mover of every thing, that I am inclined to think, it moves nothing. My reason for this opinion, is as follows. The more fluid a matter is, so much the less impulse it makes on any body it encounters. Thus we see, that water gives a much less violent shock to a wall, than any other solid body of equal size, that may be forcibly driven against it; and air, a much less than water. No building could stand against, or resist a moderate gale of wind, if the air was as solid as water. The subtil matter then, according to the doctrine of the Cartesians, being infinitely fluid, can have but little impulse, or be able to impress but little agitation or motion, on the bodies it encounters. This consequence to me, is clear and plain; because, if in proportion to the augmentation of its fluidity, the impulse of a body is lessened, when the fluidity comes to be infinite, the impulse will cease totally. From hence it would follow, that there is no body whatever, which could be moved by the impulse of the subtil matter.

LXXXI. But admitting that it has power, as the Cartesians pretend, to move the insensible particles of the mixtures; it will not follow from thence, that it furnishes us with an explanation of the present phenomenon. For in the first place, the subtil matter, if it has any impulse, exercises it on the particles of the juice the instant that it is pressed from the grape; and even perhaps did it before, while the liquor was contained within the outside skin or covering. How then does it happen, that it did not sooner excite that

tumult, which is the symptom and property of fermentation? Secondly, how can the acids and alkalies be said to produce this effect? for, according to the doctrine of the Cartesians, of whatever particles the mixtures consist, the subtil matter would set them in motion, for that there is no mixture whatever, impenetrable to its extreme subtilty. Thirdly, how can those exceeding slow fermentations, which are years before they manifest themselves, as in the instance of treacle, be attributed to the rapid and swift motion of the subtil matter?

SECT. XXI.

LXXXII. Saint Austin says wisely, that what is most surprizing, does not seem to strike us, although it is the object of our daily experience; which is a maxim, the saint applies to the wonders of nature, and falls in exceedingly *à propos* to suit with our present discussion. All the philosophers he observes, admire as portentous things, the flight of the iron to the load-stone, the pointing of this last to the poles, and the flux and reflux of the ocean. If we ask them why they consider these motions as wonderful, they will answer, it is because they cannot ascertain their causes. But we cannot help remarking, that this answer amounts to a virtual acknowledgment, that all the movements of nature are equally wonderful, with those of the iron, the load-stone, and the ocean, for their causes are equally disputed, because we are equally ignorant of them. The only difference between them is, that these motions are confined to particular or determined entities, and the others are common, or nearly common to all.

LXXXIII. I confess for my own part, that on whatever side I view nature, I equally wonder, because I find myself equally ignorant of it. The same Saint Austin, whom we have just quoted (Tract. 24. in Johan.), observes, that the ordinary multiplication of grain, which is obtained at harvest, by means of the fertility of the earth, is equally a prodigy, with that extraordinary multiplication of loaves and fishes, which was effected by the majesty of Christ in the desert. Let now the vain philosopher boast, that he is able to decipher that great mystery, only because he is possessed of a compleat apparatus of faculty phrases; such as *seminal virtue, previous dispositions, the corruption of one form, and the introduction of another,*

attraction of the nutritious juice, the conversion of it into the proper substance, vegetation, nutrition, &c. Was Saint Austin peradventure ignorant of those phrases, or of others equivalent to them? Yet notwithstanding this, he looked upon that natural multiplication of grain, as an impenetrable mystery. These phrases, only express or describe those operations which are familiar and open to our experience; but do not reveal to us, their causes, or the manner in which they are performed. The rustics, are acquainted with many more terms than we, expressive of the various operations with which nature in succession, proceeds to perfect that work. Are they peradventure on this account great philosophers? what do I explain, by calling vegetation or nutrition, that progression, by which a plant acquires its increase in bulk? does this afford me any philosophical knowledge, respecting the manner in which that operation is performed? There are two principal things to be considered in vegetation; the first is, the ascent of the nutritious juice by the fibres of the plant; the second is, the conversion of this juice into the vegetable substance; and we perceive in those two things, two great mysteries. If we ask the school philosophers, how the nutritious juice which is ponderous, rises spontaneously to the very uppermost leaves of the highest trees, they will tell us that it rises by attraction. And what is this, but placing us in the same state of doubt and difficulty, with respect to the most common work of vegetation, that we remain in, with respect to the motion of the iron to the load-stone? Both the one and the other are called attraction, but we are equally ignorant, why, or how, the highest leaves of a tree attract the juice which rises from the bowels of the earth, as we are, why or how the load-stone attracts the iron.

LXXXIV. Let us proceed to the second mystery. Who can explain to me, the manner in which a juice that is so exceedingly fluid, subtil, and fine, as to be able to circulate through the smallest channels of the fibres, is afterwards converted into the solidity of wood or grain; and this difficulty will increase, if we cast our eyes on the other mixtures, and reflect, that from another juice, or rather most fluid vapour, is generated copper and marble. Aristotle had certainly some reason for saying, that nature was the very devil: *Natura dæmonia est, non divina* (Lib. de Præsens. per somnum); for by observing her works with attention, it seems as if she did every thing by enchantment.

SECT. XXII.

LXXXV. It would be even some consolation to us under our ignorance, if only the manner, in which nature works in the interior part of her operations, was hidden from our eyes; but the most humiliating circumstance is, that it is the same, with respect to every thing that immediately presents itself to our senses. Bodies are familiar to our touch; but hitherto, we have not been able to discover, whether they are composed of indivisible points, or of parts infinitely divisible; nor to what it is owing, that one body is hard, and another soft, one solid, another fluid, one opaque, and another diaphanous. We are continually viewing colours; but we do not yet know, what sort of things colours are; whether they are mere reflexions of the light, or whether they are intrinsic accidents appertaining to the object. The light illuminates, and assists us to see; but we find our understandings greatly obscured, when we consider the nature of light. Whether we conceive it to be substance, accident, body, or spirit, none of them are applicable to its nature, and still they all seem to be so. And with how many impenetrable doubts and difficulties are we surrounded, when we consider the species which we call visible? If there is any one difficulty superior to, or any one inequality more striking than another in the mysteries of physics, I will venture to pronounce this is it. How the visible species of a star in the firmament, can in an instant, be translated from the same star to our eyes, when it must travel in that instant many thousands of miles, is beyond conception; as is likewise, how this species can exist at one and the same time, in the whole immense space between this and the firmament; it being certain, that in all this space, there is not a point, from which, by taking the view, you will not see the star. How also, contrary to the maxim of Aristotle, can many material species, differing only in number, exist or be seen, from the same point of space; it being certain, that you may from such an identical point, see distinctly, many stars at the same time? I shall omit many other objections, that are not inferior or less forcible to these I have urged, against the common opinion, that might be also urged, against the mode of reasoning of the modern philosophers.

SECT. XXIII.

LXXXVI. So that we see, our philosophy, from what we call first principles, down to our ultimate conclusions, is nothing but a fabrication, or weaving together of fallible conjectures; and that even these conjectures, terminate in nothing more than certain general notions; for the nature of all the specific things, and the greatest part of the causes respecting the lowest species, are so far removed from our penetration, that we can hardly arrive at attaining a doubtful idea of them. If we hit upon a truth, we owe the knowledge of it to experience, and this cannot be called scientific knowledge; for it is derived from self-evident principles, which may be comprehended by the most stupid of mankind; between whose manner of explaining the matter, if they attempt doing it, and ours, there is only this difference, that we define it in terms of art, and they in ordinary and common ones, which are better than the others, because they are more intelligible. This made the learned Jesuit Claudius Franciscus Dechales say, that our physics consisted of nothing but a particular idiom, which conveyed no certain knowledge of any thing. (Tom. I. tract. de Progressu Matheseos).

LXXXVII. It is much to be lamented, that those who are called professors in the schools, do not know more of the nature of things than the vulgar. But what would you think, if I was to say now, they know even less? It would seem, as if I was advancing an extravagant paradox; but notwithstanding this, I assert that the proposition is a most true one, and may be easily proved; for as experience is the only channel, through which a knowledge of nature can be conveyed; they must have the best experimental knowledge of natural things, who in various mechanical occupations, work up various natural entities; and not those who amuse themselves with speculations, and live retired in schools. A fisherman, knows something of the properties of fish; a pilot, of the winds and the tides; a sportsman, of birds and wild animals; and a husbandman, of the generation and increase of plants. But what does the philosopher know? why, he knows how to raise doubts about every thing, and that is all. Thus the school of physics is a theatre, where people are taught to doubt without end. I say without end, because it is scarce possible the period should ever arrive, when they will be able to pass from doubt to certainty. This may be clearly inferred, by their disputing at this day with the same obstinacy, the same questions they disputed two hundred years ago. If any elucidation, or

certain knowledge, has been acquired with respect to here and there a physical theorem, we are not indebted for it to the schools, but owe it to the benefit of experience, which comes from the world at large. We may thank the experiments of Torriceli, Monsieur Pascal, Otho Guericus, and Boyle, for our knowledge that the air is ponderous. If we are certain, that the blood circulates from the heart through the arteries, and is returned by the veins, we owe the discovery to the anatomical observations of Peter Paul Sarpi and William Harvey. If we are clear, that the chyle is not conveyed to the liver, but to the heart; what ascertained this truth, but the diligent and practical scrutinies of John Paqueto, Thomas Bartolino, and the Englishman Lowther? Experience has been the only arbitrator, which has settled some disputes, and banished certain errors from the schools; and wherever matters have been left to speculation and reasoning, the suit still remains depending and undetermined. One age and another passes away, and the world during the whole time, has been accustomed to hear the same vociferations, the same arguments, and the same distinctions; and has seen the obstinacy of the contending parties, transferred, as if it were by hereditary succession, from professor to professor, without the least prospect, either of victory or reconciliation.

SECT. XXIV.

LXXXVIII. From this known ignorance of ours, we may deduce a very useful reflexion, which is, to be firm and steady in observing a due subjection to the sacred dogmas of faith. The inordinate confidence we place in our own reason, is a great enemy to religion. He who estimates his own understanding at an excessive high rate, rests his faith on the edge of a precipice; for this vanity has shewn itself very glaringly, in all the heresies we ever knew. In their pursuits of other vices, mankind have taken different turns, but in this they have been all uniform; for although they have neither been all lascivious, nor all covetous, nor all ambitious; still in this instance, they have all presumed much on their own understanding. And what can be a more efficacious antidote against this mad presumption, than reflecting upon the little or nothing, we have been able to discover in matters of philosophy? how can he, who knows he cannot penetrate the mysteries of nature, dare attempt fathoming those of grace? If he reflects, there will result from that reflexion, a distrust of his own reason, and he will submit

himself obediently to authority. The philosopher Anaxagoras, whom on account of the extraordinary subtilty of his ingenuity, they by way of eminence, called the soul or spirit of antiquity, after having laboured with infinite industry in philosophy, said, that nature was all surrounded with clouds and darkness. *Anaxagoras pronuntiat circumfussa esse tenebris omnia.* (Lactant. lib. iii. Divin. Instit. cap. 28.) And I must observe, that this philosopher, who knew nature was impenetrable to his understanding, was the first of all the philosophers, if we believe Aristotle, Laertius, and Plutarch, who declared himself convinced of the indispensable necessity of a supreme intelligence, who must be the author and director of the whole. On the other hand, those who boasted, and flattered themselves that they had discovered all the mysteries and profundities of nature, for the most part, denied either the existence, or the providence of a deity.

LXXXIX. I can with truth say of myself, that next to the divine grace, a conviction of my own ignorance with regard to natural things, is the most powerful weapon I could ever discover, wherewith to overcome all those difficulties or objections, which natural reason suggests, against the mysteries of faith. I often say to myself, good God! how can I understand those wonders, which by using his extraordinary power, are wrought by the omnipotent hand, if I cannot comprehend, the common effects of his ordinary power? It is true, I am ignorant how, or by what means, a divine person can unite himself to human nature; but I am also ignorant, how a spiritual soul can be united to a material body. Notwithstanding which, this is a matter of fact, and happens within myself. Neither do I perceive, how the same water which falls from heaven, should be converted, not only into here and there a particular body, but into all the animal and vegetable substances upon earth. In the most plausible part of theological controversy, I find myself exceedingly embarrassed; for if I take the side of providence, I am assailed with the forcible arguments in favour of liberty; and if I put myself on the side of liberty, they wage powerful war against me with the arguments in favour of providence. But do not I see the same arguments, urged with greater vehemence, in the vulgar philosophical controversy respecting the composition and unity of universal space, in which, whatever sentiment is maintained, they instead of answering their opponents arguments, perplex the dispute with a multiplicity of words? If I defend, with Aristotle, the infinite divisibility of universal space, although to avoid

being concluded, I don't do it with my mouth, I cannot in my mind, help acknowledging the unity of a great number of its parts; and if with Zeno I acquiesce in the indivisibility, the mathematical arguments derived from the diagonal of a square, two concentric wheels united, and many other principles, not only leave me without an answer, but in a manner strike me dumb.

XC. I say again, if in those natural things that are open to our view, and which we touch with our hands every day, there occur a thousand difficulties, that are insuperable to, and beyond the reach of our understandings, have we not the greatest reason to suppose, that the same thing happens with respect to supernatural matters, they being totally superior to the sphere of our senses? If, maugre all my prying, I can't perceive how God does an infinite number of things, which I see him do every day, would it not be madness in me, to deny or to doubt the existence of revealed things, only because I can't descry how God executes those things?

If there was a man so short-sighted, that he could not see objects that were very near him, and should pretend that he saw those which were at the distance of a hundred miles from his eyes, or else should take it into his head to insist, that such objects, although they are well known to be *in esse*, do not exist, only because he can't see them, would not all the world pronounce him a lunatic? This is exactly the same sort of madness as that of those who deny there are divine mysteries, only because they can't comprehend or pry into them. Little vain stupid man, if the fabric of those material compositions which are ever before your eyes, and are at all times familiar to your touch, is totally impenetrable to your short and limited capacity, how can you expect to comprehend the ineffable manner, in which Omnipotence performs those supernatural wonders? You will tell me, that you can find no solution of the arguments, which the Gentile urges to you against the mystery of the Trinity, or against that of the Incarnation; and I answer, that neither can you find any, to those the philosopher urges against the composition of universal space, take which side of the question you will, and suppose it to be made up, of either divisible, or indivisible matter; but would you conclude from thence, that it is not composed of either one or the other? you certainly would not; and it would be equal, if not greater

delirium, to deny the truth of those mysteries, only because you can't answer, and remove the objections that are made to them. Is it fit, that an Infinite Power should proportion his works, to the limited measure of your understanding? or is it reasonable to suppose that God can do nothing, which you can't comprehend?

XCI. No north wind, so quickly disperses the clouds which interrupt the rays of the sun, as these reflexions clear up the doubts, which natural reason opposes to the mysteries of faith. Leave then the presumptuous dogmatists to chew their scepticism, and to make the most of the objections it furnishes against religion. But the sort of scepticism, which is precisely limited and confined to physics, is so far from prejudicing faith, that it serves to confirm and rivet it; by removing the obstacles, which the presumption of natural reason throws in the way, to interrupt the effect of that humble docility, which is so necessary, to preserve the understanding in a due state of subjection to the authority of revelation.

XCII. Those men do great injury, not only to philosophy, but to the church also, who rashly endeavour, to interest the doctrine of revelation in the defence of their particular philosophical opinions. This the heretics lay hold of, to calumniate us with converting philosophical sentiments, into articles of faith; and by this artifice, they persuade their followers, that our belief is arduous, and unnecessarily complicated. Upon this presumption, some strangers found their assertions, when they accuse us of countenancing idiotism, and mixing it with religion. It is but a little while ago, that one of them declared in his writings, that the opinions of men in Spain were not more free, than their persons were in Turkey. In order to preserve a due respect for sacred things, it is necessary, not to confound them with profane ones; whoever should begin to erect habitations in temples, would be the author of temples losing that respectful reverence that ought to be preserved in them. There are judges appertaining to the church, whose province it is, to determine what doctrines are useful, what pernicious, and what indifferent. Let us then leave the decision of those matters to them, and don't let those who sincerely seek the truth, be terrified and disturbed by those scare-crows that are set up by partiality and faction, and sometimes by the pride of those, who have given their names to

particular schools, or by the envy of others, who could not arrive at doing so much.

SECT. XXV.

XCIII. We having shewn now, that we possess no physical science or demonstrative knowledge of natural things, we may entertain a reasonable doubt whether we shall ever be able to attain any such. The most learned Valles, is clear that we shall not; because physical knowledge is confined to singular things, and alledges that from singular things, you can't derive a science. But as we have observed before, this argument is ill founded, and insufficient.

XCIV. We might lay more stress upon two authorities the sceptics alledge in their favour, which are taken from the Ecclesiastes. The first is from Chap. III. in these words. *Cuncta fecit bona in tempore suo, et mundum tradidit disputationi eorum, ut non inveniatur homo opus, quod operatus est Deus ab initio usque ad finem.* The second is more formal and precise, and taken from Chap. VIII. *Et intellexi, quod omnium operum Dei nullam possit homo invenire rationem eorum, quæ sunt sub sole: et quantò plus laboraverit ad quærendum, tantò minus inveniatur, etiamsi dixerit sapiens se nosse, non poterit reperire.* But in truth, these texts, when they affirm the impossibility of discovering the causes and reason of natural effects, may be understood to allude to the providential, and not to the natural or physical ones. In fact, this is the sense in which some fathers and expositors understand them.

XCV. Others again argue, that the desire of knowing the causes of natural effects, is natural to, and implanted in man by nature itself; and as a natural desire can't prompt a man to an impossible inquiry, it follows, that attaining the science we are speaking of, is within the compass of possibility. To this argument, Valles replies, it is absolutely possible; and although not in this life, it is in that to come, at which period, the blessed will see with God all things most clearly. This solution is attended with the following difficulty, that a natural desire can't be directed to an impossible object; neither can it terminate in a supernatural one; and the knowledge the

blessed will have of supernatural things, must be entirely supernatural, because it will depend immediately on the light of glory. Upon the whole, we may suppose, that there may belong to the soul in a state of separation from the body, abstracted from the supernatural blessing of the light of glory, a certain knowledge of all material things, in consequence of the species, infused in the course and order of nature; which is the sentiment of Egidius Romanus, Father Saurez, and others; and this knowledge being natural, may possibly be the effect of the thirst we possess for acquiring it in this mortal life.

XCVI. But we can't help remarking, that the foregoing argument does not stand in need of this solution, because it proceeds upon a false supposition not adverted to by Valles; which is, that the desire of knowing things philosophically, is a principle implanted in man by nature. If this was so, all men would be impressed with this desire, which is not the case; for the greatest part of them seem to have no passion at all for physics; and many despise philosophical speculations, as useless, vain, and by no means entertaining. It is true, that all men are desirous of knowledge, but this desire is not directed in all of them to the same object, or to the same species of objects. Generous souls are naturally lovers of truth; but the majority, are only anxious to understand those things, the knowledge of which, may contribute to the gratification of their passions.

XCVII. We have seen the little force of the arguments urged on both sides, with respect to the doubts we have been canvassing. On which account, I won't venture to give sentence in the question. Nor do I, nor can any one else without the assistance of Revelation, know the just limits of the human understanding with regard to natural things. And although the various philosophical systems which have hitherto been invented, are exposed to great doubts, or are impeachable of containing manifest nullities in them, who knows but that, in future some one may be discovered so compleat, and so well founded, that the understanding may be convinced of the truth of it. My opinion is, that if this is ever to be accomplished, it must be done by pursuing the plan and method prescribed by my Lord Bacon. It is true, that this is so laborious and prolix, that the execution of it may be apprehended to be morally impossible; for although the monarchs of two

very powerful kingdoms, France and England, for the space of more than a hundred years, have caused four hundred able men, at a great expence, and under proper regulations, to be employed in making innumerable experiments, and in reasoning upon, and explaining them, the work is not yet far advanced; when then may we expect to see it accomplished? The Academy of Sciences at Paris, and the Royal Society of London, don't amount to more than a shred of my Lord Bacon's extensive project.

On the Shew or Affectation of Learning and Knowledge.

SECT. I.

I. Science, like Virtue, has its imitators and its hypocrites; and the vulgar are as much imposed upon by the one as the other. The numbers of unlearned people who pass for men of literature, is considerable; and the false appearances they put on, becomes a copious source of errors, both particular and common. In this earthly region which we inhabit, apparent learning is as much revered, and often-times as much respected, as true. There are those, who are very expert at putting on the semblance of learned men, and of imposing themselves as such upon the world, although the portion of literature they possess is but very small; however, if they have the address to make their copied imitation of it appear an original painting, the copy will often make the same impression on mens minds, as if it actually was an original. When Zeuxis with his pencil imitated grapes, the birds flew with as much eagerness to eat the painted, as if they had been real and natural grapes.

II. In the eleventh century, Arnoldus Brixienis, who was a man of but little literature, did great mischief with his errors, both in his own country, and even in Rome itself; for as Gunterus Ligurinus observes, that besides being an elegant reasoner, he had the address of giving himself the air and appearance of a man of learning; *Assumpta sapientis fronte, disserto fallebat sermone rudes*; or as Otho Frisingens remarks, a copious verbosity passed in him for knowledge and erudition; *Vir quidem naturæ non hebetis; plus tamen verborum profluvio, quam sententiarum pondere copiosus*. Thus Vigilantius, although he was an ignorant man, by his art in gaining over to him booksellers and publishers, who were the proclaimers of his fame, so far acquired the reputation of a person of literature, that he had the assurance to write against Saint Jerom, and to accuse him of being an Origenist.

III. The vulgar, who are incompetent judges of men of letters, are apt, although it is against their own interest, to countenance and give credit to unlearned persons, whose deceptions in consequence of this encouragement become formidable. The delusion of popular ignorance is apt to magnify a very small light, into the blaze of a flaming torch; and to fancy it is as luminous, as the lanthorn placed on the top of the tower of Pharos, which Pliny says, at a great distance off, appears like a star to those who navigate the sea of Alexandria.

IV. It may not be improperly remarked, that in order for a man to be esteemed a person of learning with the populace, it is not so necessary that he should really be one, as that he should put on the pompous appearance of such a person. Arrogance and verbosity, if they are accompanied with a small degree of discernment, that helps a man to judge, when are the proper times to talk, and when to hold his tongue, and what are the subjects he should discourse, and what be silent upon: I say, if he has a little discretion to know how to conduct himself in these particulars, such management will have a notable effect. A confident magisterial air in his decisions, and an artificial gesture, which when he sprinkles about the little he understands of the subject he is talking upon, and which seems to indicate that he has an infinitely greater stock of knowledge of the matter treasured up in his inside; such affected appearances, I say, will have great weight and efficacy to fascinate the ignorant vulgar.

SECT. II.

V. On the other hand, men of true learning, are modest and candid; but these two virtues, are enemies, and also great obstacles to the advancement of their fame. He who knows most, is sensible, that what he really knows, falls greatly short of what he is ignorant of. His discretion teaches him this, and his sincerity induces him to confess it; but his acting in this manner, greatly obstructs his gaining the applause of the world; for these confessions have the same effect, that the evidence of those have, who bear witness against themselves; that is, they are readily believed; and although it is impossible for any man to know every thing, the vulgar are very

unwilling to esteem him learned, who owns himself ignorant of the least matter in his own profession.

VI. Men of learning, are also most commonly timid, because they are apt to be diffident of themselves; so that although all they say should be divine, if it is pronounced with a tremulous tongue, and a faltering voice, it makes no impression on those who listen to them. A man would gain more credit with the world in general, by talking loud, and making extravagant bold assertions, than by reasoning diffidently, and as if he was not quite clear; for the estimation due to discreet doubts has always been suppressed, and in consequence of this suppression, has contributed to countenance errors, and beget false conclusions. How has a presumptuous ignorant fellow, frequently availed himself of his imposing gesture, and the loudness of his voice? and how much by the strength of his lungs has he often concealed the weakness of his argument? But in truth, the noise made by a vociferous person, ought to render his solidity suspected; because men, like other sonorous machines, are loudest the more hollow and empty they are.

VII. If to these imposing appearances, there happens to be joined a moderate portion of literature, it has a most powerful effect to captivate the vulgar, and to gain popular applause. In the instance of Luther, who although he might truly and properly be stiled a man of learning, still the forcible manner in which he delivered himself, added to his address, may be supposed to have contributed more to the success of his preaching, than his literature.

SECT. III.

VIII. There are qualities also, which give people the reputation of able and learned men, when in reality they are quite the reverse. Gravity and circumspection, whether they are natural or artificial, contribute much to produce this effect. Gravity, says Magdalen Scuderi in one of her moral conversations, is a bodily mystery, invented to conceal the defects of the mind; and if it is carried to excess, elevates the person who wears this appearance, to the rank of an oracle. But I can see no reason why a person on this account, should be esteemed more than a man, because that the

nearer he resembles a statue, he in reality is by so much less than a man; nor why risibility, being the distinguished mark of rationality, he should be esteemed the most rational, who is the least addicted to laughter. The ingenious French author Montaigne, says pleasantly, that among the whole brute creation, there is no animal so grave as an ass.

IX. Aristotle considered melancholy as a token of ingenuity. But I can't tell why he did this; for every day's experience convinces us, that there are melancholy people who are very dull and stupid. If we were to judge of things as they appear to us at first sight, we might be easily induced to confound the stupid with the thoughtful, and to mistake the one for the other. People of dark and gloomy geniuses, have in some measure, the air and appearance of profound thinkers; but if we reflect, we shall be convinced, that insociability is not a characteristic quality appertaining to a rational being. In those who look as if they were always absorbed in thought, the negotiation or operation of the interior part of the soul seems to be inverted. Instead of the understanding surveying and contemplating the species, the species seem to seize upon, and take possession of the understanding; and instead of the mind being master of the object, the object appears as if it was master of the mind; and the species that appear to seize upon it seem as if they bound it down, and kept it confined. A person in this state should not be said to be contemplative, but ought more properly to be compared to a man that is stunned; the immobility of whose thoughts, keeps his reason in a state of suspense. I observe, that there is no brute of a more festive and sociable disposition than a dog; and that there is no one which is endued with a more noble instinct; yet, maugre this experience, I look upon the opposite extreme to excessive gravity to be the worst symptom of the two; for men who are always giggling and bantering, are generally very superficial.

X. But both the silent and the loquacious have their partizans. Some consider men of few words to be the most sagacious, and others give the preference to those who are rather prodigal in their use of them. The talking but little, is occasioned either by excessive caution, or by fear, or else is the effect of modesty, and the want of a ready flow of words; but not, as is commonly thought, does this indicate a want of knowledge. There is no

man, who if he was to speak all he thinks, or that occurs to him, would not talk a great deal.

XI. There are some, who observe an artificial medium between talking and holding their tongues, which is very well adapted to attract the veneration of the vulgar; that is, they speak with confidence upon such things as they understand, and take care to be silent upon such as they are ignorant of; but this they do, in a manner as if their silence proceeded from reserve, and an unwillingness to enter upon the subject. By these artifices, men of very moderate capacities, and whose informations are very limited, pass themselves upon the world for persons of deep penetration, and a sort of walking libraries; and although they are possessed of only a very minute or abstracted portion of whatever is the subject of conversation, they engage in the argument in very general terms, always taking care not to advance too far, and to retreat in time; which they do in a manner, as if they were tired of talking further upon the question at present, and therefore would be glad to postpone it, to be discussed more at large another time. In reality, they have said all they know, but behave in this manner, in order to have it thought they have only given a small specimen of their knowledge of the thing; although, like the painter who undertook to paint the eleven thousand virgins, but never finished more than five of them, and declared that all the rest should be supposed coming behind in procession, they would be glad, that what they have already said should conclude the debate; still, if any one, aware of their tricks, should then press them to a further discussion of the point, they either artfully turn the conversation, or affect a scornful dislike to discussing so weighty a matter before so thin an audience; or else get rid of the invitation made to them, with a disdainful smile, and treat both the proposal and the proposer, as if they were beneath their notice. These people are very ready at such sort of expedients, because they study them much, and have a great deal of practice in them.

XII. There are others again, who have recourse to ambiguous and confused expressions; which seem to say a great deal, when in reality they say nothing; and which, like the answers of the oracles of old, are applicable to all kinds of events. In fact, they may be termed the images of those oracles; for although they are listened to with attention, they are little

better than mere stocks or stones. The obscurity with which they talk is a dark shade, that conceals what they are ignorant of; and they make use of the stratagem of those who have no other but false money; that is, take care to pass it off under the cover of the night. But notwithstanding all these glaring appearances, there are not wanting weak people, who esteem their embarrassment and confusion as marks of their learning, and who think, that men are like mountains, which the more elevated and sublime they are, the more liable to be covered with clouds, and to cause dark shades which obstruct the fertility, and obscure the chearful appearance of the vallies.

Majoresque cadunt altis de montibus umbræ.

XIII. This deception, is commonly kept up and assisted by a persuasive carriage, and by certain mysterious gestures and distortions of the features; such as wrinkling the forehead, knitting the eye-brows, rolling the eyes, swelling out the cheeks, thrusting forth the under-lip, and shaking the head; all which affected tokens of wisdom and importance, are displayed and accompanied, with a kind of disdainful countenance. These are a sort of men, who have more than half the wisdom they seem to possess lodged in their features and muscles, which serve them to exhibit and express these symptoms of profundity. Tully very justly ridiculed this artifice in Pison in the following words: *Respondes, altero ad frontem sublato, altero ad mentem depresso supercilio, crudelitatem tibi non placere.*

SECT. IV.

XIV. Men, by affecting to despise others who know more than themselves, make use of the most vile and mean artifices imaginable; but notwithstanding this, it is the most sure method, to make them pass for persons of great knowledge, among people of groveling and plebeian minds. There can be no greater injustice committed, nor a vainer effort made, than that, of an envious man's attempting to transfer the merit which he has purloined from another person to himself. It is true, that a small cloud may obstruct the splendor of the sun; but this obstruction does not prevent this luminary from being the illustrious torch of heaven, nor can such a circumstance be of more consequence than a speck in the

atmosphere; nor does it require any science or learning, to enable a man to cast a thousand reflections and blemishes upon other people's doctrines and writings. When there is no reason to conclude that such behaviour proceeds from envy or malevolence, we ought to suppose that it springs from pure ignorance. I remember to have read in a Treatise intitled The Man of Letters, written by Father Daniel Bartoli, that an ass once happening to stumble over the Iliad of Homer, he in a passion tore it to pieces with his teeth. Thus, in order to outrage, abuse, and destroy a noble writing, nothing is better qualified than a beast.

XV. Being tenacious, insolent, and overbearing in argument, is also another means, which, although it is equally base and bad with the beforementioned, is sometimes very efficacious to gain a person the reputation of being a learned man.—Foolish people, such as the Megalopolitans, of whom we have an account in Pausanias, pay greater veneration to Boreas the god of the north-wind than to any other deity, and are apt to adore tumultuous geniuses as beings of supreme intelligence; and although it is absolutely irreconcilable and incompatible with it, to look upon barefaced arrogance as the child of superior learning. To this we may add another thing, that may have an effect to make them pass for wise and able persons, which is, that those who are truly learned, avoid as much as possible, all controversy or dispute with people who are of this tenacious overbearing disposition; and this prudent declining to contend with them, is often construed into fear of engaging them; as if it was necessary for men of learning, in order to shew their spirit, to encounter reptiles and venomous snakes. Cato's lamentation, or repentance, was just and generous, when he regretted having led his troops into the burning deserts of Africa, where they had no other enemies to encounter but asps, horned serpents, vipers, fiery serpents, and basilisks. The horrors of the civil war in the plains of Pharsalia, where they engaged the invincible troops of Cæsar, appeared to him less frightful, than those they experienced among the burning sands of Libya, where the most vile and abominable reptiles fought on Cæsar's side,

*Pro Cæsare pugnant
Dipsades, et peragunt civilia bella Cerastæ.*

XVI. He who can reconcile it to himself, to be inflexible in argument, and to dispute without end, and with all his might, never suffering himself to be convinced by reason, has made a large stride towards being reputed an Aristotle among the vulgar; for they, with respect to the wars of Minerva, as well as those of Mars, are apt to declare the victory, in favour of him who remains longest in the field, and, in their opinion, he never fails to come off conqueror, who has the last word. This is the way the vulgar judge; but he who would be thought superior to that class of people, must permit himself to be convinced by reason, for, if he does not, instead of acquiring the reputation of a learned man by such conduct, he lets himself down to the level of a brute.

The ingenious Doctor Luis Rodriguez, being asked what sort of a man a shallow physician was with whom he had been arguing, answered pleasantly, (although we must allow his wit favoured of the arrogance which is rather too common with the Portuguese) *He is so great an ass, that goad him ever so much, it is impossible to make him get on, or be the better for any reasoning that is bestowed on him.*

XVII. It is also a very common artifice with those who know but little, to bring the conversation to turn upon that little they are informed of. This is very easy for people of power or authority to do. I knew a person of this sort myself, who, whatever conversation happened to arise in company, was used to train the subject of it insensibly, to fall on the few points relative to it, which he had been reading or studying that day, or the day before, by which piece of management, he generally used to appear more learned than the rest of the company. Even in scholastic disputes, this art is often practised. I have more than twice in my life, seen a good theologian foiled by a novice; who, by artfully sliding some chimerical proposition into the dispute, has drawn the argument from its proper object, and caused it to fall into a sumulistical labyrinth of amplifications, restrictions, alienations, oppositions, conversions, and equipolations; the rules of which were fresh in the young man's memory, and which the old theologian had forgot. This was adopting the craft of the rogue Cacus, who, having artfully drawn Hercules into his own cave, rendered his arms and superior strength of no use to him, by blinding him with the smoak he continued to eject out of his mouth.

SECT. V.

XVIII. Besides the visionary wise people, or those which are such in appearance only, there are others who have credit given them for being men of profound knowledge, merely from the mistakes and misapprehensions of others. He who studies logic and metaphysics in the schools, together with those other matters, which under the denomination of philosophy are taught there, notwithstanding the sounding names, of these faculties, and his being supposed to know every thing, in reality, knows but little more than nothing. It is commonly said, that such a man is a great philosopher, when in truth he is no philosopher at all, either great or little. All the ten categorical principles, together with the whole contents of the eight books on physics, and their two adjuncts *on generation and corruption*, put into a logical alembic, would not produce a drop of true philosophical spirit, that would assist us to explain the most vulgar phenomenon in the sensible world. The Aristotelic ideas, have no more to do with physics, than the Platonic ones; and the physics of the schools, are pure metaphysics. All that the Peripatetics have hitherto written or disputed concerning motion, has not determined, what is the line of reflection, by which a ball returns back that is struck against a wall, or what is the degree of velocity, with which a heavy body descends by an inclined plane. He who by the ordinary metaphysical reasonings, thinks to attain a true knowledge of nature, is as mad, as he who should fancy himself master of the world, because he is possessed of a map of it.

XIX. The great advantage of these philosophers in name, if in the schools they manage with dexterity the rules of science, consists in this, that in virtue of the four species or rudiments of theology or medicine they acquire there, they are enabled to dazzle the eyes of mankind, and to pass in the world, for great theologians and doctors of physic. With respect to theology, the mistake in this matter is not so great; but with regard to medicine, it cannot be greater. By the rule, *Ubi desinit physicus, incipit medicus*, it seems as if it should follow of course, that a good philosopher is easily made a good doctor of physic. Upon this supposition, when people see a practising physician, who has twenty syllogisms at his fingers ends, ready to explain and demonstrate, whether privation is a principle annexed

to a natural entity, or whether the united mass or the whole, can be of a distinct or different texture from the parts, they fancy he is possessed of all the requisites and recommendations, that are necessary to establish and fix him as a most able, or first rate man in his profession.

XX. That most learned commentator upon Discorides, Andrew de Laguna, says, that if it could conveniently be done, it would be good policy to send those flaming young physicians, who are just come from the university, brimful of the bragadocio arrogance of *ergo* and *probo*, to practise physic in those nations we are at war with, for that it would be attended with a great saving, both of men and money, to our own country.

XXI. I can with confidence affirm, that there is no art or faculty, that can be less conducive to the acquisition of medicinal knowledge, than the physics of the schools. If all the philosophers that are, or ever were in the world, could be convened together, and remain in consultation for a hundred years, they would not in all that time, by the mere help of philosophical speculations, be able to instruct us how we should cure a chilblain; nor from that tumultuous consultation, should we obtain any maxim, that ought not to be prohibited as contraband, and refused admittance into the chamber of a sick person. Good sense joined to experience, either acquired by a man's own, or the practice of other reputable people, are both the father and mother of medicinal knowledge; nor has physics, that is to say the physics of the schools, the least share in the propagation of this species.

XXII. The reasoning of naturalists upon all kinds of mixtures, consists, in whether they are constituted of matter and substantial forms according to the doctrine of Aristotle, or of atoms according to that of Epicurus, or of salt, sulphur, and mercury, according to that of the chymists, or of the three elements, according to that of the Cartesians: whether these are composed of undivisible points, or of parts dividable *in infinitum*; whether their action or operation is from their texture, or the motion of their particles, or from some accidental virtues which they call qualities; whether these qualities are of the manifest or the occult kind; and whether they are of the first, second, or third class or species. Now what has all this to do with medicine?

I will venture to pronounce no more, or rather less, than it has with geometry or jurisprudence. When a physician goes about to cure a tertian ague, all this farrago of questions applied to the operations or effect of the bark, is of no sort of use to him. The only thing that is of any importance for him to know, is, whether his experience has taught him, that in the present state of the disorder, it will be proper to administer this febrifuge; but this he is to infer, not from the maxims, *dici de omni, dici de nullo*; but from conviction, drawn from experiments he has made himself of its efficacy in the like cases, and also from those, which have been made by physicians of eminence who have wrote on the subject.

XXIII. In no art whatever, is it of any use, to have a physical knowledge of the implements, with, or by which, a man does his business. A man may be an excellent pilot, without being able to explain the directive virtue of the load-stone to the pole; and a great soldier, although he is totally ignorant of the physical component parts of gunpowder, and the metal with which military implements are made; and he may be a great painter, without knowing whether colours consist of intrinsic accidents, or whether they are produced by the various reflections of light. Nor does the being able to argue well upon these points, conduce in the least degree, to make a man either a pilot, a soldier, or a painter. But there is no necessity for my enlarging further, in order to extirpate this common error from the world, as the learned Doctor Martinez, has fully and effectually exploded it in his two volumes, intituled *Medicina Sceptica*.

SECT. VI.

XXIV. There is another common error respecting the subject-matter of this Essay, although the opinion built upon it, is rather better founded than the one we have just been speaking of, and that is, looking upon every man as wise, able, and skilful, who has studied a great deal. The labour of study, seldom makes any great improvements, if it is not bestowed on a clear and penetrating natural understanding; and may be compared to the toils and expences of cultivation, that are employed upon a poor and nearly barren soil, which seldom produce much fruit. Among mankind, are to be found the turtles and the eagles of the human species. The last, with a speedy and

easy flight, ascend to the summit of Olympus; the first, after several days labour, shall not be able to reach to the top of a moderate hill.

XXV. Much time employed in reading, furnishes a man with a deal of matter; but the being able to penetrate, or rightly comprehend that matter, is more a gift of nature, than a production from toil. There are some, who may be termed learned from their memories, but who are not so in virtue of their understandings, and who may be compared to marble tablets with inscriptions engraved on them, which display the letters, but do not perceive them. They are a kind of mental books, that are filled with many texts, which they no more comprehend, or have a right idea of the meaning of, than the leaves on which they are written. If you observe the use or application they make of the species they have acquired, you will perceive, they deduce no proper conclusions from what they have read, nor make any observations on the things they have been perusing, that are pertinent or applicable to the subject-matter of those things. Thus we see, that from the same species or set of ideas, may be formed good reasoning and bad; just as with the same sort of materials, you may build elegant palaces, and rustic habitations.

XXVI. Thus it may also happen, that a man shall know all the works of Saint Thomas by heart, and be a very poor theologian; and that he shall in the same manner, know all the statutes of the civil and canon law, and be a very bad lawyer. And although it is commonly said, that law-knowledge depends almost intirely upon memory, or at least, that it depends more upon memory than understanding, yet I consider this as another common error. With a great many law-cases and maxims at his fingers ends, a man may draw a very bad plea; in the same manner as one, although his memory is stored with a vast variety of texts of scripture, may make a very bad sermon. The choice of the most fit to apply to the subject, depends upon the understanding, and not upon the memory. If men were obliged to do law-business by surprize, and without having time to consider and digest what they were about, a happy memory, in which were treasured up pertinent texts and citations, would be a most essential ingredient, and one, that it is almost indispensably necessary they should be possessed of. But as this, in the regular course of practice, is not the case; he who has attended properly

to the best books that have been written on the profession, and has a good general idea of them, will seldom be at a loss where to look for apt cases and authorities, wherewith to support his arguments; and as I observed before, the choice of those which are the most conducive to this end, is more the business of understanding, than memory.

XXVII. I have observed, that professional people in all faculties and sciences, have a great propensity to complain of the want of memory, and I have also observed, that they are apt to set a much higher value upon the gift of memory, than the power of reasoning; so that it appears to me, that if there were to be two shops opened, in one of which was sold memory, and in the other understanding, the man who kept the first would soon make a fortune, and he who kept the second, would not take as much money as would buy him salt to his porridge. But my opinion, was always quite different from this common notion, and I can say for myself, that I set more value upon a drachm of understanding, than upon an ounce of memory. I have been told, that I do not estimate memory at a high rate, because I have no occasion for it; but it is possible that those who told me this, may judge of me by themselves, who are not anxious about an increase of talents or ingenuity, because they fancy they are abundantly provided with all they stand in need of. I would not pretend to be endowed with a great share of memory, but I however think, I am rather better furnished in that respect, than I am in point of understanding; but I do not set a greater value on the last of these faculties on that account, nor does it proceed from my being anxious of an increase of it to myself, that I set a higher value upon understanding, than I do upon memory; but I give this preference from a persuasion, that in all the ordinary occurrences of life, understanding will afford a man more assistance, and be of infinitely greater use to him than memory.

SECT. VII.

XXVIII. We have not as yet, said any thing of authors or writers; but this semblance of learning, is the easiest to put on of all others. There is no more difficulty in writing ill, than there is in talking ill; and provided a man writes in the *bon ton*, and can flourish with the king's licence in the front of

his work, the book will go off, and the author will pass with the ideots of the world, for a learned, and an able man.

XXIX. But a person may make sure of gaining applause as an author with the generality of mankind, and this may be done in two ways, either by filling his writing with common-place observations, and just taking care to diversify and scatter them about; or else by stealing from other authors; and where there are great numbers of books to have recourse to, the danger of being detected in the plagiary is not very considerable; for there are very few who read many books, and nobody can read all that are published; so that all the hazard a person runs of being found out, is, that here and there one out of many thousands of readers, may discover from whence he made the theft; and among all the rest, he would pass for an original author, and they would acknowledge and respect him as such.

XXX. The writing from alphabetical compilations is also extremely easy. There is the Theatre of human life, the Polyantheas, and many other books, where erudition is arranged under the initial letters of the several branches of learning, and these books, by having copious indexes annexed to them, become a kind of public fountains, where all the animals of the world may drink, both men and beasts. Whatever subject a man undertakes to write upon, be it politics, morality, humanity, or history, he has only to turn to the index, which will point out to him, where he may be furnished with a copious assortment of texts, and citations, that are amassed together in these books for the use of all the world, and where he may collect whatever he has occasion for. By this management, the new author may gain himself the reputation of a man of great erudition and reading; for there are very few who can distinguish by the connection and regular series of a writing, that kind of copious erudition, which is well arranged and properly separated in the brain, and which flows opportunely from thence to the pen; from that, which a man when he is hard put to it, is obliged to collect from indexes and common-place books, with which he swells his work, and with which he heaps up in it, gross and bulky trifles, that consist of straw and chaff, collected from common-place Latin citations and numbers.

MORAL AND POLITICAL *PARADOXES.*

PARADOX I.

*The Invention of Gun-powder has been of great Use, and also very
beneficial to Mankind.*

I. If in the account which Virgil gives of the descent of Æneas into hell, he points out, as one of the most tormented and afflicted he saw there, Salmoneus that king of Elide, who in order to acquire to himself divine honours, attempted to imitate, although it was but in a very poor way, the thunder and lightning of Jupiter;

*Vidi, & crudeles dantem Salmonea pœnas,
Dum flammæ Jovis, & sonitus imitatur Olympi.*

I say, if this was the case of Salmoneus, I believe I may venture to pronounce, that the bulk of mankind, would judge the man deserving a much more severe punishment, who invented gun-powder, and contrived cannon, and would think, that he had imitated much better than the king of Elide, the noise, the flash, and the havoc made by those fiery flying meteors. In truth, the world contemplate the Author of that invention with such horror and indignation, that they detest his name. And Quevedo speaks the sentiment of them all, or they nearly all assent to the opinion he expressed of him, in the following lines:

He was of iron race and heart,
In concave metal vast machine,
Who first combustible with art
Did shut;
And then to raise a horrid scene,
By violence produc'd a flame,
Destructive as his savage mind,

And thus transferr'd his odious name
As infamous to all mankind.

II. This abomination of the inventor, arises from the world's considering the invention of gun-powder, as a most pernicious thing to the human race, as in consequence of it, the number of violent deaths has been greatly increased; but this is an error, which in the discussion of this Paradox I propose to banish from the world; and do not doubt, that by the help of a little reasoning and reflexion, I shall be able so to explain the matter, as to cause the mistake to vanish.

III. So far is the common conjecture, that the invention of gun-powder has increased the mortality of mankind, from being true, that on the contrary, it has lessened it. It is a notorious fact, established upon the faith of all antient and modern histories, that when they only used what are commonly called white arms in war, the battles were much more bloody than they are at present. It was very rare then, that the dispute was decided, when the contest lay between troops that were nearly equal in valour and discipline, till one of the parties were almost half destroyed; but at present, instead of that, the death of a tenth part, is generally sufficient to determine the victory in favour of the conquering side. I confess, that this in some measure may be owing to the superior perfection the art of war is arrived at now, compared with what it was formerly. I say in some measure, because I apprehend the greatest difference, should be imputed to the different way of fighting. In former days, when they fought principally with swords, or cutting and thrusting weapons, the troops could not engage, without being intimately mixed one with another. This mixing, conduced to irritate men's minds, and caused a greater difficulty in discerning the advantages one army had gained, and the disadvantages the other laboured under: it also created great confusion, and made it difficult for them to attend to or obey orders; and likewise made it very hard to withdraw the vanquished out of the reach of the conquerors; all which causes, concurred to make battles very obstinate and bloody. Now-a-days, it is sometimes sufficient to decide the contest, if one of the parties before their coming very near each other, is greatly disordered by the fire of their adversaries; in which case, if the

general upon weighing the circumstances, concludes he cannot repair the mischief, he orders a retreat.

IV. In the sieges of fortified towns, this difference of bloodshed is still greater; for the reduction of places, is become much more easy by the invention of gun-powder, and the slaughter attending reducing them greatly less, than it was before the use of it. The siege of Troy, which we are told lasted ten years, would probably not have continued two months, if they had then known the use of cannon and mortars; by so much the more havoc these implements make of the stone walls, by so much the less, is the havoc made of men's lives. Bombs and cannon balls, are more terrifying than destructive. All hear their noise, and but few perish by their flash. Frequently, the surprize and panic they occasion, redeem people from the mischief of them, because the garrison, by being intimidated before they are considerably wasted, entertain thoughts of surrendering; by which means, an infinite number of deaths, both on the part of the besieged, and the besiegers, are prevented.

V. It has not only been observed, that there has been a saving of men and time in sieges, since they have introduced carrying them on by artillery; but it has also been remarked, that in proportion as the fire of them has been augmented, the destruction of the human race has been lessened. Upon this experience, and with a view of preventing the effusion of blood, Louis the fourteenth during his reign, either from his own feelings, or by the advice of his best officers, directed, that they should increase the expence of gun-powder in sieges. And Spain once imitated this practice very happily, and with great success; for having observed that the siege of the town of Namur in 1695, on account of the smallness of their fire, cost them much time, and a great number of men, before they could reduce the place; therefore when they came to besiege the citadel, they for the space of seven days, kept an incessant fire against it, with a hundred and forty cannon, and a hundred mortars and royals; by which means, they obliged it to surrender in much less time than they spent in besieging the town, and when the garrison consisted of eight thousand good troops able to do duty, exclusive of sick and wounded. It is a fact, that this good purpose was attained upon that occasion, and would no doubt be attained upon others of the same sort, not

only on account of the terror, which such a fire impresses the besieged with, but because also, the continual fatigue it exposes them to, exhausts their spirits, and does not permit them, either to eat or sleep in safety, and obliges them to exert an almost continual bodily labour, in repairing the breaches, and clearing the ditches of the rubbish that is beat down into them, and also in transporting materials to make breastworks at the places attacked, together with ammunition and other things. Where the garrison is not composed of veterans, the terror occasioned by the noise of so many cannon and mortars, joined to the tumbling of the edifices, is sufficient to intimidate, and frighten them into a surrender. And the same thing will happen, when there are a great number of inhabitants shut up in a place, although the garrison should be a veteran one; as we are informed by that great master in the art of war, the Marquis de Santa Cruz de Mercenado, in the fifth book of his Military Reflexions.

VI. It being then certain, that gun-powder has prevented the loss of a great many lives in war, the mischief that has happened from it is very light, compared to this great advantage, nor is the evil of its affording people the means of putting one another to death, to revenge private grudges, to be compared with this benefit; for such misfortunes do not amount to a thousandth part of the other advantages. Nor should all the deaths that happen in that way, be charged to the account of gun-powder; for the steel in most of these cases, is commonly the instrument of vengeance, and there are many of those instruments contrived very artfully, for the purpose of giving desperate wounds to people when they are off their guard. To this we may add, that rigorous laws against peoples carrying concealed pistols, may in a great measure, prevent these cruelties from being perpetrated by the means of gun-powder; so that by making a general computation, of all the good and bad resulting from gun-powder in these respects, we shall find, that for one man that is slain by it in consequence of private piques and quarrels, the lives of a thousand are saved, that would otherwise have been lost in the disputes between princes.

VII. If we consider powder with regard to the advantages we derive from it in other respects, we shall find it to be very useful and beneficial to mankind, for it assists us to kill game, to extirpate wild beasts, to smooth

rugged surfaces, to blow up lime-stone, to open passages through craggy mountains, to stop the progress of fires, and is likewise useful in a thousand other instances.

VIII. From all which it may be inferred, that the inventor of gun-powder, instead of meriting the execrations that are fulminated against him, is deserving of the thanks and applause of the world. Who this inventor was, agreeable to the general opinion, may be seen in the twelfth Discourse of the fourth volume of the Theatrico-Critico^[1].

PARADOX II.

What is commonly called Clemency in Princes and Magistrates, is pernicious to the Public.

IX. Clemency, as it is explained by Moralists and Theologians, is a virtue; but as understood by the vulgar, it is a vice. This different acceptation of one and the same word, may be easily accounted-for. If we advert to the doctrine laid down by St. Thomas, we shall there find, Clemency explained as not inconsistent with severity. But I ask whether in the idea of the vulgar, these two qualities are not very different? They are clearly so; because that which is termed severe, they without the least hesitation deny can be clement. Therefore the signification which the vulgar give to the word Clemency, is different from that which is given to it by learned and wise men.

X. Severity then in the opinion of these last, is an habitual inflexibility of the mind, which will not relax in the punishment of crimes, whenever the dictates of right reason require they should be punished. Clemency also, as explained by them, is an habitual disposition to lessen the punishment of crimes, whenever the same right reason dictates that they should be lessened; *Quando oportet, et in quibus oportet*, says the angelic doctor, and it is from his doctrine that this definition is taken. It is clear, there is no opposition, but rather an agreeable harmony between these two qualities; and it is also clear, that the vulgar look upon that inflexibility of the mind in which severity consists, as diametrically opposite to clemency; and thus

they call obdurate, rigorous, inexorable, and austere, all those, who possess that inflexible habit.

XI. That prince or magistrate is clement in the opinion of the vulgar, who is to be wrought upon by the intreaties of friends, the tears of the guilty, and the cries of his orphan family; and who indulges the softness of his own disposition, and is induced from these motives, to mitigate the punishments, which the law has allotted to crimes. But in reality, this is not Clemency, but injustice. It is rather a vileness, and a weakness of mind, which is cloaked under the name of Clemency. He is a protector and an encourager of wickedness, who from such considerations, and without any other motives, slackens his hand in the punishment of crimes. He is an indirect tyrant of the state, and becomes an abettor of all those evils, which are occasioned by the daring rashness of delinquents; and he is also a multiplier of them to an excessive number, by his forbearing to deter bad men from committing them by proper examples; and it is for this reason, that we said in the title of this paradox, that that which is called Clemency in princes and magistrates, is prejudicial to the public at large.

XII. Who then should be deemed truly clement? Why he who after duly weighing and considering all the particulars and circumstances of a case, is convinced by the dictates of right reason, that he ought to lessen the punishment assigned to a crime by the common law, and does lessen it in consequence of such a conviction. This is all agreeable to the doctrine of St. Thomas before quoted; and from hence it may be inferred, that the exercise of clemency can never be arbitrary, as is generally imagined. I mean, that the lessening a punishment which the law prescribes for a guilty person, can never be supposed to depend upon the mere will of a prince or magistrate. He ought only to lessen it, when he finds after maturely weighing the circumstances, that it should be lessened, for if they would not justify his doing it, he should not lessen it at all. There is no middle way. Clemency is a moderating virtue, which if exercised with excessive zeal may become vicious. I am very well aware, this is giving much less extension to the virtue of clemency than the general opinion allows to it. But that is nothing to the purpose, for this is the true and sound doctrine respecting it.

XIII. The just motives for lessening punishments in various cases, are many; for example, the antecedent merits of the convicted person, the use he may be of to the public, his known ignorance of the consequence of what he did, or his having committed the crime inadvertently; any serious inconvenience that might result to the public from his punishment; or any convenience that might be derived to the state from moderating it, &c.

XIV. That great Asturian hero, Pedro Menendez de Aviles governour of Florida, acted on various occasions, and in matters of the utmost importance, contrary to the orders that had been given him by the king. For each of these transgressions, according to the strict letter of the law, he deserved to be punished capitally. The king, and a king who was so zealous of his authority as Philip the second, pardoned them all; but not totally, for we may estimate as a partial punishment, his delaying so long to confer on him the rewards due to his signal merits, during which interval, that eminent man experienced not a few pinchings and inconveniences. The king acted clemently by proceeding in this manner; for it would have been unjust, cruel, and savage in many respects, if he had adhered to the letter of the law in punishing him. The state would have lost a most profitable subject, some signal merits would have gone unrewarded, and so ill judged a precedent, would have been productive of great disadvantages to the public, because other commanders, who might afterwards find themselves in circumstances where strictly obeying their orders would be attended with pernicious consequences, would obey them notwithstanding, for fear of the punishment. Even without the assistance of so bad a precedent, that dread, occasioned the ruin of the grand Armada, fitted out by the same king Philip for the chastisement of England.

XV. I suppose, that Peter Menendez, having always been successful when he acted contrary to his orders, conduced much to excite, if it was not the sole cause of the king's lenity and benevolence to him. But even the plea of this merit, was of no avail to that valiant youth the son of Manlius Torquatus, who after returning home victorious and crowned with laurels, was by order of his own father put to death, for having contrary to his orders, fought with, and defeated his enemies. This was acting with improper severity, for although the offence by the strict letter of the law,

should have been punished capitally, the putting him to death was savage, cruel, inhuman, and barbarous. The ardour and fire of his youth, ought to have been admitted as a plea to mitigate his fault; but there was much more room for shewing lenity to him on account of the zeal he manifested for the public good, by taking the advantage of a favourable conjuncture, which it was impossible for the Consul to foresee when he gave him his orders, and thereby, doing so eminent a service to his country. But the ferocious and sour virtue of the obdurate Manlius, neither weighed circumstances, nor attended to, or was influenced by the dictates of morality; and thus, he unjustly deprived his country of the life and services of a youth, who gave the most promising hopes of his becoming one day a great commander.

XVI. When the circumstances of a case do not afford just motives for departing from the letter of the law, there is no room for the exercise of clemency; as the departing from it in such an instance, would be injustice, and it is impossible that the same action, should be conformable to one virtue, and contrary to another, because at that rate, it might be both good and bad at one and the same time. Thus in these cases, there is no other course to take, but that of applying the punishment the law prescribes, without regarding the clamours of weak-minded people, who are capable of censuring such conduct as over-severe or harsh; for acting in this manner conduces to the public good.

XVII. Annon, the pious archbishop of Cologne, did, in the eleventh century, cause the eyes of several of his judges to be put out, for having given an unjust judgment against a poor woman; but he left one of the number with a single eye, in order that he might be able to lead the others about. I suppose that such an example, could not fail to fill the whole city with horror; but although many might exclaim against the cruelty of it, it was still just and useful, as the blinding of those few judges, might contribute to open the eyes of an infinite number of others, and cause them to look attentively, how they pronounced sentences in future.

XVIII. The case I am about to mention, is more singular still: when Count Evkembaldus the sovereign of Burdan, was in a very weak and infirm state, a complaint was preferred to him against a nephew of his, who

had violated the chastity of a young lady, and upon the charge being clearly proved against him, he, as he was a zealous lover and assertor of justice, ordered him to be put to death; but those who were directed to execute this sentence, evading doing it, upon a supposition that the Count must soon die, somebody informed him of the omission; and as he saw clearly that in his present weak state, although he should repeat his orders, they would not be executed, he artfully contrived, by declaring that his indignation against the youth was appeased, and that he was disposed to forgive him, to get him brought into his sick room, where coaxing him under some pretence to approach his bed, he seized him by the collar with his left hand, and with a poignard that he held concealed in his right, stabbed him into the throat, and killed him on the spot. Many seemed scandalized with this act; but there is no reason to suppose, that it was displeasing in the eyes of God.

XIX. This inviolable integrity in the administration of justice, is no indication of obdurateness, but is rather compatible, with the greatest tenderness and compassion the human heart is susceptible of; although, where effective clemency cannot be admitted, there may be room for the introduction of the affective.

Briante Prieneus, one of the seven wise men of Greece, was observed to weep very bitterly after condemning a guilty person to death; upon which, somebody present asked him why he lamented, since it was in his own power to pardon the man; to this he answered, *You mistake, for it is by no means in my power, and therefore I weep. His life is a debt that is due to justice, and this tenderness of mine, one that I owe to nature.* It is said of Vespasian, that he frequently wept and lamented in tears, the deaths of guilty persons, whom he himself had justly condemned.

XX. To him who has a heart so delicate, that the gentleness of his disposition is apt to degenerate into weakness and debility, I will prescribe an admirable remedy, which shall comfort and strengthen his heart, and not lessen or abate its softness. This consists, in directing his attention to contemplate another object, and in making that the object of his compassion. Let us figure to ourselves a judge, at the crisis when he has just passed sentence of death on a notorious rogue, who has been guilty of many cruelties and outrages, and when he is assailed with the cries and intreaties

of the convict to be merciful to him; and let us also consider him, when those intreaties are afterwards repeated with the enumeration of all those particulars, that are most likely to excite tenderness and compassion, such, as imploring him to commiserate his distressed wife and children, and to look with an eye of pity, on the miserable and afflicted state of an unhappy man, who is all contrition and repentance, and whose ignominious death, will bring disgrace and affliction on his innocent friends and relations. Upon hearing this, he may be apt to say to himself, taking away the life of a man is a terrible thing, and may perhaps revolve in his mind, all the horrors of a person leading to execution, and also those he may feel, when he is just on the point of suffering the punishment he has been condemned to by the law. These reflections may possibly stagger his constancy, and dispose him to pardon the offender.

XXI. But if he would turn the eyes of his understanding from contemplating this tragical picture, and view another, which is much more tragical, and which is drawn and depicted from the circumstances that appeared in evidence at the trial; he might there see representations of cruelty and outrage in a variety of shapes, and distress and misery of various kinds, which have been brought by the depredations of this savage fury, on numbers of innocent people, who are all then praying that justice may be done on this violator of the laws of God and man; this enemy to the peace, security, and happiness of mankind. When a judge contemplates such things, ought he not to be more affected with the cries and lamentations of these sufferers, than with the intreaties and affliction of a wretch, who has been the author of so many evils, and whose misery and distress has been brought upon himself, by his own outrages and misdeeds?

XXII. It may be replied to this, that these mischiefs can't be remedied by the execution of the man, and that putting him to death, will be only adding a new tragedy to those already exhibited; but have patience. Although it is true, that this will not remedy the mischiefs that are past, it will deter others from committing the like offences for the future. The pardoning crimes, operates like a contagion; and the impunity of a delinquent, inspires others with boldness, and infects them with the inclination of becoming delinquents also; on the contrary, by punishing

such a man, you strike terror into ill-intentioned people, and prevent the distresses of thousands; and although you can't remedy the misfortunes of the innocent persons who have already been injured, you may prevent the like mischiefs from being done to an infinite number of others. Let a judge now weigh all these things, and then determine, whether advancing the public good, and doing justice to these injured people, ought not more to excite his compassion, than that devil in human shape who awaits the execution of his sentence; and finally, I must observe to him, that if he lets such a miscreant go unpunished, those very innocent people whom he has injured, will implore the justice of heaven against him for having pardoned so horrid a wretch.

PARADOX III.

What is termed liberality in princes, is for the most part injurious to their subjects.

XXIII. I consider liberality, not only as a virtue, but as a most noble one, that is so much the more worthy to be treasured up in the breast of a man, by so much the more his station of life is exalted. It is certain, that although all vices are vile and base, and all virtues noble, there are vices, which in an especial manner deserve to be stigmatized with the epithet of sordid, and that there are virtues, which shine forth with a superior splendor and dignity. Among the first sort, avarice should be classed, and among the second, liberality.

XXIV. From hence it may be inferred, that avarice, which is always a vile quality, in princes, is a superlatively vile one, as this meanness of spirit, is unworthy of the elevated dignity of a throne. Vespasian was a prince of admirable endowments, he was a great warrior, politician, and magistrate, and was besides temperate, discreet, and affable; but his avarice was a dark shade, which obscured all these perfections; so that the most a person can do who reads his history, is not to abhor him, but he never can bring himself to esteem or love him. He, to increase his revenues and fill his coffers, went to the extreme length, of laying a tax on the excrements of the human body;

but the matter out of which the tax arose, was not so noisome and stinking, as the tax itself.

XXV. Although, it does not follow from hence, that prodigality, which is a vice diametrically opposite to avarice, is not a great blemish in princes; for in truth, it is more blameable in them than in private people. A private prodigal, wastes his own substance; a prince, the substance of other men. A private person by his extravagance, hurts himself; a sovereign, by his, injures a whole community; so that, although the two vices are unlike in themselves, when centered in princes, they produce with respect to the public, the same effects. The avaricious prince impoverishes his people to enrich himself; and the prodigal one, impoverishes himself to enrich others. What the first heaps up, is buried; what the other amasses together, is dissipated; and by attending closely to this object, you will find, that prodigality is the most pernicious vice of the two; because what a prodigal sovereign squanders in needless largesses to enrich particular people, does not return, or if ever it does, it is very late, or it is by some rare accident, that it ever returns again to the public stock; whereas what an avaricious prince hoards up, may be serviceable in the days of his successor, and may greatly contribute to lessen the burdens of his subjects.

XXVI. But how shall we define what should be termed prodigality in princes? Why by calling all that such, which is commonly stiled liberality. The vulgar, and even those who are superior to the vulgar, allow of a large extension, to the arbitrary and voluntary expences of princes. It is commonly understood, that when a prince from caprice, or from particular affections for a subject, makes him a present, that the donation should be proportioned to the power and grandeur of the person who bestows it; but I consider the thing in quite a different light. Whatever considerable sum of the public money is expended, which is not laid out, directly or indirectly, to advance the public benefit, is injustice and profusion. That which comes out of the pockets of the public, should be expended in such things as are beneficial to the public. Would it not be most unjust providence, to apply, for the sake of indulging the caprice or ostentation of a sovereign, that which is contributed by millions of people, to enrich a particular person, who by some chance accident, in a matter that was of little importance to

the community, has done something that was agreeable to, and that has gained him the favour of his prince?

XXVII. Alexander ordered his treasurer to give to the philosopher Anaxarchus, any sum of money he should desire. He requested a hundred talents; and the treasurer informed Alexander of the excessive demand of the philosopher; to which Alexander answered, *he has done very right, for he well knows that he has a friend who is both able and willing to give him that sum*; and ordered the treasurer to pay him the money immediately. Is this liberality? It is true, that it is celebrated as such in an infinite number of books; but I say, that it was no such thing, for that it should rather have been termed mad prodigality, which is the legitimate child of vain-glory; and that it was not only prodigality, but cruelty and tyranny. With those hundred talents, he might have administered to many distresses; and if a prince has superfluities, he should lay them out for such purposes. But refraining to administer to the hunger and necessities of numbers of poor people, to satiate the gluttonous cravings of an avaricious philosopher, was a glaring act of partiality, which rendered it doubtful, which of the two was the most unjust person, Anaxarchus for demanding such a sum, or Alexander for gratifying him in his demand.

XXVIII. The same Alexander, being requested by his friend Perilus to furnish him with a sum of money to portion out his daughters, ordered fifty talents to be delivered to him. To which Perilus modestly replied, that ten would be sufficient; to this Alexander answered, *that is not a matter for my consideration, for although ten talents would be as much as would answer your purpose, it would not be a present suitable to my grandeur*. I find this celebrated by many writers, as a noble and magnanimous act, and Alexander's saying upon doing it, as a commendable and a well-pointed one; but in my opinion, the act was an act of madness, and that his saying upon doing it, was a very weak and trifling one. Does the grandeur of a prince consist in extravagances and profusions? Does it display grandeur, to take from many what is absolutely needful for them, to furnish a few others with superfluities? It does not, but rather favours of injustice, baseness, and tyranny; and they only can call such behaviour magnanimous, who have lost the use of their understanding.

XXIX. A thousand crowns in specie, were one day presented to Don Alfonso, the fifth king of Arragon of that name, and the first of Naples. A person who was standing by at the time, said, *How happy it would make me, if all that money was mine!* To which the king answered, *Take it then, for I am desirous of making you happy.* Was this magnanimity? I know that it has been cried up as such; but I say, it rather shewed weakness of mind, and a want of proper resolution, to resist an absurd impulse of vain-glory. I suppose also, that it was from the want of thinking or reflexion, that the king was guilty of that profusion, and that he was hurried into it by his vanity, which suggested to him, that making the man a present of the money would blazon his fame, and manifest to the world, that he had both the disposition and the power to make a man happy. But I would ask him (and this is a question that might be put to all the princes in the world) whether, if it is an act of greatness to make one man happy, it is not a much superior one to make a great many so? If it is glorious in a sovereign to make an individual happy, is it not beyond comparison more glorious, to make a whole nation so? And there is no doubt but this might be done, if a prince would avoid all profusion, and regulate his conduct by a discreet œconomy; if he would curtail all superfluous expences, be a check upon the avarice of his ministers, or else deliver the administration of his affairs into the hands of none but men of integrity, who are capable of proportioning the contributions to be paid by his subjects to their abilities, and who should be careful, not to over-burthen the husbandmen and manufacturers; for these are the people, who by their labours, are the principal instruments of enriching a state, and whenever they find the weight of the taxes, squeeze out of them the greatest part of their earnings, they will leave off work, and betake themselves to an idle and vagabond life. To sum up the whole, a prince, by conforming to the precepts that are dictated by justice, religion, and prudence, and by not bestowing on any one in particular, more than his necessities demand, or than is due to his merits, will become the common father of his people; and by extending his paternal care, and dispensing his generosity with an equal and impartial hand to them all alike, will be able to make them all happy.

XXX. The royal treasury, may be compared to the ocean. It receives its pecuniary contributions from all the monies in the kingdom, as the ocean does its stock of waters from all the rivulets, fountains, and streams in the

whole world. The royal treasury then, should do by the kingdom, as the ocean does by the world; that is, after permitting those waters to be exhaled in vapours, return the same stock in refreshing rains, to fertilize the earth. It would argue a great defect in the sovereign providence, as the stock of the ocean is supplied by all the waters of the world, if he was only to permit a return of them, to fertilize here and there a district, in consequence of which, all the rest would become barren. Just so, would it be an intolerable absurdity in human government, to apply the money of the public treasury, to which the whole kingdom has contributed, in prodigal donations to enrich a few individuals, and by with-holding it from all the rest, leaving them in distress and misery.

XXXI. The reigning emperor of China at the beginning of this century, was with respect to the matter we are treating of, one of the greatest examples, that ever has, or perhaps ever will be exhibited to the world. I rely on the authority of Father Contanein for the truth of this assertion, and on the account he gives of this emperor in his letters, which are dated at Canton the latter end of the year 1725, and which are copied into the 18th volume of edifying and curious letters of foreign missionaries; but I only at present, have before me an extract from them, which is inserted in the second volume of the Memoirs of Trevoux of the year 1728.

XXXII. That prince, laboured incessantly to promote the good of his subjects. The object absorbed all his thoughts, and occupied his whole attention. Every day of the year, and all hours of the day, were days and hours of giving audience and expediting business; and he scarce devoted any to amusement or recreation. To provide for the conveniences of his own person, he used the riches of his treasury with great moderation; but he expended them with a truly royal magnificence, when he administered to the necessities of his people. He procured punctual information, of the state and condition of all his provinces, and took care to know, which of them was opulent, and which indigent; this he did, to the end, that he might succour with the greater ease, such of them as were in distress. If any town had been desolated, either by an earthquake or a conflagration, if any province had suffered by an inundation, or by any adverse accidents, had been prevented from yielding their usual produce, or if by any other chain

of accidents, a province happened to be impoverished, he immediately remitted large sums, to repair their buildings, and to relieve their poor. All his subjects experienced to flow from his bowels, a balsam of tenderness, compassion, and paternal love, which healed all their calamities, and relieved all their distresses.

XXXIII. In the same year, 1725, some provinces suffered greatly from excessive rains, and the emperor took measures to relieve the distresses that had been occasioned by them, which, in order to do more effectually, he sent instructions written in his own hand, to the principal men and mandarines of all his empire, which began thus; *This summer, extraordinary and uncommon quantities of rain have fallen, and the provinces of Peking, Chantong, and Honan, have been inundated by them. I feel much for the distresses of my people who inhabit those provinces; and have it much at heart to relieve them. Their afflictions are continually present to my imagination both night and day, nor can I enjoy sound sleep, or tranquillity of mind, while I know that my subjects suffer; and as it is absolutely necessary to send immediate relief to those vast numbers of poor distressed people, I command you great men and magistrates of my empire, to choose some trusty and able persons, such as are capable of executing my instructions, and who prefer the public good to their own private interests; and dispatch them to the three before-named provinces, to distribute to the afflicted inhabitants of them, the tokens of my benevolence and compassion. Let them scrutinize the most obscure and remote corners, to find out all the poor and distressed, to the end, that no one may remain without proper succour and relief. I know that it has sometimes happened, that in the making these sort of distributions, acts of injustice and partiality have been committed; but I will take care that the conduct of those you send shall be watched, and do you look to it also.*

XXXIV. There is another testimony in the before recited letter, which does honour, not only to the generosity and benevolence of this prince, but to his heroic disinterestedness also. It having been customary with him for a great many years, to release a certain province from paying a part of their annual tribute; which he did because he thought it was just and necessary. The inhabitants, to express their gratitude for this kindness and generosity

to them, meditated erecting some monuments to his honour, and had actually begun the work, which the governor of the province informed him of. To which information of the governor's, the emperor gave the following answer: *What you acquaint me with, is totally disagreeable to me, and what I by no means approve. When I conceded this grace to the inhabitants of the province you preside over, I had no other motive or view in doing it, than that of acting justly by, and making all my subjects happy alike; but had no intention of procuring to myself a vain honour. Such expences are superfluous and unnecessary, and can never be of any use or benefit to me; and as I have sent proclamations through all my empire, exhorting my people to practise frugality and œconomy, I wonder how they could presume to think of running into these needless and mad expences, or how you could permit them to do it: it is also to be apprehended, that the inferior officers who are generally the collectors of money for these purposes, may be tempted to put part of it into their own pockets. Prohibit therefore immediately, the proceeding any further in this matter, and with respect to the edifice and monument of stone, I forbid the erecting of it; and I repeat again, that when I do these acts of grace and favour, it is not with a view of acquiring a vain reputation, but because I think it just and necessary; and to the end, that every one may be enabled to do his duty in society, and that the tranquillity and happiness of my subjects in general may be established on a solid foundation.*

XXXV. All the conduct of this prince was of the same tenor. He with a most sagacious penetration, attended to the proceedings and conduct of all the mandarines; and gave them instructions, that they should apprise him of all that occurred to them, which might conduce to promote good government. He made many regulations, that were just and wise; he established premiums for the husbandmen, who had distinguished themselves by their industry, and the improvements they had made in agriculture; and for the manufacturers and mechanics, who had signalized themselves by their diligence and ingenuity; he made provision for the widows of virtuous citizens, and settled rewards that were to be paid to such children, as distinguished themselves by their filial care of, and tenderness to their aged parents, &c. and is this Prince who was so perfect in his morals, and so great a pattern of policy, the same man, who proscribed Christianity throughout all his dominions? I fear we must be obliged to

answer in the affirmative, and to contemplate with astonishment, the inscrutable secrets of the divine Providence; and to exclaim, *Oh! how incomprehensible are God's judgments, and how untraceable are his ways!* But the blindness of this emperor in matters of religion, should not preclude our recommending him as a signal pattern, of that sort of œconomy and liberality which should be practised by princes.

XXXVI. I have said of the *œconomy and liberality*, for both these virtues, are found to be perfectly compatible with each other, and to be admirably reconciled together in the practice of that prince. The true and proper effect, and essential operation of liberality, according to the doctrine of St. Thomas, consists, in proportioning our affection for money in such a manner, that our excessive love of it shall never obstruct our laying it out for such purposes, as are just and laudable. Thus the epithet of liberal, does not belong to him who expends it to indulge his whim or humour, or from motives of ostentation, or to gratify his affection for some particular people, he is desirous of enriching; for all this is prodigality: but he deserves to be called liberal, who applies money to promote and accomplish virtuous ends, and who is always ready to part with it, when it may be made conducive to promote such purposes. There is field enough within these limits, for the exercise of the virtue of liberality. He is liberal, who succours the poor, rewards the deserving, supports by generous donations men of ability, and who constructs useful public edifices, and, in general, all money that is laid out for the benefit of the public, may be said to be expended on objects of liberality, which liberality, if judiciously extended, may be deserving of the epithet of magnificence. These two last virtues are distinguishable from each other, by the first being limited not to exceed the bounds of moderation: and the other being permitted to extend to the expending vast sums, provided the money is laid out from laudable motives, and on objects, that are really conducive to the public utility. Thus Louis the Fourteenth did an act of magnificence, in building the Hospital of Invalids at Paris, and did a much superior one, by constructing the canal, that makes a navigable communication between Languedoc and Bourdeaux; because the great expences that attended executing these works, were incurred to promote the public good, and actually were great advantages to the public at large; but the palaces that were built by Nero and Caligula, which occupied as much ground as two large towns might have stood on, do not deserve to be called

works of magnificence; because the vast sums that were laid out in constructing them, were by no means expended to promote the public good, but only to gratify the ostentatious vanity of two profuse men. The emperor Adrian was magnificent, by forgiving at once, the payment of all that was owing to him for the sixteen preceding years, by Rome, Italy, and all the imperial provinces; but Alfonso the Twelfth of Castile was prodigal (provided the story that is told of him, and circulated in so many books respecting this matter is true) in expending a vast sum of money, for the redemption of Baldwin, emperor of Constantinople; as the first case concerned the whole Roman empire; and Spain was not at all interested in the last.

XXXVII. Finally, A prince may exercise not only his liberality, but his magnificence also, by bestowing great donations, upon here and there an individual of signal and superior merit, (but I would be understood to mean such merit as has been beneficial to the state) because in doing this, his attention may be supposed, as not confined to rewarding the virtue of a single man; but to the exciting the emulation of many. And considered in this light, what Spain gave to Colon, did not exceed the bounds of justice and moderation; what she gave to Cortez was little; and what she gave to the great Captain next to nothing. When a prince or nation should exercise magnificence, if the donation they bestow is not equal to the merit of the person it is conferred on, or the dignity of the party who confers it, it can never deserve to be termed liberality, for there is no middle way in these cases, and if the gift does not amount high enough to be deserving of the epithet of magnificence, it declines to the other extreme, and sinks into parsimony.

PARADOX IV.

There is more Objection made to promoting People to important Employments on account of their Youth, than there ought to be.

XXXVIII. As in the use of their potential or intellectual faculties, some give early tokens of dulness or perverseness, and others display bright tokens of commendable and useful talents; which render it probable, that

the state would prevent great mischiefs by the early chastisements of the first, and that it would reap great benefits by the timely favouring and encouraging the virtuous bodings of the second; as there are young men, who in prudence and understanding excell those who are much further advanced in years. If such, in the prime of their lives, were promoted to fill posts of importance, the state would for a long time, reap the benefit of their good administration: whereas, by deferring their advancement till they grow old, this benefit is but of short duration. The most learned and penetrating society of Jesuits, at the age of thirty-eight, raised to the high post of Prepositor-general of their Order, Father Claudius Acquaviva. Who can entertain a doubt, but that in so extensive a society, there were many men further advanced in life, possessed of all the requisites for the well-discharging the duties of such an office? But notwithstanding this, Father Acquaviva, at this age, was preferred to all the others, to fill this important station; this was done, either because he possessed the requisite qualities for discharging the duties of the office in a more eminent degree than the others, or because, although they might be equal to him in talents, there was a probability of their enjoying the fruits of his excellent government for a much longer time, than they were likely to enjoy the benefit of the services of those who were further advanced in life; in which expectation it turned out afterwards that they were not disappointed. The famous Servita Fray Pablo Sarpi was made Provincial of his Religion, at the age of twenty-seven. The portentous talents of that young man, gave just reason for their electing him, and their judgment in doing so, was afterwards confirmed by the sentiments of the republic; who appointed him, contrary to their ordinary practice, at about the same period his own society conferred this honour on him, their Counsellor. It is true, that this extraordinary favour of the republic, was very injurious to the religious character of Father Sarpi, for he engaged with such warmth in defence of the state, against the pretensions of the Apostolic See, that only in his habit of a friar, he preserved the appearance of being a Catholic.

XXXIX. He who at thirty, has the discretion that people commonly have at fifty, when he arrives at the age of forty, will have a discretion, much superior to that of the bulk of mankind. And this exceeding will be greater still, if from the age of thirty, he begins to exercise his talents in public business, as he will go on improving them more and more every day by his

practice. Why then should not the state encourage the cultivation of a soil, which is capable of producing so much for their service? Or why should they lose any part of the copious fruit, that may be produced from such a stock?

XL. To this reasoning I shall add, that when there is an equality of intellectual talents, the middle age should be preferred to the more advanced, for that is the time, in which the faculties of both body and mind are in their full vigour, which are things of great importance, in the good administration or execution of public business. What might be gained by the experience of a man who has had a great deal of practice; if he happens only to begin to act at a time of life when his faculties are on the decline, may be overbalanced by the loss that would be sustained by the languid execution of his designs. I am persuaded that the miscarriages of Cyrus, Pompey, and other famous warriors in their advanced state of life, who had always been conquerors in the days of their youth and vigour, was owing to no other cause but this; and am of opinion, that they attributed to a falling-off of fortune, what only resulted from a decadence of their robustness, and the activity of their minds.

XLI. It may perhaps be objected to what I have been advancing, that it is in but here and there a rare instance, that this doctrine of mine will apply, for that it is very uncommon to meet in an early time of life, with a man of equal ability, to one who is more advanced in years; and that if I only mean to recommend, that in these rare instances, they shall have the preference given them, I have been beating my brains to little purpose; for that this is already an established practice; and there is no man who has observed the world with any attention, who is not convinced, that we have had here and there an instance, of a young man being preferred to one more advanced in years?

XLII. But to this, I reply in the first place, that admitting, in these particular cases, what is just and proper has been done, it does not follow from thence, that the doctrine we have been laying down is useless; for although our arguments on this head may be unnecessary as an instruction to those who have the dispensation of employments, they may be of use by

way of reproof or correction, to grumblers and complainers. A young man scarce ever attains any honour or preferment, but a thousand old unprofitable ones murmur at it, and not only a thousand old ones who are useless, but the majority of the young ones also; who by being of about the same standing in life, are stimulated to shew resentment, by the fire of emulation.

XLIII. We shall answer secondly, that a young man's excelling old ones in understanding and judgment, is nothing near so uncommon as is generally imagined, but is rather a thing which we experience very frequently. There is scarce a community consisting of twenty or thirty individuals, where we do not see a particular young man, better informed, and more capable than many of the old ones. This proceeds, from the temperaments of men having generally a greater effect on the faculties of their minds, than their ages. The exceeding that a man experiences in himself in this respect at fifty years of age, and at thirty-five, is seldom very considerable, and will rarely be perceptible, if he has not passed the last fifteen years, from a life of much indolence, to one of much application. On the contrary, the exceeding there is between some men and others on account of their different temperaments and geniuses, is enormous. We every day see those, who from their aptness at acquiring every science and faculty, will become adepts in them, in a quarter part of the time, that others are obliged to employ in hard study to attain them.

XLIV. From this great difference in the temperaments and genius of individuals, spring those prodigious advances of some young men in literary attainments, which are frequently not equalled, by those who have studied till eighty. Such for example as John Pico de Mirandola, the Scotch man Jacob Creighton, the Spaniard Fernando de Cordoba, Gaspar Scioppius, Hugo Grotius, Spanolito who is now the wonder of Paris, and others. We might add to these examples many more, which are not so well known, but which are not less extraordinary; but we shall content ourselves with pointing out two, that are the most striking ones. The first is Gustavus de Helmfeld, the son of a Swedish Senator, who at ten years of age, could speak twelve Languages, the Swedish, the Muscovite, the Polish, the French, the Spanish, the Italian, the German, the Flemish, the English, the

Latin, the Greek, and the Hebrew; he besides had a knowledge of philosophy, and a tincture of that of theology, and also understood some parts of the mathematics.

XLV. But the prodigious child, that was born at Lubec in 1721, and died in 1725, exceeded every thing we have hitherto heard of. His name was Christian Henry Heneiken. The relation I shall give of him, is taken from the first volume of the Memoirs of Trevoux of 1731, into which it was copied, from the accounts we have had of him from various authors of the same city and country, all of whom were esteemed men of the greatest credit and veracity. This surprizing child, at ten months old begun to speak, and at twelve, understood the principal events contained in the Pentateuch; and at thirteen, those of the history of the Old Testament; at fourteen those of the New; and at two years of age, he answered pertinently to any questions that were asked him, concerning antient or modern history, and likewise, with respect to geography. He very soon after, began to speak Latin with ease, and immediately upon that, French also. Before he entered into his fourth year, he knew the genealogies of the principal houses in Europe, and explained sensibly and judiciously, many sentences and passages of holy scripture. He soon after this, learned to write, at a time when he was scarce able to hold the pen. He abhorred all other aliment except milk, which must always be that of the nurse who first begun to suckle him; so that he was not weaned till a few months before his death. He was of a very delicate frame, and frequently visited with sickness. And at length, on the 27th of June, 1725, he died; filling with astonishment all those who knew him, at the constancy and heroic resignation he shewed through the whole course of his sickness, till the period, that he surrendered his soul into the hands of his Creator.

XLVI. I am very well aware, that the circumstances of this history may have been exaggerated, but I do not find any thing impossible in them. Who knows the ultimate bounds, to which the ability of man may extend? Perhaps there may be none assigned to it, but that it may continue to increase more and more, and that the limits of its extension have never yet been fixed or assigned. With respect to his essential perfection, all philosophers and theologians agree, that no creature was ever so perfect, but

that God may create another still more perfect. Why then with regard to accidental perfection in the same species, do we not see the same thing? Our gross mode of reasoning, is apt to confine possibility within the narrow limits of our experience. That which we never saw, we imagine can never happen; as if in the little which God has been pleased to present to our view, is displayed the utmost extent of the omnipotent power. Setting bounds to possibility, is setting them to the operations of the all powerful.

XLVII. I agree, that our assent to the existence of a thing, should not be extended so far, as not to be confined within less bounds, than the immense space of possibility; letting our belief be regulated by probability, is much safer, than letting it extend to possibilities; possibility, can only be measured by the extent of the divine power; probability, may be guided and limited by the force and credibility of testification. Thus he would act prudently, who with respect to the history of the child of Lubec, should reject a good portion of the circumstances, but still admit enough of them, to render the story a most wonderful one, the like of which, had never been known in former ages; as it is not probable, that the authors who were the countrymen of the child, should lie exorbitantly, in a case where if they had deviated greatly from the truth, it was in the power of thousands of living witnesses to have convicted them of the imposture.

XLVIII. From the beforementioned examples, and an infinite number of others that might be instanced, may be inferred, the enormous difference there is between some souls and others, which difference, is owing solely to a difference of temperament; compared to which, that produced by a disparity of age, is very small, even if we compute it from early manhood, to the period of decadence. The result of my own observation, with the exception of here and there a rare instance, is, that those who at thirty years old are dull, will always be dull; that those who at that age are imprudent, will always be imprudent; and that those who at thirty, in matters of argument or conversation, do not talk pertinently, or reason well, will never reason well. I do not however deny, that cultivation will not improve both men and plants, but it will never make thorns bear grapes, or brambles figs.

XLIX. I think there seems now to remain but one objection for me to answer out of all those that have been urged to what I have advanced; and that is, that although admitting some have a great natural superiority of talents compared to others; still, the impetuosity and fire which commonly prevails in the flower of a man's youth, is apt to have a pernicious effect on his conduct. Granted. But besides the exceptions to this rule being infinite, as we every day see youths, who are very sedate and discreet; and to which we may add, that there is another passion very predominant in old people that is productive of much more mischief in public affairs, than the fire or impetuosity of youth; I mean the vice of avarice, in the operations of which, there is not a moment's relaxation; on the contrary, the passion of anger, is only apt to be excited by particular accidents, and when the ebullitions of choler abate or subside, they are succeeded by long intervals of calm and tranquillity. Anger is a transitory fury, or a fever of short duration, whose attacks are sudden, and last but for a little while, and which in a moderate space of time vanish and disappears; but avarice is a harpy, that has nested itself in the heart, and is a dropsy of the soul, which increases and grows worse every day. The first, now and then disturbs and irritates the moral temper of man; the other, vitiates all his actions, because its venomous influence ever exists. The first grows weaker every day, and is enfeebled by its own exertions; the other, is incessantly gaining new strength, and becoming more powerful, *vires acquirit eundo*; so that avarice contrary to the ordinary course of nature, is by so much the more vigorous, by so much the older it grows; it is a passion, that not only acts in cold blood, but acts with so much the more activity, the colder the blood grows; and hence it is, that its mischiefs are not only greater than those produced by anger; but they are also much more incurable. Thus if the infirmities incident to youth, are pernicious in those who occupy posts of importance in the public line, those that are attendant on old age are much more so.

PARADOX V.

All Handicraft Trades Should be hereditary.

L. In Lacedemonia, which was one of the best-governed states of antiquity, it was an inviolable law, as Herodotus informs us, that the son of a husbandman, should be bred a husbandman, and the son of a taylor, a

taylor, and the same with respect to all other occupations where people earn their living by their labour. The same practice prevailed in Egypt, and prevails among the idolatrous people of Indostan at this day.

LI. I know very well, that to enforce the importance and propriety of this paradox, the authority of these, and other such examples that might be instanced, is weak, compared to the infinite number of opposite ones that might be produced against it. Therefore, it will be absolutely necessary for us to have recourse to reason, to supply this defect of authority.

LII. It appears to me, that two advantages of great importance would be derived to the public by trades being hereditary; the first of which is, that it would contribute greatly to the perfectioning of arts. When there is no other relation or connexion between the master and his pupil, than the latter being the apprentice of the first, the master is seldom very anxious about instructing his scholar; and what is more, he is not well disposed to communicate to him any particular secrets of his art, which he has acquired by virtue of his own penetration and experience; and generally, contents himself with instructing him in what is commonly known and practised. But there is no reserve of this sort, when the instruction is conveyed from the father to his son, for paternal love cannot assent to it; and hence it is, that where the skill of the master is equal, he will be better taught who learns of his father, than he who is instructed by a stranger.

LIII. By this total translation of skill from father to son, and by the same trade being continued in, and handed down to their posterity, there would without doubt result this benefit, that the art of it would be improved, and that it would continue advancing every age, nearer and nearer to perfection. It is very common, for one artist to make some improvements on what he has learned from another; and it is very common also, for him who has made them, to let them die and be buried with him; on account of its being against his interest, to communicate them to other people. This impediment to making his improvements known, seldom subsists between father and son; because the father, most commonly considers the interest of his son as his own; and consequently, communicates to his son all he knows. If the son from his own ingenuity, makes improvements on what he has been taught

by his father, and hands them down to the grandson; and they all continue to do the same by their successors; the arts by this means, would proceed to approach nearer to perfection every generation.

LIV. There are two other circumstances that are very worthy of being attended to, which would contribute to recommend this political system for improving the arts; the one is, that children would begin to learn much sooner than they commonly do. In the house of an artificer, where the son is destined to follow the trade of his father, he will scarce have done sucking the breast of his mother, when he will begin to drink the milk of his father's instruction; in consequence of which, not only time would be gained, but his application to the business would become more natural to him. The other circumstance is, that the state would avoid the loss of many good artificers, which is occasioned by the inconstancy of the tempers and dispositions of youth. Those, who by continuing to work at the trade they first began to learn, would become excellent artificers, by going on to change their occupations continually, would never arrive at being more, than mere novices, or beginners; and this mischief would be avoided, by obliging the son of every handicraft person, to follow the trade of his father.

LV. The second convenience that would result to the public by trades being hereditary, is, that it would ascertain the ranks and classes of people in a state; it being no uncommon thing, for disputes to arise between families, about birth and precedence; which inconvenience would be remedied by such a regulation, as in such a case, the rank of most people would be ascertained, and the cause of these disputes in a great measure removed.

PARADOX VI.

Torture is a very fallible means for discovering the truth in criminal Cases.

LVI. I enter upon this essay, begging permission of all tribunals of justice to speak my sentiments on this head. I venerate the laws and the use of them; but as I am treating here of such as are purely human, I apprehend I may be permitted to reason on the tendency or propriety of them. Nor is

the torture being admitted in the ecclesiastical courts, as a mode of enquiry in the examination into crimes, any objection to my doing this, for as the learned Canonist Francis Schomer observes, this practice is not conformable to the antient discipline of the church; and he quotes other authors in support of this opinion, and says further, that in a long course of time, it by little and little, came to be transferred from the secular tribunals to the ecclesiastical ones: *Predetentim à curiis sæcularibus ad ecclesiasticas pervenisse* (Schmier in Supplem. ad lib. v. Decret); so that doubting of the propriety of inquiring by torture in the ecclesiastical courts, amounts to no more, than disputing whether the antient or modern practice of those courts, is the most consonant to reason.

LVII. Besides, this being a matter which in its own nature is open to controversy, there are two notable circumstances in it, which encourage me to enter upon the discussion of the point. The first is, that very many people think as I do on the subject, and that among those very many, there are more than a few of the very judges, who practise ordering the torture, in cases where that practice has been long established. Their sentiments in theory do not correspond with their conduct, but notwithstanding this, they act as they should do, for they have no authority to dispense with the laws, but are only ministerial under them. The second is, that the most learned father Claudius Lacroix, has preceded me in publishing the same sentiments I entertain on this subject, in his first volume of Moral Theology, lib. iv. &c.

LVIII. Countenanced and encouraged by so illustrious an author, whose opinion in questions of morality is so much respected, and so generally assented to by all Christendom, I shall enter upon the undertaking boldly, and shall argue with spirit in support of his opinion and my own. The extent of the question is but short, and whenever reason sets its foot within its boundary, it soon arrives at the end of it.

LIX. It cannot be denied, that the not confessing in the torture, depends upon the spirit and fortitude of the person tortured, to resist the rigour of that trial; and I ask now, is this spirit and fortitude produced by the innocence of such a person? It clearly is not, but rather by the robustness of mind or body which that person possesses. The torture then, can be of no

use to ascertain the guilt or innocence of any one, but it is the bodily strength or weakness of mind, of whoever is unfortunate enough to experience its rigour, that determines the point.

LX. Nero having unjustly repudiated his wife Octavia, and married Poppea, she not content with having usurped the bridal bed, and crown of Octavia, but in order to deprive her of life and honour also, accused her of having had a criminal correspondence with a slave. For the purpose of ascertaining the guilt of Octavia, six of her female servants were put to answer by the torture; and what was the result of this? why, that some of them declared her guilty, and others denied it. Did not they all know that the accusation was false? and do not all authors agree that they did? But what did this signify, if, in the torture, their pain, and not their veracity, influenced them to confess she was guilty of the crime? Whoever has fortitude enough to endure the rigour of the cord, will deny the fault he is tortured to confess, although he was really guilty of it; whoever has not, will confess a crime he never committed. The effects of the torture on the servants of Octavia, tended to prove the fortitude of some of them, and the weakness of others; but for ascertaining the truth were of no use.

LXI. It seems then, that the innocent are exposed to equal danger from the torture with the guilty. A terrible inconvenience this, but the worst part of it is, that the danger is not equal, but greater on the side of innocence. I may be told that I am advancing a new paradox, and I confess that I am, but if I am not greatly mistaken, I shall make it appear, that I have been advancing nothing that is not strictly true. It is a known fact, that those who are daring enough to commit great crimes, are much more hardened, robust, and ferocious people, than those who lead a quiet and regular life. It should follow then from these premises, that they are better calculated to endure the rigours of a torture, than the tranquil and inoffensive part of mankind; and it will also follow from thence, that these last will be more likely to confess themselves guilty of a crime they are falsely accused of, than the others will be, to confess one they have really committed. This is a remark made by father Lacroix, and the following are his words: *sequitur per torturas sæpè everti justitiam, quia innocentes, qui sæpè sunt impatientes*

dolorum, coguntur se fateri nocentes; e contra nocentes, qui plerumque sunt ferociores, tolerata tortura se probant innocentes.

LXII. On the contrary, those of quiet and peaceable dispositions, are generally people of more delicate feelings than the others, and especially, if their mode of living has corresponded with the native excellence of their tempers. From whence it should follow, that the probable result of putting one of these to the torture, would be, that he would sooner confess himself guilty of a crime he never committed, than one of the others, would own himself guilty of one he had actually perpetrated.

LXIII. I consider the sentiment of Plato to be a true one, who says, that great vices, no less than great virtues, require great powers and fortitude to exercise them. The serenity with which Geronimus Olgiatus, Balthasar Gerardus, and Francis Ravillac, the first for the assassination of Galeazus Maria duke of Milan, the second for that of William prince of Orange, and the third for that of Henry the Fourth of France, fully demonstrate, that those who dare attempt great and dangerous things, are capable of enduring great torments.

LXIV. I shall conclude this subject, with the striking, and most efficacious testimony of Father Frederic Spe, which throws all the light upon this matter that can be wished for, or desired. This learned German Jesuit, when he is treating of the little dependance there is to be placed on the confessions of people accused of witchcraft, which confessions have been extorted from them by the rack; quotes, in support of what he advances, the testimony of the Baron de Leibnitz, and Vincent Placcius, whom he supposes to be the author of the anonymous Treatise, intituled, *Cautio criminalis in processu contra sagas*.

LXV. Father Spe, when he is speaking of the confessions that have been made by supposed witches and wizards, when put to torture, delivers himself thus; the numbers of lyes they will tell, both of themselves and others, is incredible. All which the judges are desirous they should say is true, they confess to be true; and they own every thing they would have them own of themselves, compelled to it by the violence of the torture; and

they do not afterwards dare to retract what they have said, for fear of being tortured again; thus these miserable wretches, go to their deaths attesting a falsehood. Father Spe concludes in this manner, I am certain that what I have said is just and right, and I appeal for the truth of it, to the judge of that supreme tribunal, where all mankind, both quick and dead, will one day be sentenced to rewards and punishments, proportioned to their merits and demerits.

LXVI. I declare that when I first read this account, I found my whole frame pervaded by a deep melancholy sensation, that resembled somewhat like horror mixed with pity. He who gives the relation, is a learned, grave, exemplary, and sound divine, instructed in this case, not by conjectural reasoning, but by certain informations, acquired in the sacred confessional chair, and taken from the mouths of those, who were on the point of being led to execution as guilty people; and this he declared from repeated experience he had had of those matters, during a long course of years. What can be objected that is deserving of any attention, to so conclusive a testimony?

LXVII. The certainty Father Spe had, of the almost invincible force of the torture, to oblige innocent people to confess themselves guilty of crimes they never committed, is very forcibly illustrated, in a vehement declaration to the judges, with which he concludes his Essay. He says to them, why are you so solicitous in searching out, and why do you take so much pains to discover, people who are guilty of witchcraft? There is no necessity for your giving yourselves all this trouble, for if you want to discover more, I can tell you how to furnish yourselves with them, without labour or difficulty; take the first Capuchins, the first Jesuits, and the first men of any other religious order, that you shall meet by chance, and put them to the torture; and if you shall be desirous of making them do so, you will find that there is scarce one of them, who will not confess that he has been guilty of the crime of witchcraft. If any of them should deny it, repeat the torture two or three times, and I will engage that you will bring them to. Pluck out their hair by the roots, exorcise them, and repeat to them the common-place cant that the devil has hardened them; and as if you was convinced this suspicion was just, behave to them, with determined inflexibility, and you will

perceive, that there is not one among them, who will refuse to submit. I have pointed out to you already how you may be supplied with an ample stock of sorcerers, but if you want more still, take the prelates of the churches, and all the canons and doctors belonging to them; and by the application of the same means, you will be convinced, that they may be all brought to confess themselves guilty of witchcraft; for people who have been bred up in so delicate a way, will hardly find themselves able to resist the rigours of a torture. But if you are not satisfied with this supply, and should want more still, I would recommend to you, to put one or two of yourselves to the torture, and you will find that you will confess also in the same manner the others have done; and if immediately after this, you were to torture me, you would be satisfied, that I should do the same. And by this mode of proceeding, you may make witches, wizards, and magicians, of all mankind.

LXVIII. I am ready to admit, that so vehement a declamation, should not be made to all judges indifferently, but that it should be addressed only to such as proceeded with the little caution those did whom Father Spe alludes to; although it is certain, that in accusations for witchcraft, there is more danger of innocent people being brought to capital punishment by the torture, than there is by charges of any other nature. Every man of discretion knows, upon what ridiculous grounds, the common people spread reports of folks being guilty of witchcraft, and with what ease the world believe them, and how ready they are to testify to the truth of such reports. In consequence of which, if the accused persons happen to be brought before judges, who, like the rustic vulgar, are filled with the notion or belief, that there are multitudes of witches and wizards, they immediately have recourse to the torture, and innocent people are tormented like delinquents. From whence it follows, that those who are falsely accused, from their inability to resist the pain of the rack, assent to the interrogatory, and against their consciences own themselves guilty. To this number, we may add many others, who own themselves guilty from delusion or infatuation. This delusion is contagious, and multiplies and spreads exceedingly, whenever there prevails a rage for finding out witchcrafts; for the numbers of these delinquents are increased, in all places where there are officious inquisitors after sorcerers, just as the numbers of possessed people are, where there are plenty of persevering and absurd exorcists.

LXIX. But notwithstanding that in such accusations, on account of their being so frequently ill-founded, the hazard of innocence being oppressed by the anguish of the torture becomes greater than in other cases; still the same danger subsists, though not in so great a degree, with respect to all those who are falsely accused of any other serious crimes. I mean, that if any one from want of fortitude, confesses in the torture the crime of witchcraft, which he is quite innocent of, he may in the same manner, and by the same means, be brought to confess himself guilty of a murder, a sacrilege, a robbery, or any other great offence he never committed. Thus the experience of the learned Jesuit, with respect to the fallibility of the torture in the examination into the truth of witchcraft; identifies and proves its fallibility, in the inquiry into the certainty, of whether any other person has been guilty of a crime he stands accused of.

PARADOX VII.

It should be the Duty of Magistrates, to oblige every individual in a state to earn his own living.

LXX. This was one of the laws of the most prudent Solon; and it was so inviolably observed in Athens, that Athenæus tells us, the two philosophers Asclepiades and Menedemus, were convened before the Areopagus, to give an account how they got their living; and they obviated the charge that had been brought against them, of leading an idle life, and having no visible means of subsistence; by proving, that they earned two drachmas every night, by grinding in, or turning a horse-mill; and Herodotus says, that before the days of Solon, this same law had been established in Egypt by King Amasis.

LXXI. There is no doubt, but the same establishment would be useful in all states. Why do I say useful? It would be laudable also, and of the utmost importance. By a careful examination into, and attention to this matter, communities would be freed from the nuisance of great numbers of drones, and poisonous reptiles. There is scarce a populous town or city, where you don't see great numbers of people, who without any estate or income, and without employing themselves in any honest or useful occupation, live well

in their houses, and appear genteelly and handsomely dressed in the streets. And what are the funds by which they support all this? why some of them support themselves, by thefts and robberies; others by the vile practice, of prostituting the beauty of their wives for hire; others by money they borrow of a thousand people, without any intention of ever paying it; others by tricks and cheats of various kinds, with which they impose upon innocent and unguarded people; and indeed, if the mask was to be taken off from all those who are said to live by their ingenuity, it would be found, that almost all of them, live by foul practices and roguery; and the mask would be taken off, by adopting the before proposed examination and enquiry, and by providing, and rigorously executing punishments proportioned to such evils, the body politic, would be purged and cleansed from an infinite number of vicious humours.

PARADOX VIII.

A great part of what is expended in Alms, is not only thrown away, but does mischief.

LXXII. The following sentence of King David's is a remarkable one; *Blessed is he who exercises his attention and his understanding, for the relief of the poor and the needy.* Beatus qui intelligit super egenum, et pauperem.—He does not say, blessed is he who to succour the poor, exercises his love, his compassion, or his charity, but blessed is he who exercises his attention or understanding on this object. There is doubtless the appearance of a mystery in this, but the mystery is, that alms, unless distributed with understanding, discretion, and judgment, do no good.

LXXIII. A hand that is precipitate in giving, such a one for example, as Claudian represents that of Probus to have been,

*Præceps illa manus fluvios superabat Ibèros
Aurea dona vomens—*

who relieved many poor, but at the same time supported many people in idleness; and such conduct not only supports, but it creates or breeds up many such, for wherever alms are distributed copiously and without

discretion, there will be found many people, who without this assistance would betake themselves to work to procure themselves a livelihood, who at present live idly, and omit their own industry, which they are enabled to do, at the expence, and by the profusion of others. The evils that result to a state from such imprudent dispensations, are sufficiently serious; it loses by this means, the work of a great many useful hands, and the numbers of the indolent, vicious, and profligate, are greatly increased by it.

LXXIV. It is said of a man who distributes great quantities of alms, that he gives with both hands; but he should remember, that according to the directions of our Lord and Saviour, he should only give with one; he says, when you dispense alms, don't let your left hand know what your right hand does: *te autem faciente eleemosynam, nesciat sinistra tua quid faciat dextra tua*. This implies, that it is with the right hand only that alms should be given. If I am told that I dwell too much on the literal meaning of the word, I answer it was never my intention to do so; for I am inclined to think, there is a profound implied sentiment couched under this description. It is the constant stile of holy writ, to use the expression right hand, to symbolize or define good works, and the expression left hand to signify the contrary; and hence it is, that in many parts of it, when it speaks of the hand of God, it never mentions it in an emphatical manner, without the addition of the adverb right, which is used to convey an idea, that the operations of God are just and holy. Christ then requires, that alms should be given with the right hand, by which expression, we may suppose he meant to signify, that there are good and bad alms, and that he approves the first, and condemns the others; and also to prohibit our giving with both hands, for that this would be proceeding without choice or distinction, and would also confound the good with the bad.

LXXV. I know some divines are of opinion, that we should bellow our charity, and that it is not incumbent on us to examine scrupulously into the necessities of those we bestow it on; but this is no argument why a state should not take proper precautions, to prevent those from receiving the benefit of charitable christian donations, who ought to get their living by their labour, and to exclude all such from being partakers of this

benevolence, who surrender themselves up to indolence, and a voluntary and vicious state of poverty.

PARADOX IX.

Death should not be dreaded, on account of what there is simply in the thing itself, or on account of what a person feels at that crisis.

LXXVI. There is a fear of death, which is well founded and salutary; another, which is ill founded and pernicious; and another, which ought to be indifferent, because it is natural, and which only by being carried to too great an excess can become vicious. He fears death reasonably and profitably, who contemplates it as a transition from hence to eternity; he fears it naturally, who considers it as the termination of his life; and he fears it unreasonably, who viewing it simply with regard to what there is in the thing itself, abstracted from all that has preceded, and all that is to follow it, supposes that it will be attended with excessive pain.

LXXVII. This apprehension, although it is very common both among the ignorant and the learned, I look upon to be chimerical and vain, and to have no foundation in truth, therefore, I shall not scruple placing it in the catalogue of vulgar errors. I don't mean to treat here, of the pains incident to diseases which dispose or lead to death, which without doubt are often very severe; but I only mean to enquire or examine, whether it is probable we feel any, or any very sensible pain, at that moment when the soul is separated from the body. It is generally thought, that at that instant, a more intense pain is felt, than can be produced by all the torments a man is capable of inventing. Authors exaggerate this pain in books, orators in pulpits, and all sorts of people in conversation, and reason upon it in the following manner. If, say they, tearing a nail from the finger, or a finger from the hand, produces a pain so acute, that a man is scarce able to endure it, what must he feel, or how will he be able to bear that which tears the soul from the body? Here the strict union between these parts is described in the most feeling manner, in order to enhance or aggravate the sensation that must be experienced at their separation; and they compare it to the final parting of two fast or fond friends, who have lived long and happily

together; or to two integrant parts of an animated body, the division of which is the more painful, the more firmly and intimately they are united together. To heighten this description, they add, that this pain pervades all, and every one of the parts of the human body, both internal and external, because the soul is torn equally from them all alike; which is a universality, that cannot exist in any other pain, because he who is roasted alive, or thrown into a great fire, does not feel pain in his entrails, at the time his external parts begin to scorch: agreeable to this mode of reasoning, they conclude, that the pain which is felt at the moment of death, is enormous, beyond all imagination and description.

LXXVIII. I see this matter in quite a different light, and look upon all this aggravated degree of pain to be imaginary, and consider the reasoning by which the existence of it is attempted to be proved, as fallacious and delusive. It is confounding our ideas of objects, to suppose that the division of integral parts, is analogous to the separation of the soul from the body. The pain in the first instance, consists in the forcibly breaking their connexions, or in the first shocks of the violence, which disposes to their disunion. In the separation of the soul from the body, there is no insertion of material fibres to be torn out by the roots, nor any division to be made of connected substantial parts. From whence then can bodily pain be derived?

LXXIX. The using the same words promiscuously, causes mankind to run into infinite errors, and especially if the application of them is made to things that are fundamentally very different. The expression *tearing the soul from the body*, fascinates or misleads many people in the business we are now treating of; the phrase should be understood in a figurative sense, and we are apt to construe it in a strict or literal one. In consequence of which, as we know we can't tear from our bodies, the most minute shred, without feeling great pain, or even extract any foreign substance, that has been violently introduced into, and sticks in any part of our frame, without being liable to the same sensation; led away then and betrayed by the sound of the expression, we are apt to imagine, that something similar happens in the separation of the soul from the body; but the soul is a pure spirit, that can neither adhere, or be made to adhere to any body whatever, nor can it be bound to it by ligatures, nor united with it by fibres, nor fixed to it by any

kind of fastening, nor intangled with it by any kind of roots or insertions; and finally, the mode of its union to the body, is incomprehensible to all our philosophy or understanding; and consequently, a description of its disunion cannot be given in the words of any language. There is no doubt, but the term *tearing from* is metaphorical; and that we might with less impropriety, although we never can with propriety, in describing the separation of the soul from the body, say, that it evaporated off, it was dissipated, or that it had been exhaled, than that it had been torn away; for its disunion from the body, is performed by a movement that is supremely insensible, because on the part of the corporeal substance, there is not the least resistance made to its flying off. Vapours are continually exhaling from every part of our bodies, without giving us the least pain. And why is this? Why it is, because that on account of their thinness and delicacy, they find no opposition in their passage, either from the solids, or the pores of the skin. What obstruction then can you suppose the soul meets with, in its exit from the body, which is infinitely more subtil and thin, than the finest vapours?

LXXX. Let us contemplate the thing in another point of view; and admit that the soul at the time of its being snatched away from the body, caused a violent shock, like that of tearing to pieces all the entrails, and inverting the whole interior organization. I say that even supposing this, the pain that it would occasion would be very slight, or next to nothing; and the reason is, because that in those ultimate stages of life, all the faculties are so extremely languid, and the operations of nature so feeble and remiss, as hardly to be perceived; and the sensation of pain, which is one of those faculties, being in the same state with the others, and the agent which is to stimulate them being equally feeble with the rest; although in the time of vigour, it was capable of exerting a force that had the power of producing great pain, in the present state of things, it is not capable of giving a shock that can excite any very acute sensation, nor if it was, is the subject it is exercised upon capable of perceiving, or being violently affected by it.

LXXXI. I am inclined to think, that a few moments before death, there comes on a kind of half death, or stupor, that is something like a lethargy or swooning, and that in this interval, there remains no kind of recollection or reflexion; and it is probable, that the morning of our life, and the evening of

our death, are preceded by a sort of crepuscules or twilight, which brighten and grow clearer as the day of our life comes on, and which darken and grow more obscure, as the total night of our death draws near.

LXXXII. Hitherto we have been treating of natural deaths; but violent deaths, which do not happen till three or four days after receiving the injury that occasions them, may be considered to come under the description of natural ones; as we may suppose those people die in the same way, that those do who are carried off by an acute disease.

LXXXIII. Sudden violent deaths, which are so much dreaded, are the least painful of all; and indeed I was about to say, that people in such cases, hardly feel any pain at all, or at most but an instantaneous one; because the operation of the cause which induces them, takes away in a moment, all sense of feeling. It is well known, that those who have fallen from a great height, and have lain a considerable time afterwards as if they were dead, have, when they came to themselves, affirmed, they did not feel the blow they received upon coming to the ground. The great Chancellor Bacon, tells a story of a gentleman who was very desirous of knowing what people felt who were hanged, and in order to be satisfied about it, determined to make an experiment of the thing on himself. For this purpose, he fastened a rope to the ceiling of a room, in which rope he made a noose, and after adjusting it to a proper length, he stood upon a stool, and fixed the nooze round his neck, in the expectation, that after suffering himself to be suspended, he should be able to recover the stool again; but the good gentleman was a little out in his calculation; for if it had not been for a friend who was present, to whom he had communicated what he intended to do, and who cut the rope in time, the experimental philosopher had been as dead as if he had been executed by the hands of the hangman. The account he gave of this matter was, that from the moment his body became suspended by the rope, he lost all sense and recollection; that he had not the least remembrance of the stool, or apprehension of the danger he was in, nor even any sensation of pain, arising from the suffocation that was brought on.

LXXXIV. This, I firmly believe, is what happens to all those who are executed by the hands of justice, whether they are hanged, strangled by the bow-string, or beheaded; and in general, to all those who suffer violent deaths that are as sudden as these are; for they can only feel a momentary or instantaneous pain, and the instant they receive the fatal blow, they are from that time, to the separation of the soul from the body, mere effigies of men, and have no more sensation of pain, than stocks or stones; and notwithstanding, that between the intermediate space of their receiving the blow, or in case they are hanged, of being turned off, they are seen to make some convulsive motions, these motions are merely mechanical, and are by no means governed by the will, or directed by reason.

LXXXV. We will not exempt, as coming under this general rule, even those who are burnt alive. This is a sort of punishment, which strikes all the world with extreme horror, because they generally conceive, that from the instant a person who is executed in this way, is thrown into the fire, to the time of his resigning his last breath, he feels the excruciating torments of the fire. But I am of opinion, that he feels nothing after the first minute that he is committed to the flames; nor can I conceive, that his perception of pain can endure even so long as that.

LXXXVI. I think I have tolerably well proved, what I at first asserted; but as the reader may object, that this paradox ought to have been classed among physical matters, instead of among moral and political ones, I will endeavour now, to remove this objection; which I hope I shall be able to do, notwithstanding the decadence of the faculties, and the want of sensation at the moment of death, are properer objects for philosophical, than moral speculations. I shall begin with observing, that we ought to distinguish between the matter of the proof, and the essence of the subject we are handling. The subject in the present case, consists in a theoretical proposition, that death with respect to what there is simply and merely in the thing itself, ought not to be feared, or that the fear of death, considered in this manner, is not reasonable or well-founded; now thus considered, the question is purely a moral one, because it directly combats an inordinate passion of the soul. The proofs of the truth of the proposition, appertain to philosophy; but this is what we see happen every day with regard to other

moral questions. When the question is, whether a marriage should be dissolved on account of imbecility, all the proofs in the trial are purely physical, &c.

LXXXVII. But the question more immediately appertains to morality, on account of the end for which I proposed it, than it does with respect to its own proper matter; for this end is a point of morality of the most serious importance. There is great necessity for banishing this panic terror, and this dismal apprehension of the tormenting pains of death. It is very common to see dying people (and I speak of what I have known and experienced myself) extremely distressed by this idea, not so much on account of the dread of the exquisite pain itself, as on account of the consequences that may result from it. They figure to themselves, that the pains which terminate this mortal life, are so extremely intense, that they will occasion them to lose all patience, and prevent their submitting themselves to the will of Providence with the christian resignation they ought; and are also apprehensive, that it would cause them to break out into furious acts of desperation. This anxiety has such an effect on them, that it prevents the operation of those christian dispositions, that should accompany a man in his last moments, and which are so necessary to promote his dying a good death; and besides this, they even put him in danger of distrusting the Divine Goodness at so critical a period. I have seen many who were in their perfect senses, and who had been people of good and exemplary lives, who have been greatly distressed by this idea;

O genus attonitum gelidæ formidine mortis!

LXXXVIII. I suppose that sentence of St. Paul's, *Fidelis autem Deus est, qui non patietur vos tentari supra id quod potestis*, in English, *God is good and just, and will not suffer you to be tried or tempted beyond your strength*, would be an excellent antidote for this malady. The thinking otherwise of the Deity, would not be contemplating him, as a most merciful father, nor as a just God, but considering him as a cruel tyrant, who at the moment on which your eternal happiness depends, should afflict you so severely, as to cause you to commit acts of desperation. What faith, and the light of natural reason teaches us, is, that his goodness never permits the rigour of the trial, to exceed the power of the soul to contend with and resist

it; and as I observed before, this reflexion is an excellent antidote against the malady we have been speaking of; but with all this, if it is not assisted and enforced by the persuasive eloquence of an able friend, or a good pastoral director, it is apt to lose some of its efficacy, and not to quiet the fluctuations of the mind so thoroughly as could be wished, and especially if not attended to in time; and I therefore think that it would be necessary, whether sick or well, for all people to remain in a firm persuasion, that these excruciating pains in the article of death are imaginary.

Appendix to the foregoing.

LXXXIX. I have sometimes observed, that those who attend on dying people have been much dejected upon finding them in their last moments, make some very irregular and extraordinary motions, and have been afraid and believed that those agitations, had proceeded from some impatience that had seized them. But let them not be uneasy on this account; because it is most likely, that these motions are merely mechanical; and that in case they should not be so, there is no mischief to be apprehended from them; for in that proximate state to death, if people are not deprived of their senses, the use of their reason is so feeble, or so confused, that very little, if any, of that free will can be exercised by them which is necessary to constitute sin, or at least any serious sin; for no intoxicated man, nor any one at the instant he awakes from a profound sleep, can be in a more stunned or stupified state, than a dying person at such a crisis.

XC. Finally, both with respect to the matter of this appendix, as well as with regard to that of the subject of this Essay, I shall proceed to give a last, and most efficacious consolation, against the apprehension, that the extreme pains of death are likely to endanger the loss of people's souls; for admitting that those pains were real, and as severe as they are represented to be, is there any danger that the dying person who is oppressed by them, should fall into the serious sin of impatience, or that he should incur the guilt of any other mortal crime? To this, I resolutely answer that he could not; for the same reasoning that states the pain to be so insufferably intense, removes all the hazard of sinning, because it must disturb the understanding to such a degree, as to deprive a man of all free will. This is a consequence

resulting from all passions that are excessively violent, as is agreed by all philosophers and theologians. Virgil, who had great judgment and penetration in these matters, represents Choræbus, who had been totally deprived of his senses by grief for the imprisonment of his beloved Cassandra, as divested of all free will or reflexion also, in the following lines:

*Non tulit hanc speciem furiatâ mente Choræbus,
Et sese medium injecit moriturus in agmen.*

PARADOX X.

The desire of posthumous fame is vain and futile.

XCI. There is no appetite or craving of man can be more irrational, than that, which is directed to an object he can never taste of or enjoy; and such a one, is the desire of having his name become famous in the world after his death. When a man is dead, every thing here that respects himself dies also; and what advantage can it be of to him after his decease, that all the world burst forth in acclamations and applauses of his great deeds and talents? The smoke of all this incense vanishes in the air, nor can the least particle of it, touch or affect him to whom it is offered. He feels no more of the praises of his virtues, than a statue; nor is he any more sensible of the celebrations of his grandeur, than an edifice that is erected to perpetuate it. If his works were pleasing in the eye of God, and he is in the regions of rest, he may feel the satisfaction of having left a good example to the world; and all that passes out of that sphere, let the celebrations of the world be what they will, can be of no avail to him. He will either despise, or be totally ignorant of the eulogiums that are bestowed on him by mortals. What convenience or what satisfaction, do either Alexander and Cæsar now enjoy, from being applauded through the globe for the two most illustrious warriors of the world? Homer and Virgil, from being celebrated as the two most elegant poets? and Cicero and Demosthenes, from being admired as the two most eloquent orators? They are perhaps entirely ignorant of all that is said of them here; and if they are permitted to know it, it is most likely, that such knowledge tends more to torment than please them. Empedocles was certainly a great mad-man, if, according to what some have said of him, he

precipitated himself into the flames of Mount Ætna, in order that the world upon not finding his body, should imagine he had ascended up to heaven, and would worship him as a deity. This philosopher however, as he was a follower of the Pythagorean system, and believed the transmigration of souls, might expect, that by his being placed successively in a variety of bodies, he should hereafter view with great pleasure, the worship and adoration that was paid to him in this world; but what enjoyment of this sort can a man hope for, who believes that when he leaves these regions, he shall never return to them again? And what can it be to such a man, whether he is worshiped or forgotten? Thus the emperor Adrian was much madder than Empedocles, who without believing in the doctrine of transmigration, erected temples and altars, and appointed priests, making provision at the same time for maintaining them, and providing victims to be sacrificed to his infamous little idol Antinous. Of what service could all this be to that disgraced and unfortunate boy? And we may make the same observation, on the apotheosis and ridiculous deification of the Roman Emperors. Vespasian, although he expected this farce after his death, would be played over with respect to him, treated the thing with the scorn it deserved, by saying to those who surrounded him when he was near his end, *I feel as if I was going to be converted from a man into a deity.*

XCII. That mankind should be desirous of seeing themselves applauded, and their names honoured while living, seems very natural, because they may find a gratification in it; but that they should be anxious for posthumous honours, which they can neither taste or enjoy, seems to bespeak a disordered imagination, and a distempered brain. Ovid paints Sappho, as feeling great satisfaction, at seeing her muse celebrated by all the world.

*At mihi Pegasides blandissima carmina dictant;
Jam canitur toto nomen in orbe meum.*

Thus far he expressed himself very properly, because he spoke in the name of Sappho while she was living, and might be supposed to be gratified by, and pleased with the aromatic fumes of those acclamations. But he reasoned very ill, when speaking of Hercules and Theseus, he reckoned as a

balance for the loss of those heroes, the applauses the world would bestow on their memories:

*Occidit et Theseus, et qui tumulavit Orestem;
Sed tamen in laudes vivit uterque suas.*

XCIII. The eulogiums of the dead, can only be enjoyed by the living. The relations, the friends, and the country of the deceased, divide among themselves the whole fragrancy of this grateful gale, nor can the least breath of it reach the region, which is inhabited by those who depart hence. There remains to the dead but one happy lot, and that is derived from, and depends on their having died well. *Beati mortui, qui in Domino moriuntur.*

PARADOX XI.

There is no man of a clear and good understanding who is not a good-intentioned one.

XCIV. I believe that all the mortals in every quarter of the Globe, will be struck with surprize, at hearing me broach this paradox, and will look upon it as one of the greatest chimeras in ethics, that ever entered the head of man; for there is scarce any one of the least observation, who cannot affirm and attest, he has seen and known people of very good capacities, who were very perverse and ill-disposed. But I, in opposition to all this, assert that I never met with such a one; and I not only make this declaration, but declare further, that I think it next to impossible that there should be such a man, and that if by chance such a one should be found, he ought to be considered as a monster.

XCV. But in order that we should not misunderstand or confound things, I think it necessary for me to explain, what I understand by an evil-intentioned man. By an evil-intentioned man, then, I mean such a one in whom those vices reign, which are most pernicious to society, that is to say, malignity of heart, unforgivingness, turbulence or restlessness, and a desire of usurping other peoples property; and in general, all sly and crafty persons should be enumerated in the catalogue of evil-intentioned men, such for

example, as are attentive to nothing but their own gratifications and emoluments, and who have not the least concern for, or who are totally indifferent about the good of their neighbour, or the welfare of the public.

XCVI. The deformity, the baseness, and the dissonance from natural reason, there is in a person's doing a voluntary injury to another, is so strikingly represented to a man of a clear and a sound understanding, that except in here and there an instance, where some violent passion intervenes to disturb and disorder the reason, it seems impossible, that a person should voluntarily commit acts that are directly injurious and offensive to his neighbour. And it may be from this principle, that we have seen some who have been reputed as atheists, who, notwithstanding their erroneous belief that there is no such thing as future rewards and punishments, and that they expect no recompence for their good actions, or chastisement for their bad ones hereafter; as members of human society, have behaved well, or at least have done no civil mischief to it; I mean that they have conducted themselves like quiet peaceable people, and have lived contentedly upon their own patrimonies, or on what they have lawfully acquired, and have shewn themselves averse to all violence and injustice. Such among the antients, was Pliny the elder; and such among those of more modern date, was the Englishman Thomas Hobbes.

XCVII. The genuine and true reason of this is, that the existence of a Supreme Being, although it is most plain and clear, is not with respect to the human understanding, self-evident, or, as the Theologians explain the thing, is not *per se nota quo ad nos*; but is made infallibly evident, by deductions drawn from other principles; and where a deduction of this sort is absolutely necessary, a person may suspect, that it is very possible now and then some fallacy may creep in. But the deformity of such vicious actions as we have been speaking of, is evident of itself; and whenever such actions are represented to the understanding, it clearly comprehends their baseness, and the operation of them is odious and apparent in the eyes of every man, unless as we observed before, some circumstance intervenes to disturb his reason.

XCVIII. To this it may be urged by way of objection, in the first place, that in order to perceive and be convinced of the turpitude of those actions, there is no necessity for a man to be possessed of a bright understanding, as a middling one, or even one below the middle class, would be sufficient for this purpose; so that this reasoning will prove the turpitude to be plain to every understanding, high, middling, and low, or else that it is evident to neither of them.

XCIX. To this I answer, that although the thing may be known with intire certainty to every one of them, there is a great difference between the knowledge and comprehension of one man, and that of another. Two understandings that are unequal, notwithstanding they may both know, and be thoroughly persuaded of the same truth, may be struck with it very differently; and in proportion as one of these understandings is the most clear, that one will know it more distinctly, more strikingly, and with a more refined degree of penetration; and in proportion as the other is less clear, that other will perceive it less distinctly, and more confusedly. In consequence of this inequality of understandings, objects make a more strong or a more weak impression on the soul, and have more or less influence to excite in it, these, or those affections. The same infinite goodness of God which is known to the blessed, is known with infallible certainty by the worldly also. How then comes it to pass, that the first love him necessarily and intensely, and that we worldly mortals are so lukewarm in our love of him? This is occasioned by no other cause than the following, that although our knowledge of him is evident, that of the blessed is the most clear, and ours the most obscure; and in proportion as the understanding knows a good or an evil with greater clearness, the will is moved with a greater impetus, to love the first, and abhor the last.

C. This may be very opportunely and aptly explained, by the operation of any corporeal sense; for he whose olfactory nerves are very quick, will be more offended with an ill smell, than one in whom that sensation is more languid and remiss; and although this last may be able clearly to distinguish the ill smell, he will be less disgusted at it; nor will it appear so hateful to him; and this happens from no other reason, than that the olfactory perception of the first is very clear, and that of the second rather obstructed

or more dull; and although not only the man who has a very quick and delicate ear, but also he who has one that is more obtuse, may evidently perceive the dissonance of three or four voices which are totally discordant; the last will bear it without being much affected, and the other will be almost distracted by it; and this is all derived from the cause above-mentioned.

CI. And it happens just so with regard to intellectual perception. The deformity of vicious actions, which are self-evident, is apparent and clear, not only to men of perspicuous understandings, but to those of the most inferior capacities, provided they are not quite stupid; but by the first perceiving them with a lively clearness, and the others somewhat confusedly, they produce a kind of horror in those who have that clearness of perception, which does not permit their wills to embrace such objects; while those to whom they do not appear so disgusting and unpleasant, may be betrayed into grasping turpitudes, under the disguise of delights and pleasures. But I would not have it understood that I mean to insinuate by what I have said here, that there ever is a suspension, or obstruction of the operation, of every, or any man's free will.

CII. It may be objected in the second place, that there are entire nations, among whom it cannot be denied, that there are to be found many men of excellent understandings, who hold robbery, deceit, and even cruelty to be lawful, and that consequently, they cannot consider these things as turpitudes, or have a just sense of their baseness. To this I answer, that our assertion with respect to a good understanding, does not allude to one placed in such a situation. The general error of a nation in any matter whatever, is like a dark fog which bewilders people, causes them to mistake their way, and perplexes the clearest understandings. If in early infancy, when the rational faculties are weak, children are familiarized to, and brought up under the influence of deceitful prejudices; and when they come to years of maturity, they are accustomed to reverence a common error as irrefragable authority; if it should happen afterwards, that a ray of light breaks in upon them, which discovers to them the truth, they timidly fly from the elucidation, distrust their own reason and reflexion, and are apt to

suspect such elucidation to be a delusion, and to suppose that it would be criminal in them to regard it.

CIII. I answer secondly, that it is not known with any degree of certainty, that men of excellent understandings, who are educated and brought up in those nations we call barbarous, are infected with all the errors that prevail in those nations; and with respect to myself, I am fully persuaded they are not. We know that various eminent men among the Gentiles, in matters of religion, thought very differently from the populace. It is true however, that there were but few of them who had sufficient resolution to speak out, as they for the most part disguised their opinions, from motives of fear or policy. We ought also to admit, that among the barbarous nations of these times, there are still to be found men of this character. Nor is there any necessity for limiting such an opinion within the bounds of mere conjecture; for there are various historical relations, which bear testimony to actions of heroic virtue, that have been lately done by some of the individuals of those very nations, where maxims of inhumanity are prevalent at this day; of which if it was necessary, a long catalogue might be adduced.

CIV. In the third place, it may be objected, that experience teaches us, there is scarce a country or populous city, where you may not find some people of perspicuous understandings, and who, although they are of wayward dispositions and depraved inclinations, are subtil and penetrating. To this I answer resolutely and determinedly, that I defy any man to produce such an instance. I have known and conversed with many of those people, who have been esteemed men of good understanding, and perverse dispositions, but have always found the common opinion of those persons to be extremely erroneous. The vulgar frequently look upon persons of very superficial talents, as men of great understanding; and upon hearing them talk off hand, although there is nothing solid in what they say, and observing in them a readiness at expressing themselves, and more especially if they deliver themselves with confidence, and a magisterial air, most people are apt to give them credit for being men of admirable understandings; when in truth, there is hardly one out of a hundred of them, who can penetrate more than skin-deep into the objects he converses upon.

There is another very common deception in this matter, and that is, looking upon cunning people as men of depth or penetration, when they are as palpably different from one another, as light is from darkness. I call those cunning people, who are solely attentive to nothing but their own concerns, and who by all sorts of under-hand ways and means, and by all kinds of little arts and deceits, are endeavouring to promote their own particular interest. Can these be called men of sublime understandings? To do this, requires but little depth or penetration, as all that is necessary to accomplish such ends, is low craft and roguery; and there is hardly any capacity, be it ever so mean, which cannot comprehend and apply such trivial artifices: every one may arrive at doing this; but a noble understanding, discerning the baseness of them, abominates such practices; although the vulgar, to whose bastard dispositions they are better suited, embrace them with eagerness. Dissimulation, so far from requiring an exalted understanding to support and carry it on, requires none at all, for we see some irrational animals who exercise it with great address. The foxes are very expert at it, but that does not in any wise render their nature superior to that of brutes; and I repeat again, that I never knew an understanding that had any thing of the elevated or sublime in its composition, that did not abhor all duplicity and fraud.

CV. If we see this matter in the other extreme point of view, we shall find it liable to great equivocation. It frequently happens, that a man of very pure virtue, who has somewhat of native dryness or bluntness in his composition, appears to those of rude and uncultivated capacities, as a person of a depraved disposition. Those who are zealous lovers of truth and justice, are accustomed, not always to accommodate themselves to those courteous condescensions; by which people acquire popular acceptance; as by attending to the substance of things, they are apt to overlook forms and ceremonies. Words from their mouths, signify what the sound and sense of them express: they confider courteous dissimulation as a treacherous enemy of virtue; and are ignorant of the art of painting vice in counterfeit colours, for the sake of pleasing or flattering any man; but on the contrary, are always careful to describe it so, as that it may appear in its true native shape, and in all its deformity. The more prevalent lying, deceit, and perfidy is, the more they loath and nauseate it, and are observed to be more strict and severe in their reprehensions of it; and besides, they never look

smilingly, but upon those in whom they perceive a clean mind. This unpleasing integrity, is regarded by the bulk of the world, as a kind of misanthropy, or malevolence towards the generality of mankind; and the number of those, who busy themselves in painting such men, as impracticable, perverse, and ill-intentioned is infinite; for they are pleasing but to very few, as there are but few who are pleased with them; so that either from the malice of their opponents, or from the want of a proper knowledge of the world in those that are indifferent with respect to their opinions of them, it easily comes to pass, that a man of exalted and sincere virtue, is often looked upon by a whole town, as a person of intentional malignity.

CVI. Whoever is upon his guard not to fall into one of the two beforementioned errors, and has capacity to distinguish true virtue from false, and a clear and good understanding, from a cloudy, crooked, or crafty one, will be convinced as I myself have been convinced, that there never fails to be much virtue lodged in the person, in whom you find a real good and clear understanding. I would not however be understood to insinuate, that all men of great genius and capacity should be saints; for meritorious virtue, or such as entitles a man to inherit eternal life, is the child of grace, and not of nature. Neither would I be understood to say, that all kinds of moral virtues should be resplendent in such a man as I have been describing, but only those, whose opposite vices, at first sight, and without the assistance of any reasoning or reflection, are manifest and apparent; and whose deformity, at a glance, strikes the eye of every beholder; nor would I even be understood to assert so much as this, without some limitation and exception; for every vehement passion during the time it lasts, will make the most prudent man act like a mad one, and the most acute one, like a fool; but abstracted from the intervention of such accidents, it is my firm opinion, that every man of a clear and good understanding, is an honest and a good-intentioned one.

THE
GREAT AND MASTERLY
AUTHORITY of EXPERIENCE.

SECT. I.

I. There arrived in the kingdom of Cosmosia, two famous women, who were very opposite to each other; but both with the same design, which was that of obtaining the absolute dominion over that empire. The first was called Solidina, the other Idearia; the first was learned, but simple; the second ignorant, and ostentatious. The people of the country, were ignorant like the last, and simple as the first. Hence, Solidina thought to win them to her by kindness, and by instructing them; and Idearia, to subdue them by craft and imposition. Idearia opened a public school, and promised in pompous language, that in a very short time, and with little or no trouble, she would make all those extremely learned and wise, who chose to attend her lectures. The greatness of the promise, joined to the imposing appearance of the new doctress, mounted in the professional chair, together with her great volubility, and flow of mountebank rhetoric, soon filled the school with pupils. She began her lectures, which all consisted, in laying before her auditors in new and unusual language, the chimeras, contained in the extensive field of the imagination. And, oh wonderful to relate! either Idearia had somewhat of enchantment about her, or else there was something very singular in her method of applying her artifice and cunning; for in a few years after opening her school, she persuaded those miserable people, they perfectly understood all that could be learned.

II. Solidina, pursued a course diametrically opposite to that taken by Idearia. In an humble garb, and without any parade or ostentation, she went from house to house, and familiarizing herself with all men, taught them in plain and easy language, true and useful documents. The most retired cottages, and the most humble work-shops, were schools suited to her doctrines, for she found in all of them, sensible objects, which examined by

the help, or auxiliary aid of the understanding, served the purpose of books for teaching and explaining her lessons; and so far was she from inspiring an indiscreet presumption in her disciples, that she ingenuously told them, that all she taught, was a mere trifle, compared to the infinite deal there is to be learned; and that to arrive at a moderate knowledge of things, required infinite labour and application. This modesty of Solidina's, was very prejudicial to her, because at the very time she made this declaration, Idearia, was boasting and blazoning in her school, that in a concise mode, and with very little trouble, she would make all her auditors universally learned; the consequence of which was, that the pupils of Solidina one after another began to drop off, and go over to Idearia, in hopes that in her school, they should arrive at the summit of learning *per saltum*. What contributed much to forward this defection, was, that Idearia also spoke of Solidina with contempt, calling her base, vile, mechanical, and stupid; by which means, the poor tutress became abandoned by all the people of rank, and was obliged to retire from the city to the villages, where she applied herself to instructing poor husbandmen, in that sort of knowledge, that was necessary and useful, for the cultivation and improvement of their lands.

III. Idearia by the banishment of her rival, now becoming triumphant, entertained thoughts of establishing an absolute and despotic sway over her disciples; and to accomplish this purpose, she published an edict, by which every one was required not to believe in future, aught he should see with his eyes, or touch with his hands, but only to credit such things as she should be pleased to order him to believe; requiring further, that he should look upon it as an indispensable obligation, always to defend with invincible obstinacy and unremitting vociferation her doctrines, against whoever should presume to contradict them. All heads bowed obedient to this tyrannic decree, and people began firmly to believe many maxims, which before they had found a difficulty in assenting to; such for example, as that the truth can never be found out or ascertained but by means of fiction; that there is a mode of coming at the knowledge of things, which may be taught to a child in four days; that mankind are all alike, which is a rule that will hold good with respect to every other species, and if you know what one of a sort is, you know what they all are; that insensible and inanimate things, have their desires, their prejudices, and their affections, the same as animate ones; that that body, which is the most brilliant, and most heating of any, has nothing

igneous in its composition; and that on the other hand, there is another very large body, which is purely igneous, that is neither luminous, nor heating, nor does it stand in need of any pabulum to sustain it; that all living creatures have a large portion of fire in their composition, without excepting even the fish, although they are always in or under the water, nor the turtle, whose blood is positively cold.

IV. These and many other such-like portentous particulars, did Idearia teach to, and impose on her credulous disciples; who all received and embraced them as infallible truths; but at length there arose in the very school of the doctress herself, a contentious schism, or scandalous disagreement, which was begun by one Papyratius, a man of subtil and animated genius, but a great lover of novelties. This man introduced new and not less astonishing dogmas than the others; such as that all the living creatures in the world, man only excepted, have no more sensation or feeling than stocks or stones; and that in every man, there is but a very small portion of the body which possesses the presence of the soul; that the extension of the world is infinite; that the motion of sublunary bodies, is equally semper-eternal with that of the cœlestial ones; that the imaginary space, is really and truly a body; that every thing upon the face of the earth, is in so continual and rapid a motion, as in the space of every twenty-four hours, to travel some thousands of leagues; that we in all things, should give credit to our imaginations, but none to our senses; for that the representations made by these last, are apt grossly to deceive us; and that neither is the swan white, or the crow black, nor is fire hot, or snow cold, &c.

V. These novelties, and others of the same sort, although they were condemned from the beginning by the majority of Idearia's disciples, did not fail to attract a number of votaries, sufficient to form a new school. The two parties inveighed against each other with great bitterness, and one side reprobated as absurd errors, what the other maintained as conclusive dogmas.

VI. This division, after long and obstinate disputes, in which the arguments were so nearly poised in equilibrium, that neither side could

claim the victory, in the course of time, opened the eyes of many, and made them doubt whether they had not lightly, or even blindly, admitted as articles of human faith, doctrines, that were uncertain, and exposed to be terribly contested. They observed, that the arguments with which each side attacked the opinions of their opponents, were beyond comparison stronger, than the reasons with which they defended their own. From hence they inferred, that both the one and the other were evidently doubtful, and might very probably be false. This brought to their remembrance the poor and neglected Solidina, and made them reflect, that she proved all she advanced, by sensible and plain demonstrations. This remark becoming every day more general, and more adverted to by men of the first-rate ingenuity in the schools; they took the matter into consideration, and the result of their deliberations was, a determination to recall Solidina from her retirement, and to bring her back to the city; which after having done with solemn pomp, they erected a magnificent school for her, where she has continued to teach from that time, with a reputation that has increased daily; and her being favoured with the countenance and protection of some illustrious personages of high rank, has greatly contributed to advance her credit, and particularly the encouragement she has met with from the princes, Galindo and Anglosio, who are both great lovers, and patrons of Solidina.

SECT. II.

VII. This History, which was printed in French, was given me to read by a stranger who was on his travels; but as soon as I had read what I have just related, he snatched the book out of my hands, and asked me if I comprehended what I had read? The question gave me to understand, there was something mysterious in the History, and that under the plain letter, was conveyed some signification, different from what the words expressed. I answered him, that I had not understood it in any other sense, than what the language seemed to imply; but that if he would permit me to read it over again with a little attention, I might perhaps be able to comprehend its meaning. He gave me leave, and then upon reflection, first on the nature of the doctrines it mentioned, although they were not pointed out very clearly, and secondly, on the allusion of the names given to the personages who

were introduced into the scene, I found it was not very difficult to decypher the riddle, which I interpreted in the following manner.

VIII. The kingdom of *Cosmosia* is the world, which is the signification of the Greek word *Cosmos*. *Solidina* is experience, and *Idearia* imagination. These names are very properly suited, both to the characters and doctrines of the parties. Experience *solidly* proves her maxims, by sensible and clear demonstrations; and imagination, grounds her opinions on the vain representation of her *ideas*. Solidina was a long while banished, and during all that time, Idearia reigned triumphant; for from the period that Pythagoras reduced all philosophy to his numbers, Plato to his ideas, and Aristotle to his precisions; and for many ages afterwards, the world knew no other but an ideal physic; nor did it pay the least attention, to the experimental or solid. In the first maxims of Idearia, we perceive many of the dogmas of the peripatetic doctrine, and in her second, many of the Cartesian ones; and they gave the name of Papyratus to the broacher of this last system, because the French word *Carte*, signifies the same as the Latin word *Papyrus*, from whence the appellative was derived. Des Cartes is ranged among the auditors of Idearia, because he was disposed instead of less, rather more than the Peripatetics, to regulate all physics by imagination and ideas. At length the mistake of this method was perceived, and Solidina was recalled from the villages to the city; and experiment and observation, which for a long time had only been in use among the rustics, and employed in the business of husbandry, to increase the product of the earth, improve the mountainous land, and to encourage the propagation, and mend the breed of herds and cattle; were not long since introduced in pomp, and countenanced by some courts, in the neighbourhood of which, academies were instituted, to examine nature by their help; and as the most celebrated of these, were the Academy of Sciences at Paris, and the Royal Society of London, which were founded under the protection of the kings of France and England, they called the two princes who were the lovers and patrons of Solidina, by the names of Galindo, and Anglosio, which were derived from the Latin words *Gallia*, and *Anglia*, which in that language signify France and England.

IX. The stranger approved of the whole, and every part of my explanation, and assured me, that by the context that followed what I had read, the intention of the author of that mysterious History, could be no other than what I had described it. But I told him I was not inclined to approve, nor could I approve of the whole and every part of its content; for that I observed in it some latitudinary expressions, which seemed to reflect upon, and were derogatory of the peripatetic doctrine, so I only acknowledged to him, that in the most essential parts, the thing seemed just and right. He replied, that as I was a Spaniard, and a sectary of the peripatetic school, my objection and the reasons for it might be admitted. Upon saying this, he took his leave of me, and pursued his journey, leaving me in a disposition to meditate on the subject of our conversation, and lay before the public, such reflections as should occur to me respecting the matter of it, which may be seen in the sequel of this discourse.

SECT. III.

X. The first thing that offers itself to our consideration, is the little or no progress, which natural reason, unassisted by experience, has made in the examination of the affairs of nature in the course of so many ages. Nature is as little understood in the schools at this day, as it was in the academies of Plato, of Lyceus, and of Aristotle. What secret have these academies developed? or what diminutive portion of this most extensive country have they discovered? What utility has the labours of so many men of excellent ingenuity, as have cultivated philosophy in the reasoning and speculative way, produced to the world? What art, either liberal or mechanical, of the many that are necessary for the service of man, or the good of the public, do we owe to speculative invention; and I might even say, what small advancement in any such art, has been derived from it? What document of the schools, has conduced to instruct a husbandman in the cultivation of his lands? They there talk much of causes, effects, productions, and dispositions of matter, but all this has not yet produced any maxim that can be relied on, for the most advantageous mode of cultivating the land, in order to dispose it to the production of this or that particular plant, nor to instruct us at what time it should be sown, nor in the least to inform us with regard to many other circumstances, that should be attended to in the raising it. Schoolmen, after the example of Aristotle, treat largely of qualities,

which they place in a predicament apart, or by themselves; without having by this means, discovered any qualities at all, either in the mixtures, or the elements; but on the contrary, Aristotle has rather been miserably mistaken in those he attempted to point out, by his rules of proportion and combination; I mean in those that appertain to the air, and the water, as we have proved in another part of our works; and it would not be difficult to prove the same thing, with regard to those he has assigned to the earth. If perchance he has been right in ascertaining the qualities belonging to fire, (although in our *Physical Paradoxes* we have denied his position with respect to its being hot in the highest degree) I say, if he has been right in this particular, it was not because philosophy had penetrated the secret, but because the thing was manifested to us by our senses.

XI. These are the organs, by which all the natural truths we are able to obtain a knowledge of are conveyed to our spirits. Even in the mathematical faculties, which affect to confide every thing to theoretical demonstrations, they cannot, except in the two elemental ones, arithmetic and geometry, safely advance a step, without holding before them, the light of experience. It was this taught the geographer, the situation and position of the divers parts of the globe; the navigator, the directive virtue of the load-stone; the static philosopher, the weight, the descent, and acceleration of motion in their descent, of heavy bodies; the mechanic and engineer, the augmentation of power by machinery, or the multiplication of purchase; the astronomer, the movements and course of the stars; the musician, the consonant and dissonant intervals of music; the optician, perspective, or the effects of vision when the eye surveys distant objects; to the catoptrician, and dioptrician, the laws of reflexion, and refraction.

SECT. IV.

XII. And it is very worthy of remarking, that even after experience had made those first discoveries, on which the theory and practice of arts are founded, those primitive lights were in general found to be insufficient to enable the understanding to make further advances in them, but it was rather found to be necessary, that the same experience should continue as

they proceeded on, to direct their steps, and correct their errors. I will explain what I mean, by two examples taken from navigation.

XIII. The first regards the direction, or pointing of the magnetical needle to the pole. This admirable property, which was totally unknown to the antients, was discovered in the thirteenth century, and immediately applied to the improvement of navigation. Upon its first discovery, the philosophers, according to their wonted custom of pretending to discern the causes of things, imputed this effect, as derived from an occult sympathy with the pole, contained in the very essence, form, and substance of the loadstone; and as this is supposed to be invariable, they concluded, that the direction must infallibly be invariable also. They continued in this good faith for about four hundred years, at the end of which long period, Crinon, a pilot of Dieppe as some say, or Cabot, a Venetian Navigator as others believe, was the first who observed the declinations of the magnetical needle from the true north, that is, that it did not always point directly to the pole, but declined in different places, sometimes towards the west, and at others towards the east. The philosophers heard this novelty with great disgust, because it gave the lye to some of their most established maxims, and therefore they set about to contradict it with all their force. But in the end, they were obliged to submit to repeated experiments, authorized and confirmed, by the testimonies of people of undoubted credit.

XIV. Having afterwards discovered, that under the meridian of the Azores or western islands, there was no variation of the magnetical needle at all, the astronomers and geographers, thought they had found out a fixed station, whereat to commence the first meridian, which had before been counted arbitrarily, from whatever place they chose to begin it. But this idea soon vanished, for a little while afterwards, they discovered two other meridians, where there was no variation; the one at a head-land near *the Cape of Good Hope*, to which, on this account they gave the name of *Needle Cape*; and another, at the spot where the city of Canton is situated. Upon this, they thought they had found out a certain principle, whereon to ground a compleat system for calculating or computing variations, by graduating them for the intermediate stations, in proportion to their greater

or less distance from the mean space between the two places where there was no variation.

XV. But as nature frequently mocks the ideal propositions that are fabricated by the brain of man, this time of rejoicing lasted but a few years, for they discovered, that this declination of the magnetic needle, varied more or less at the same place at different times, and that this change of variation was perpetual. This discovery, not only demolished the antecedent imagined rules, but nearly took away all hope, of their ever in future, finding out any certain one for their guidance; and this, notwithstanding many men, eminent for their skill in physics and mathematics, have long, and still do, labour at accomplishing it.

XVI. In this instance, may be seen the fallibility of the most plausible reasonings unaccompanied by experiments. And we shall see the same, in the other we are about to mention, which relates to the flux and reflux of the sea.

XVII. As the flux and reflux of the sea, is evident and apparent to all those who inhabit near the sea-coasts, and the course and changes of the Moon to all mankind, it was easy and natural, to observe the correspondence there was between the movements of the one and the other; that is, that the rising and falling of the tide, keeps pace with the rising and setting of the moon; and it is probable, that the first people who remarked this, fancied that by this single observation, they had discovered the system of these admirable movements. But this delusion was but short-lived, for they soon afterwards observed, that within the space of the same lunar month, there was a great disproportion between the rising and falling of the tide, and that it flowed higher, and ebbd lower, at the new and full moon, than it did at the quarters. But when they had advanced this step, it is likely they gave themselves credit for having discovered the whole secret, and supposed, that they knew all that was to be known of the matter; and when they observed there was so exact a correspondence, between the motions of the tide, and the phases of the moon, they did not entertain the least doubt, but that they were totally influenced by this planet. But experience also undeceived mankind with respect to this error, for they found out, that there

was another variation of the tides, which did not depend upon the revolutions of the moon, but upon those of the sun; that is, that supposing other things to be equal, they rise higher, at, or near the equinoxes, than they do at the solstices. This gave them to understand, that the moon did not reign so despotic over the tides, as to exclude the sun from all share in the dominion of them.

XVIII. But after all this was ascertained, they found, that by trusting the bringing of ships into barr'd or tide harbours, to the mere combination of the before-nam'd observations, they fell into very dangerous errors; for that there are two other variations, which are very considerable, and especially one of them, and which can't be comprehended under any general rule whatsoever; the one is with respect to the time of full sea; and the other, regards the quantum of rise of the water. The time of full sea varies at different ports, even under the same meridian, and does not happen in all of them at the same hour. They vary also with respect to the magnitude of them, because there is an enormous difference in the height they flow at some ports, to what they do at others, for the water will rise more than ten fathoms upon some coasts, and not above a few feet on others, and in others again, the rise is hardly perceptible.

SECT. V.

XIX. If the rules deduced from experimental observations are so fallible, that it is absolutely necessary in order to avoid all error, to pursue the thread of them so scrupulously, that reason should not venture to advance a step, without the light of an experiment appropriated to the business it is in search of; I say, if these rules are not to be relied on, what confidence can we place in those maxims, which derive their origin from our arbitrary ideas?

XX. Nature conforms to, and is governed by the idea of its great artificer, not by that of man; and it is strange temerity in man, to presume he can comprehend the idea of such an artificer. I have sometimes thought, that if we were told, that there were numbers of those luminous bodies in the heavens which we call stars, although we could not see them, every one

would imagine, such a disposition and collocation of them in the sphere, as best suited his own fancy. One would conceive them distributed into various regular sets of figures, such as triangular, hexangular, circular, &c. which formed so many different constellations; another, that they were composed of a beautiful suit, of well ranged and harmonious groups; another, that they were disposed after the manner of flowers, which he had seen growing in parterres in some garden; another, that they were formed in such a position, as resembled the shape of various images, either mystic or natural. In fact, there would be no body, that would not attribute to them some most beautiful resemblance, or imitation, of those things which he had ever seen, that seemed most pleasing to him, either in art, or nature. Notwithstanding which, they would all deceive themselves, and be greatly surprised, upon the firmament afterwards being displayed to their view, to find the stars placed and ranged in a position, quite different and distinct from all they had imagined.

SECT. VI.

XXI. Thus it frequently happens, that men think in one manner, and God operates in another. Men suppose, and they suppose right, that the works of God are all executed with order, and in proportion; but although they suppose well, they reason ill; because they think there is no other order and proportion, than that which comes within the reach of their comprehension. The works of God, it is true are wrought with proportion; but with a sublime proportion, which is much superior to our rules. It is blind temerity in man, to imagine that God in his works, is to conform himself to our human gross ideas of proportions.

XXII. It was a confidence in such proportions, that caused Pythagoras to err egregiously in his dimensions of the magnitude, and his calculation of the distance of the heavens, which he thought to compute, by the numerical series of the intervals of music. And others were not much wiser than Pythagoras, who by fancying they had found some special perfection in the number four, chose to stamp it on, and regulate all the operations of nature by it. From hence came the four elements, the four first qualities, the four

cardinal points of the globe, the four quarters of the year, and the four humours of the body.

SECT. VII.

XXIII. If even in those consequences, which to all human appearance, we deduce immediately from the truths which nature herself presents to our senses, we are sometimes liable to be mistaken; what dependance can we place in those, that are founded on principles, which without consulting nature, are established by our fancy? What consequence to all appearance could seem better inferred, than that of the repugnance of nature to a vacuum, founded on the palpable experiment of the water rising in the pump? But by the light of innumerable other experiments, the mistake was discovered; and it was found out, that the true cause of that, and all such-like phænomena, was the weight of the air.

XXIV. We see with our eyes, whether we explain the cause of it as proceeding from an intrinsic quality, or from attraction, or from impulse, that gravity precipitates bodies with a swift movement towards the earth. It seems most natural to suppose, by reasoning from the famous axiom, *sicut se habet simpliciter ad simpliciter, ita magis ad magis*, that to a duplicate gravity, there would correspond a duplicate acceleration of motion. But the fact differs widely from this proportion.

XXV. It is plain, that air is much more subtle and thin than water. Who from hence, would not infer as an unavoidable consequence, that air must penetrate all bodies which water penetrates? Notwithstanding this, we see that water penetrates paper, which air does not penetrate, or it penetrates it in so small a degree, or so slowly, that we look upon it as next to no penetration at all.

XXVI. Who judging by general or common principles, would not conclude, that wet hay was much less liable to take fire than dry? But experience teaches us, that by putting together quantities of hay too moist, it will take fire of itself, which could never have happened, if the hay had been put together sufficiently dry, or more thoroughly made.

XXVII. What maxim is better established, or more generally assented to among the naturalists, than that a vivifying heat of the blood, is indispensably necessary for the preservation of life? But with all this, Father Plumier, a learned priest of the order of Minimus, in a voyage in the Pacific ocean, for want of water, found himself under a necessity of drinking the blood of turtles, and declares, he found it as cold as the common water in Europe. (Mem. de Trevoux, an. 1704, tom. 1. page 175.) Who, agreeable to the laws of ratiocination, can say otherwise, than that the third quality, resulting from the mixing three or four fœtid things together, must be fœtid also? But experience manifests to us, that this consequence is not infallible. Mr. Lemerî bought a certain quantity of the gums galbanum, sagapenum, and opopanax, and also some bitumen of Judæa, which he put all together into a retort, and found that there resulted from the mixtures, a strongly scented oil, greatly resembling musk; now the bitumen in smell, does not in the least resemble musk, and the other three drugs are absolutely fœtid. (Hist. de l'Academ. Royal, ann. 1706.)

XXVIII. If a philosopher, destitute of all other information, except that which he had acquired in the schools, should be told that two liquids, which to the touch appear cold, should upon being mixed together, not only heat and boil, but also emit a flame; he would be in a terrible agitation, and armed with his infallible conclusive principle, *that nothing can communicate a quality which it does not possess*, would exclaim against the proposition. But let him exclaim as much as he will, it is a certain fact, that from mixing a pure acid, with the essential oil of an aromatic plant, there will result that commotion.

XXIX. We know that water, is much heavier than air; and we also know, that the vapours which arise from the earth, are nothing else but particles of water, very minutely divided, and consequently, that they must be much heavier than particles of air of the same size. We know also, that a liquid cannot ascend above another, when they are put together, except when it's particles of equal size are lighter than those of the other. From these premises, it should follow as an infallible consequence, that these vapours cannot rise superior to the air we breathe. But let this consequence appear ever so infallible, experience convinces us that the fact is otherwise.

XXX. Nobody is ignorant, that the aromatic species, such as the clove, the pimienta, and the cinnamon, are hot or heating, and that the regions nearest the poles are cold, and those nearest the equinoctial hot. From these premises, what naturalist would not infer, that the use of these species would be less pernicious to the inhabitants of the cold regions, than to those of the hot? But experience is ever demonstrating to us the contrary. For a scruple of clove is more offensive to, and has a greater effect on the first, than a drachm has on the last.

XXXI. And similar to this, has been the experience of the Hollanders, in some of their voyages to the East Indies. It happened once, that upon passing the line, the greatest part of the people belonging to a ship were taken sick, and that more than half of them died, and that those only recovered, to whom in their sickness they gave great quantities of brandy. The medical men found great difficulty in believing, that these people could owe the preservation of their lives to drinking to excess of a liquor, which if not taken sparingly and with moderation, is found to be very pernicious to health. But in the end, they found themselves obliged to yield to experience; and admit of the liberal use of brandy, which afterwards delivered with equal success, all the afflicted from their illness.

XXXII. It is then absolutely necessary to submit to experience, and if we are not disposed to abandon the real road of truth, to seek for nature in herself, and not in the deceitful image which our fancy forms of her.

SECT. VIII.

XXXIII. I am not ignorant that there are some schoolmen, who represent the application to experiments, as disgraceful to, and beneath the dignity of philosophy. But this is a most absurd error, for at this rate, studying the imaginations of men, would be a more honourable occupation, than ruminating on the works of God. In books of theory, we contemplate human ideas; in natural entities, divine ones. Let reason now determine, which is the most noble study.

XXXIV. The prince of philosophers, Aristotle, thought differently in this respect to the present school ones, for he said, we should not disdain to examine with our senses, the most trifling works of nature, for that we should find in them all, marks of sublime wisdom, and just and beautiful ingenuity: *Aggredi enim quæque sine ullo pudore debemus, cum in omnibus naturæ numen, et honestum, pulchrumque insit ingenium.* For so it is, that in the most humble plant, in the most vile insect, and in the most rude rock, we see the traces of an omnipotent hand, and the marks of infinite wisdom.

XXXV. Besides this, it is a matter of great importance to him who is in search of truth, to chuse the path that leads directly on to his object, and not to pursue that which will conduct him out of his way, although it should to the eye, appear the most desirable, and the most pleasant to walk through. There is no doubt, that a physician in a college, mounted in a professional chair, and reading a lecture, makes a more pompous figure, than when he is attending to, and observing the dissection of a dead body in an hospital; but it is by attending to the dissection, that he attains a knowledge of the situation and arrangement of the internal parts of the body, which he could never have acquired, by reasoning or disputing all his life long in a school. Ideal gold, is possessed in imagination by lazy people, who while they are sleeping in their beds, dream it is showered down on them; but the true ore, can only be acquired by force of labour, and digging in the mines with diligence and industry. And in no other manner, otherwise than in appearance only, can we attain more, than that shadow of truth, which we call probability, for these are the largest advances we can make, by the force of our imaginations exerted in the retirements of the closet; and the truth itself, is only to be found by scrutinizing sensible objects, and searching in them, for the hidden secrets of nature.

SECT. IX.

XXXVI. There is another objection to experimental observations, which is not more reasonable, or better founded than that we have just mentioned, which is made by some superficial schoolmen, who say, that these sort of enquiries, do not require reason and perspicuity, but only eye-sight, diligence, and memory; on which account they condemn them, as things not

well calculated for the exertion of invention, and ingenuity. But how little do these people know of the nature of those physical experiments, or of the manner in which they are made, which have employed the attention of so many learned and sublime spirits, of France, Italy, England, and Germany. In order to discover whether any deceitful appearance has crept in, they repeat every experiment many times over; and invent many ingenious methods of examination, to find out, whether the phenomenon proceeded from that cause, which at first sight it seemed to be derived from, or whether it was the effect of some accidental, or occult one. They make exact and nice combinations of their experiments, and invent ingenious methods of comparing them one with another, and then weigh in a most delicate balance, both the analogies, and the differences between them, in order to derive with almost mathematical certainty, the consequences to be deduced from them; and they peep into those crevices of nature that are nearly imperceptible, that they may discern through them, her inmost secrets; and I beyond comparison, find more delicate ingenuity, and more perspicuity, in many of the experiments of the famous Boyle, than I do in all the abstractions, and reduplications, of the most subtle metaphysicians.

SECT. X.

XXXVII. It is certainly and indispensably necessary, to make experimental observations with the most exquisite attention, in order to avoid our being deceived by them, as our forefathers have been, and as many people are at this day; who, confiding in a superficial and careless experiment, have precipitated their conclusions, and without giving themselves time to reflect, have deduced consequences, from the first information of their senses. From the ascent of the water in the pump, by considering the thing inattentively, was inferred the repugnance of nature to a vacuum. To what labours did Torrizeli, Pascal, and other sublime geniuses submit, what a variety of experiments did they make, and with what ingenuity did they combine and compare them, in order to undeceive the world, and convince mankind, that the true cause of this phenomenon was the weight of the air!

XXXVIII. From the flame flying upwards, was inferred the chimerical region of fire, immediately adjoining to the heaven of the moon. In our essay on Physical Paradoxes, we related the ingenious experiment with which my Lord Bacon proved, that the flame did not ascend from any natural propensity it had to fly upwards, but because it was compelled to it, by the lateral pressure of the air.

XXXIX. I was once in conversation with some school philosophers, and our discourse happening to turn upon physical matters, I took occasion to propose a question to them, which was, whether cold water (and the same may be said of every other liquid) was more subtil and penetrating than hot, which it seemed evidently to be, by the following experiment, that when we poured iced water to drink in summer time into a glass, we perceived the glass wet and moist on the outside, which had the appearance of the water having transuded through the pores of the glass; and as this did not happen when the water was luke warm, or in a temperate state, it was inferred that such water, was not so subtil and thin as cold. As the bystanders seemed converts to the force of the before-named experiment, it gave me a good deal of trouble, to undeceive and convince them of their mistake, although at length I accomplished doing it, by making it evident to them by various most clear experiments, that the moisture which adhered to the outside of the glass, did not transude through its pores, but that it consisted of the coagulated vapours, of the circumambient air, which being in a warm state, condensed and were converted into water, upon their coming in contact with any very cold body, and that they became more condensed the less porous this body was. For this reason, the vapours that are raised by fire, condense as soon as they come to the head of the alembick; and for the same, if we breathe upon a bright iron bar, or any other metallic body, that has a smooth or polished surface, and is very cold, the vapour we breathe from our mouths, when it comes in contact with that body, will condense, and adhere to it; and it is also for the same reason, that in frosty nights, we see the inside of the glass of windows moist, when they are dry without, which appearance, I have known surprise many people, who thought that that humidity was a portion of the external air, which had penetrated through the glass; and it is likewise for the same reason, that our breath, and the breath of other animals, is visible in cold weather, it being then sufficiently condensed by the circumambient air, to make it become

apparent. But the most convincing argument I made use of in the dispute, and which I advanced as a conclusive one, was, that if you covered the exterior superficies of the glass with paper, there would not the least moisture adhere to it, in the whole course of a frosty night, and it is clear that the paper is not impenetrable to moisture, but can only prevent the external air, from coming immediately in contact with the glass.

XL. As it relates to the subject of the impenetrability of glass by liquids, I cannot forbear in this place taking notice of another very common error, which has originated from drawing conclusions hastily from experiments, without making proper reflexions on them. It is generally believed, that the zest of the rind of a lemon penetrates through glass, which opinion is founded on the persuasion, that if you press out the zest on the outside of a glass, the liquor that is within will taste of it. I concluding this penetration to be impossible, after meditating upon the matter, easily discovered the cause of this error; which is as follows, upon pressing out the zest, some of the particles are apt to fly and rest on the edges of the glass or very near it, and adhere to that part, which in drinking the lip is applied to; and thus the palate perceives the taste of the zest, which is communicated to it, from the edge of the glass. To determine whether this was not the case, I squeezed some of the zest in the ordinary way against the outside of the glass, and then turning it half round, applied my lips to the opposite side, and did not perceive the least taste or flavour of the zest of the lemon. And it will happen the same to any one, who will give himself the trouble to make the same experiment.

SECT. XI.

XLI. The evidence of our senses alone, then, is not sufficient to enable us to derive just conclusions from experiments; for caution, reflection, judgement, and reason, are always necessary; and sometimes in so great a degree, that all the exertion of human talents and ingenuity, is not equal to an adequate examination of the phenomena. Sir Isaac Newton, an Englishman, who was a genius of the first rate, and a member of the Royal Society of London, in the beginning of the present century, published various tracts upon optics, in which, he displayed a great novelty to the

philosophers and mathematicians; that is, that all colours exist and are contained, actually and formally, in the rays of light; which are constituted of heterogeneous particles, of unequal refrangibility. He proved this singular opinion, with many experiments of exquisite invention, contrived and considered, with no less exquisite delicacy, and in fact, made more than a few converts to his opinion in many parts, but more especially among the English mathematicians; and a Mr. Gauger who was one of these, enforced the Newtonian doctrine, with a great number of additional experiments; but in a short time, one Senor Rizetti, wrote against this new system, and stated many experiments in favour of the old opinion, and even pretended, that those which had been quoted by Gauger, militated against the Newtonian sentiment. Gauger replied to this, and paid Rizetti in his own coin; that is, he not only defended the consequences which he deduced from his own experiments, but turned those derived from the experiments of Rizetti against the doctrine of the Italian. It is certain that experience, as applied to many objects, opens a most extensive and fertile field for exercising the ingenuity of man, and that nature, even to him who seeks to know her in that way, is in various instances impenetrable and inaccessible.

XLII. But we should acknowledge that in general, the difficulties are not so invincible, as not to be surmounted by reason and industry, and that the deceptions which sometimes result from experiments, proceed either from the want of proper diligence and attention in making them, or from the want of adequate ingenuity, to judge of, and decide upon them.

XLIII. This is very frequently the case with respect to medicinal observations; and from hence springs that enormous disagreement, with respect to the opinions that are alledged to be founded upon them. One says, that he has experienced the efficacy and salutary fruits, of such a medicine, in such a disease; and another insists, that he has administered the same medicine, in the same disease, and under the same circumstances, and found it to be pernicious. One of the two must be mistaken, and I believe it is not very uncommon for both of them to be so; for it might possibly happen, that the medicine neither did good nor harm, and that the administration of it, was a mere matter of indifference, and the thing itself what we call a chip in porridge. But from what could this difference of

sentiment and assertion arise? Why from one of them seeing that his patient, after administering the medicine to him grew better, and from the other perceiving, that his after taking it grew worse; although it might possibly happen, that neither the one grew better, nor the other grew worse in consequence of taking the medicine, but from a very different cause. The diseases of the two patients, from their different constitutions, or internal temperaments, (which is for the most part, a thing impenetrable to physicians) might be in such a state, that the one was disposed to abate, and the other to increase.

XLIV. And although the patients may be numerous, with respect to whom this experience is alledged; and admitting that they neither all die, nor all recover, every one according to his pre-occupation, will impute to the remedy, either the happiness of those who did well, or the misfortune of those who perished; and unless they shall resolve, to make a computation of the good or bad success of the practice of those who use that medicine, with that of those who never administer it, and compare them carefully together, this doubt will probably never be cleared up.

SECT. XII.

XLV. This inattention, is without doubt what produces, and what keeps up in the world, men's estimation for an infinite number of useless things, on which they bestow the noble epithet of remedies; and this false notion, has filled the medical books and the apothecaries shops, with an infinite deal of trash, the reading of which fatigues the memory; and the taking of which, ruins the health and constitution of the patient. You will find accumulated in them, a great number of remedies for slight diseases, which if left to nature, would cure themselves; and although all practitioners do not approve of the same remedy for the same disorder; does this prove, that the patient of each got better by taking this or that thing? I say does this prove it? for neither the imprudent sick person, nor those about him, hardly ever reflect, that there were many others besides him, who were affected with the same disorder, and who without taking any medicine at all, recovered as well, and as quickly as he did?

XLVI. A benign or favourable sort of catarrh, as is very frequently the case, becomes epidemical in a city; for which, some call-in the doctor and take medicines, and others do not; and such is the blindness of those who have had advice as it is called, that they believe they owe the recovery of their health to the doctor, although they evidently see, that all those recovered as well as they, who had no advice, and who took no physic.

XLVII. The delicate fine lady, who whenever she feels a slight pain in her head, sends for the doctor, is fully persuaded, although the pain continues for the space of twenty or thirty days, that the capital pills he prescribed for her, removed it; but it does not occur to the poor lady, that many of her friends and neighbours, who at times have the same sort of pains in their heads, and who take nothing for them, get well as soon, and frequently sooner, than she with the help of all her pills.

XLVIII. It is very common for those who have pains in their teeth and jaws, (and I say the same of all those affections which come and go of themselves,) to fancy, and give the preference to a particular remedy, which they are of opinion relieved them; but it is proper to observe here, that every one applauds his own, and reprobates as insignificant, that which is used by other people. From whence does this arise? some will say, that as peoples temperaments are different, it may happen, that in the same species of complaint, the remedy which does good to one man, may have no effect on another; but this is a common and inefficient evasion, and such a one, as overturns and levels to the ground, the use of all medicinal applications; for if this was the case, as all individuals have distinct temperaments, no less than distinct faces, it would be necessary to contrive a distinct medicine for every single individual, and in all his disorders, to apply particular remedies, different from those, which in the same species of complaints, should be administered to every other person.

XLIX. The cause then of this opposition of sentiments, is most probably as follows. The first time a man is attacked with a pain in his teeth or jaws, it is very common for him to make use of various applications, for besides those which are ordered by the doctor, all those who have felt the same pain, recommend, some one thing, and some another; and as the sensation is

acute, the poor patient anxious to be relieved, proceeds in succession, to apply all those remedies, at length, perhaps at the end of eight, ten, or fifteen days, the pain abates; and as there is not one of those days, in which the patient has not made use of one or other of these receipts, happy is that which he used the last; for to that he attributes his relief, and reprobates all the others as insignificant. Another diseased person, proceeds to take his medicines in regular order; but the course of this regulation, frequently depends upon the casual concurrence in opinion of the people who are consulted, or the power that one of them has of prevailing over, and leading the rest; from whence it frequently happens, that this man of persuasive eloquence, advises the using a medicine in the first instance, which some other practitioner might think it advisable to administer last. The result of which is, that in case the patient does well, his recovery is imputed to the application of the last remedy; and hence it comes to pass, that one reprobates a medicine which another applauds, and so *vice versa*. Thus all the reputation the remedy acquires, be it what it may, depends upon the casual application of it, just at the time nature was disposed to terminate the disorder by a favourable crisis, and hence it is, that the relief came to be attributed to the medicine. And notwithstanding that afterwards, upon another occasion, by using the same medicine in the first attacks of the same complaint, they did not experience the same effects from it; still the same prepossession is continued in its favour, that was formed on its supposed success on the first trial; and although the patient, did not find himself better for several days after taking it in this last instance, they persisted in imputing his recovery to the favourite receipt, and in thinking that without the assistance of it, the disease would have been more tedious, and more severe. Nor is it of any avail to remove this prejudice, their seeing that others, who neither make use of that or any other remedy, do not suffer more prolix, or more acute pain than those who apply it; for although they see they will not perceive it; and if they do perceive it, they will not attend to or acknowledge it.

SECT. XIII.

L. There is a certain delusion, that is very fortunate for the apothecaries, and very fatal to the sick, that is derived from a persuasion in many of these last, that their diseases would be eternal, or incurable, without the help of

medicine; although they see every day, others cured of the same diseases, without this aid; and although the disorder was of so slight a sort, that all those who did nothing but leave nature to itself, recovered from it, he who called in a physician, upon his getting well, thanked the doctor for his cure, when in reality, he perhaps did nothing but delay it. I protest that when catarrhs have been epidemical, I have in various instances observed, that they lasted as long with those who took medicines, as they did with those who took none; and the only difference that I could perceive between them was, that the last recovered their natural state of robustness as soon as the catarrh left them; and that those who had taken physic, did not recover it till several days afterwards.

LI. There is another mistake said to be built upon experience, under the deception of which, many shelter themselves, and pretend, in opposition to those who differ from them in opinion, to justify the necessity of plentiful evacuations. The rigid Helmonists, or sectaries of Van Helmont, reprobate as prejudicial, purging and bleeding, in every case whatever; and in opposition to this doctrine, is urged the common experience, deduced from the cases of an infinite number of patients, who have been purged and blooded, and have all done well; and that there have been some weeks in sickly times, in which one physician has ordered a hundred patients to be purged, and as many to be blooded, and that not one of them has died. How then in opposition to this experience, say the Galenists, can any one pretend to insist, that bleeding and purging do harm?

LII. I do not undertake to defend the Helmonists, nor do I consider their opinion, as more probable than the contrary one; but I say, that from the experience alledged, nothing can be concluded to their prejudice. We should suppose, that those who exclaim against purging and bleeding, do not think them so pernicious, as to be fatal to all those who undergo these operations; for even supposing them to be injurious, they are not capable of destroying a man in perfect health, nor one that is visited by a slight disorder. I believe, that although an intemperate use of them kills many, they are chiefly those who are affected with some obstinate disease; for in these cases, when nature is much weakened by struggling with a powerful disorder, if you add a fresh enemy for her to encounter in the imagined remedy, you complet

her overthrow, and lay her prostrate on the earth. But those who are affected with a disease that is not dangerous, (and there are many of this sort, which in appearance seem violent ones,) generally preserve a sufficient degree of strength, to resist both the distemper, and some unnecessary bleedings and purgings; and although these may weaken the natural faculties, and retard the cure, they very seldom are known to deprive the patient of life.

LIII. That the number of mild disorders, greatly exceeds that of the dangerous ones, is very plain and certain; what then can we find extraordinary, in those who are attacked with the first sort getting well, notwithstanding they may have been injudiciously purged and bled? out of a hundred patients that a physician visits in a week, seldom more than one or two of them labour under dangerous disorders. It is customary with many people to send for the doctor, upon being attacked with a slight inflammation, a catarrh, an ephemeris or fever of a day, a fluxion of the eyes, a repletion of the stomach, and such kind of indispositions; and if he happens to be one of the common or vulgar sort, he seldom fails to purge and bleed. But admitting that the purging and bleeding were never so improperly administered, does it follow from thence, that the patient must die? Or why should this be an unavoidable consequence, if in the same situation, he had been stabbed with a poignard that had not penetrated very deep, or that had not wounded a vital part, he would have escaped with his life?

LIV. As I observed before, I am neither a partizan of the Helmonists, nor a favourer of the Galenists; but I look upon it as a thing certain, that an intemperate use of purging and bleeding, has destroyed many men, and especially, if administered when nature is much weakened; notwithstanding which, I am persuaded that they are serviceable in many cases. Whether they are absolutely necessary, and whether patients could not do well, if other remedies were substituted in their places, is a point I will not take upon me to determine; and especially, when I have fresh in my memory, the declaration of a physician, whose authority has great weight with me. This is Lucas Tozzi, a famous Italian physician, who in his exposition of the third aphorism of the first book of Hippocrates, *Habitus Athletarum*, &c. after combating and controverting the usefulness of bleeding, with

seemingly very efficacious and persuasive reasoning, he asserts, and offers to prove, that it is not necessary in any one of the diseases, in which the Galenists insist it cannot be dispensed with; and in answer to the experience they alledge of its utility, he quotes his own; and says, *if they appeal to experience, and insist that it is evident from thence, that many people have been cured of various disorders by blood-letting; I reply, that in the hospital of the Anunciada at Naples, where I have officiated as physician many years, I have cured in a short space of time, and without taking from them one drop of blood, hundreds and thousands of patients, who have been attacked with some one or other of the following disorders, frensies, pleurisies, quinsies, inflammations of the liver, spitting of blood, erysipelases, and all kinds of fevers.*

LV. What shall we say to this? Lucas Tozzi was not only a great theorist, but also a most expert, successful, and much esteemed practitioner, and as such, his advice was solicited with anxiety by people of the first rank. If he without bleeding cured all those disorders, which in the common opinion stand most in need of that evacuation, and not only cured them, but did it in a short time, what diseases must they be which cannot be cured without blood-letting?

LVI. And it is very remarkable, that in the same manner the Galenists endeavour to deduce from experience the necessity of blood-letting in many diseases, they endeavour also to establish the preference, that should be given to drawing blood in particular disorders, from particular veins, such as the hepatic, and cephalic. Anatomy however, makes it as clear as the sun at noon-day, that this preference is not grounded on any solid foundation, and that the cephalic vein bears no more relation to, nor is any more connected with the head, than the hepatic, nor the hepatic with the liver, than the cephalic; and that all the veins of the same arm, communicate indifferently with all parts of the body, as the laws of circulation demonstrate; and as that observation which was pretended to be derived from experience, was founded in mistake, it is not improbable, that that which is generally alledged in favour of bleeding, may be so likewise.

LVII. What appears to me is, that the rule so much cried up by the Galenists, and which establishes the necessity of bleeding, in pleurisies, is liable to so many exceptions, that we ought not to regard it as a general one; and we have observed in another part of our works, that in some epidemical pleurisies, it has been found to be evidently injurious. It is but a little while ago, that a learned French physician, whose works are mentioned in the Memoirs of Trevoux, wrote strongly against bleeding in winter pleurisies and peripneumonies; and his reasoning had great weight with some eminent physicians of Paris. I can certify, that in the last winter 1731, when many people were attacked with pleurisies in this country, it was generally remarked, and we received informations to the same effect from all quarters, that those who were blooded died, and that those who were not did well.

LVIII. I would have it understood, that I wish all I have said, with respect to the utility or inutility of purging and bleeding in general, should be considered as the sentiments of a man, who takes no side in the dispute, but who proposes what he advances problematically, and with a view of convincing the world, that in order to avoid being misled, they should examine with great attention and exactness, any observation that is alledged to be founded on experience, for this is the principal object I had in view, in writing this discourse.

LIX. It would argue weakness and want of judgment, for any one to esteem a thing as a remedy for a particular disease, without reflecting and making a remark, the propriety of which must stare every man in the face, and that is, comparing the success of the practice of those who use that medicine, with that of those who do not make use of it. There have been physicians, who have declared themselves enemies to the administration of the bark; but notwithstanding this, there is no body at this day, who disputes its febrifuge efficacy in intermittent cases; for experience shews, that it will stop the fit, although it may return again afterwards, and although in the opinion of some people, the medicine may leave some bad impression on the body. If purging and bleeding would have the same effect, in some sort of fevers, we should all agree in attributing to them a febrifuge virtue; notwithstanding that some practitioners, might give the preference to other

remedies, as more safe, and better adapted. But this is so far from being the case with respect to these evacuations, that their efficacy of removing fevers, is at this day strongly contested, for experience has not yet manifested, that they are remedies for these disorders, in any degree or manner, that is not exceedingly doubtful or disputable.

SECT. XIV.

LX. As far as the nature of the case will permit us (for all things are not capable of being mathematically or decisively demonstrated), we should endeavour to discover the truth, by imitating the diligence and attention which many English physicians exercised, in examining into the eligibility of the precautionary remedy called inoculation, practised by the Turks, to elude the fatal effects of the small-pox. This is a subject, that has been much talked on in other kingdoms for some years; although in Spain we have hardly heard it mentioned. The event of communicating the infection in this way, most generally is, that the person inoculated has the distemper very lightly, and is hardly ever obliged through the whole course of the disease, to keep his bed.

LXI. The account of this precautionary method, was brought to England by one Maitland, who was surgeon to Mr. Wortley Mountague, his Britannic majesty's ambassador at the Porte, and was from thence communicated to the other nations of Europe. Maitland had observed, that this practice was almost universal in all the cities of the Levant, and that it was attended with wonderful success. He communicated these remarks to Mr. and Mrs. Mountague, and they had fortitude enough, upon the faith of what he had told them, to make the experiment upon a son they had with them of six years old, which being attended with success, they repeated it upon another of their children when they returned to England. Many were animated with these examples and these accounts; and the practice of inoculation, began to be much approved, and much adopted in England, notwithstanding it was greatly opposed, and objected to by some people there, and by many in other countries; and especially by the physicians of Paris, who exclaimed strongly against it.

LXII. But as this contest was such a one, as ought not to be determined by speculative reasoning, an appeal was made to experience, which appeal, was prosecuted in such a manner by those who were advocates for the practice, as seemed to exclude all doubt and perplexity. Physicians who resided in different parts of the kingdom, were requested to give accounts of the success of their practice by this method, and to transmit them to London, which when received, were printed and published. By these accounts, two facts were established; the first was, that inoculation freed the patient from the danger of a second infection; and the second was, that those to whom the small-pox was communicated in this way, very rarely died; and although it is true that some few of them miscarried, they were mostly such, as were afflicted with some other disorder at the time of their inoculation, of which they would have died, if they had not had the small-pox; but these altogether, did not amount to but very little more than a tenth part of those, who perished by catching it in the natural way; for by the best calculation they could make, one out of eight died of the last, and not above one in eighty of the first.

LXIII. This is the account I have read of this matter in the Memoirs of Trevoux; to which some may object, that it is possible the relations of the cases, were not published so faithfully as they were received. To this I answer, that it is probable great pains were taken in making the enquiries, and great fidelity observed in publishing the accounts, as the Memoirs of Trevoux tell us, that upon the strength of their authority, some of the royal family of England were inoculated.

LXIV. And let it not be insisted in opposition to this, that if the fact had been so well established, it could have admitted of no contradiction. They know little of the human passions, who think this remark a solid one. Those who contradict, either through blind prepossession, or from motives of emulation, interest, or envy, seldom submit, or will own themselves convinced by proofs that are self-evident; nor is there any species of evidence, that can bar every door, at which a false subterfuge may steal in, nor against the intrusion of a thousand sophistical objections, introduced by those, who are under the dominion of the beforenamed passions. I speak

feelingly of this matter, having had woeful experience of the truth of what I assert, since I first began to write for the public.

LXV. In reality, some of the objections that were made to inoculation, were the most ridiculous in the world. Some rigid Presbyterians, made the cause of religion interested in the question, and asserted, that that practice, was an affront to the sovereignty, and an opposition to the decrees of God; and one of them declared in his preaching, that it was a diabolical invention, for the devil by inoculation had communicated the small-pox to Job, which was the distemper that so sorely afflicted that holy Patriarch. Into what absurdities, does a violent ardour hurry a man in a controversy! Of all sensible human beings, there are none so nearly allied to mad men, as passionate disputants.

LXVI. Towards the conclusion of this virulent contest, a very curious circumstance came to light, which was, that this precautionary method which had made so much noise, and which had generally been supposed to have been brought from Turkey, had for ages before, been in use in the island of Great Britain itself; for that it had been practised from time immemorial, in the southern parts of the principality of Wales, where they communicated the infection in two ways either by rubbing some of the variolous matter hard on the skin of the person to whom the distemper was to be given, or else by dipping a needle into that matter, and pricking his skin with it; and as they gave the person from whom the matter was taken a small sum of money, this was called buying the small pox; and they produce very authentic testimonies, that scarce any one of those who came by the distemper in this manner died, and that there was no instance of a person who had acquired the disease in this way, ever having had it a second time.

SECT. XV.

LXVII. We shall conclude this discourse, by pointing out three capital errors, which are derived from want of proper attention in making experiments. The first is, that of taking for the effect, what in reality is the cause, and taking for the cause, what is nothing more than the effect. The

second consists in taking for the cause, something that comes in by accident, and which has no influence whatever. The third consists, in between two effects proceeding from one and the same cause, taking one of them for the cause of the other. I shall give examples of these three errors, in observations appertaining to medicine, which are said to be derived from experience, because mistakes in this branch, are generally attended with worse consequences, than those in other ordinary physical matters.

LXVIII. It happens, that a man feels an ardent and extraordinary thirst upon himself, without being able to assign any apparent cause of it; he drinks water to great excess, and in a few hours is seized with a fever, or an acrid fluxion. It is common in such cases, to attribute the indisposition to the excessive drinking of water, and to apprehend, that was the cause of the disease; but this was so far from being the case, that the indisposition was rather the cause of his drinking to that excess. But I would have it observed, that I speak of an instance, in which the thirst was not brought on by any manifest or apparent cause, such as the having used any violent exercise, or having been exposed to some great heat, either of the sun or fire, or having been a long time without drinking, or having eaten something very salt. I say that agreeable to this state of the case, it is very clear, that the thirst must have arisen from some internal cause; but the question is, from what cause? Why it could be from no other, than a morbid disposition, that had begun to prevail within his body; or let us express it in another manner, and say, that it proceeded from an acrid or salt humour, which had just begun to get in motion, and by vellicating the fibres that produce the sensation of thirst, had by that means excited it. Every preternatural and extraordinary effect, demands a preternatural and extraordinary cause to occasion it; and if we suppose that the thirst was such an effect, and we cannot assign any external cause to which we can attribute it; we must conclude that it proceeded from some preternatural internal cause; which in all probability, must be the morbid disposition.

LXIX. The want of this reflection, occasions great errors to be committed in physic, for by running counter to the course of nature, you must unavoidably mistake the road that leads to the cure. The consequence of mistaking the effect for the cause, is administering as a medicine, what in

reality is poison; for it is clear, that if the physician apprehends the humidity and coldness of the water to be the cause of the distemper, when in truth it was produced by the acrid, salt, peccant, or inflammatory humour, I say if this should happen to be the case, and he calculates his prescriptions to correct the humidity and coldness, he by so doing, will inflame and increase the original disorder.

LXX. This species of error, is not confined to the case we have instanced, but is capable of being extended to a vast expanse. I am much inclined to think, that all the extraordinary and vehement emotions, both of the irascible, and libidinous kind, which precede distempers a small distance of time before they make their appearance, and for which you can assign no special external cause, are the effects of those distempers, in their original stages; I mean, that in the first agitations of such cases as we are speaking of, a person is apt to fall into violent passions upon very slight provocations, such as he knows by his own experience, were never used to agitate or have that effect on him, and either immediately, or within a few hours afterwards, he is seized with a fever. People are apt to suppose, that the passion was the cause of the disease; but I say, that the disease occasioned the anger; for if this man, agreeable to his natural disposition, was never subject to such violent starts of passion, upon such, or even greater provocations, it necessarily follows, that that which he experienced upon this occasion, must have been the effect of some preternatural internal cause, which lay concealed within him, most probably the first fermentative movements of the peccant humour, which soon afterwards produced a fever. In reality, it is easy for any one to perceive, and I have remarked it many times both with respect to myself and others, that the irascible passion, is much more apt to be inflamed upon slight occasions, in those first stages, or almost insensible beginnings of such indispositions as tend to be somewhat serious, than at other times.

LXXI. I do not however deny, that the ardour of passion may excite a fever; for this without doubt, may have a great share in producing such an effect, and we may reasonably suppose that it has, and especially in such people as are of a choleric disposition; but when this is not the case, we should rather suppose, that the passion tended to augment the indisposition

which followed the sudden fit of anger, and which would have made its appearance without the intervention of that anger, although it might have been attended with milder symptoms. And the same we say of anger, is applicable to sadness and fear also, for they conformable to the language of the philosophers, are passions appertaining to the irascible.

LXXII. We may reason in the same general way, upon the effects of the amorous passion. All vehement desires for ordinary objects, which frequently present themselves to a man's view, and which are totally unusual to him, and for which we cannot assign any special external circumstance that should excite those desires, we should conclude, that they proceed from some preternatural internal disposition. The indulgences or gratifications of these extraordinary desires, are always attended with actions of excess, to which are generally attributed the indisposition that follows them; but in truth, the indisposition which laid concealed, irritated the appetite, and was the cause of the excess, and not the excess of the indisposition.

LXXIII. The following mistake also, is very frequently fallen into. A person who has always been indifferent about this or that particular food, we will say lettuces for example, all of a sudden takes a great fancy for them, and will eat two or three large ones. If he is afterwards attacked with a pain of the head, and defluxion upon the breast, or a diarrhœa, the fault is sure to be laid upon the lettuces, which are accused as the cause of all the mischief; but in reality, the mischief had before crept into the constitution, and had induced the extraordinary desire to eat the lettuces.

LXXIV. I would not however, be understood to insinuate, that eating of any thing to excess, does not frequently occasion or bring on various diseases; for I only mean that my position should be understood to allude to a desire that is vehement, and unusual to the person, and for which, you can assign no visible external cause that should excite it; for under such circumstances, there is a necessity for concluding it was owing to some internal cause, that was sufficiently powerful to merit the name of a morbid disposition; which is an appellation it could not have deserved, if the appetite although extraordinary, had not been excessive.

LXXV. I am confirmed in the truth of the remark I have just made, by the reflexion, that a diversity of appetites, must undoubtedly proceed from a diversity or alteration of temperaments; and it follows of course, that every alteration in the temperament, must be attended with an alteration of the appetite. It is easy to discern, that no sick person, preserves his appetite in the same even state, in which it continues when he is in health; and this not only with respect to the quantity he eats and drinks; but with respect to the quality of his nourishment also; nor is this confined to the objects of his palate only, but extends to those of all his other feelings and inclinations, both internal and external.

SECT. XVI.

LXXVI. The second capital error committed in making experimental observations, and which is more common than the first, is that of mistaking for the cause, something that intervenes by accident, and is neither cause nor effect. There is scarce any sick person, who does not fancy he knows what has been the cause of his disorder, which he generally imputes to something particular that he has done, or some alteration he has made in his way of living a little before he was seized with his distemper; although the thing to which he imputes it, bears no allusion to, nor any proportion with the disorder that afflicts him. The having eaten an olive more than it was usual for him to eat, or having fasted a quarter of an hour longer than his usual time, or having drunk two spoonfulls more than common, or abated twenty yards of his ordinary walking, together with some other particulars, that are equally trifling and insignificant with those we have mentioned; but notwithstanding the futility of these observations, he is apt to impute to such causes the disorder he labours under, without reflecting, that this machine of ours, from the weakness of its own texture, is sufficiently exposed to its breaks, ebbings and flowings. The humours of the body, even when the influence of all external causes, and every thing that depends on our own free will, are regulated with perfect uniformity, are nevertheless exposed to various alterations. The heterogeneous nature of them as they respect one another, and also considered with respect to every particle of each of them separately, must necessarily conduce to their being in different states. If those superstitious spirits, who are such idolaters of their health, that with respect to their own regimen they would weigh even atoms, would

well consider this, they would free themselves from the continual anxiety in which they live, and which is more pernicious to them, than those very indispositions they are so terrified at, and which they are at so much pains to escape.

LXXVII. But the most common accusation of all, is that which charges the weather as the cause of our disorders. He who commits no excesses, and cannot assign any other cause for his being out of order, lays the blame upon the weather, and even he who does commit them, to avoid criminating himself, lays the blame on the weather also; which be it hot, moist, dry, variable, or settled, people are never at a loss to find out some pretence whereon to ground the accusation. If in July, as is customary at that season, we find it very hot, we say the heat is the cause of our disorders; but if the weather at that time happens to be more benign and temperate than usual, they still lay the blame upon it, alledging that such a temperature of air is not natural to the season. The same sort of charge is brought against the cold in winter, whether it is intense or moderate. If at that time of the year, the weather is variable, there is scarce any one who does not find fault with it; neither is it exempted from blame if it is settled, for then they say that change of weather is indispensably necessary to our constitutions; and that any kind of weather which lasts a long while is hostile to them; that long cold occasions constipations, long heat dissolves and weakens us, long wet suffocates us, and that long drought burns us up and consumes us.

LXXVIII. I have often remarked, that all our misfortunes are imputed to two common enemies: our spiritual ones to the Devil, and our bodily ones to the weather. There is scarce any one, who, in order to extenuate his own guilt, does not say, he was tempted by the Devil to commit the crime he has been guilty of. He is as irrational who thinks, that if there was no Devil to tempt us, we should never sin, as he is who thinks, that if the weather was regulated in a most perfect form and order, we should never be sick. Within ourselves, and in the very essence of our being, lies the origin of all our ills, both spiritual and temporal; and our nature is swayed by its own weight, towards both the one and the other evil; although we can never be led into the first without our own consent; but the other species, may frequently be brought on us against our will.

SECT. XVII.

LXXIX. The third error committed in making experimental observations, although it is not so common as the two first, is pretty often incurred. If he for example, who on account of his having used some violent exercise, drinks small liquors to excess, and afterwards finds himself feverish, imputes his being so, as is very frequently the case, to his having drank such liquors to excess; the generality of men, seldom reflecting upon any excesses, but those of the appetite; but with all this, violent exercise is much more likely to inflame the blood, and disturb the humours, than drinking to excess of small liquors, and therefore it would be much more rational, to impute the fever to the violence of the exercise, than to drinking small liquors to excess.

LXXX. I believe that from the blunder of mistaking two effects of the same cause, the one for the cause, and the other for the effect, arose the opinion, which is so common among medical people; that all fluxions wherever they fall, the gout not excepted, descend from the head; and it being very common, for those who are affected with acrid fluxions that fall on any part of the body, to feel pains and heaviness in their heads; I suppose that from thence sprung the notion, that all fluxions originate in, and are derived from the brain; but there are not a few modern physicians who are of a contrary sentiment, and in my opinion they are right.

LXXXI. In the first place, I do not know why the vicious humours, from whence the matter of fluxions are derived, should make the grand circuit through the head, before they fall on any particular part that is at a distance from it; as they could, by being mixed with the blood in their circulation through the veins and arteries, be derived immediately from them, on any member or part of the body.

LXXXII. Secondly, if such a vast quantity of humour, as is discharged by some fluxions, was to be lodged in the brain, it seems to me that it must render a man quite stupid, and that organ incapable of exercising any of its functions.

LXXXIII. Thirdly, it is not easy to point out the channel, through which the humour passes from the interior part of the head. Many pretend to say it passes through the Ethmoides, or Os Criboso; but Sneider denies this, for that you cannot find any cavities or perforations in this bone, through which those humours could flow, and especially if they are pituitous and clammy, as the antients supposed them to be; and to this we may add, that this bone is entirely covered or lined by the meninges, and the interior tunic of the nose. It is true, as Doctor Matinez observes in his treatise on anatomy, that its upper part is very porous, and that from thence it came to be called the os criboso, or spungy bone; but as these spungy pores do not pervade the whole bone, and are not pierced through it, setting aside the obstacle it would meet with from the membranes or tunics that surround it, the humour could not pass that way. If it is insisted, that it flows through the nervous ducts, I ask how it comes not to be attended with obstructions, that must produce dangerous consequences?

LXXXIV. Fourthly and lastly, that through whatever channels you suppose the humour to pass; how does it happen, that neither in them, nor the parts immediately connected with them, it does not excite any sensations, but that sore feelings, are only perceived at the part where it vents itself? Is it not totally incredible, that a very acrid humour derived from the head, and which flows to the stomach, to the breast, to the intestines, and even to the extremities of the feet, should produce no sensations in the intermediate parts between the head, and the place on which it falls? This difficulty, which occurred to me many years ago, I have often mentioned to medical people, but never could get any solution of it that was satisfactory to me.

LXXXV. If by way of objection to what I have here advanced, the argument we have mentioned before is urged, that defluxions on any part of the body are generally attended with a pain in the head; I answer, that it cannot be inferred from thence, that the humour descends from the head. In the first place, for I have observed it many times with regard to myself, these fluxions are often not attended with any pain in the head at all; and to make the inference just, they should always be attended with one. Secondly, although a pain in the head should constantly accompany a defluxion, the

objection made would easily be removed, by saying, that this appearance is fallacious, for that both the pain and the defluxion, are effects of the same cause, and not one the cause of the other. In fact, reason tells us this is the case; for the acrid humour, which when separated from the mass of blood, falls on, and discharges itself at this or that place, while it continues in the circulation, has nothing to hinder it from venting some portion of its poison on the head, and exciting pain there; and especially, as it is generally supposed, the nidus of the humour which flows in defluxions, is in the glands, and the brain holds the first place in that class; for which reason, Hippocrates and Wharton, call it the great and principal gland.

LXXXVI. If it is replied to this, that in every defluxion which is somewhat violent, although we do not feel what may be properly called pain in the head, we at least perceive a heaviness in it; which renders that organ unfit for its operation. I confess that this is so, but to the confession, shall beg leave to add two remarks. The first is, that this is not a symptom peculiar to fluxions, for that the same thing happens in many other disorders; but the generality of physicians do not conclude from hence, that those diseases originate in the head. Secondly, neither is this heaviness or inaptitude, peculiar to the head, for upon observation, you will find that fluxions and many other disorders also, have the same effect upon the other members of the body. Whoever is affected with a violent defluxion, either on the throat, the breast, the stomach, or any other part, will find that his whole body, and every member of it, is more heavy and listless, than when he is in good health; and that all his limbs are less fit for action; and that they all with a very little exercise of them, become soon tired. Thus we have no reason to attribute a heaviness as peculiar to the head, in the attacks of fluxions, when we see it is common in those attacks, to all the other members of the body; and it has been for want of making this reflexion, that the world have conjectured all defluxions were derived from the head.

A
DISPLAY of the INTELLECTUAL FACULTIES
OF THE
VARIOUS PEOPLE,
With REMARKS on the TALENTS of the
DIFFERENT ONES.

SECT. I.

I. There is no doubt, but the different temperament of air in countries, induces a sensible diversity in men, brutes, and plants. In plants, the difference is so great, that the same which in one country is innocent and salutary, in another is noxious and poisonous, which, as we are assured, is the case with the Persian apple. The diversity between brutes, is not less than that between plants, both with respect to their size, fierceness, strength, and other qualities; but besides what is obvious and clear to the observation of all men in this matter; we are assured there are countries, where many sorts of animals, degenerate totally from the characteristic property of their species. If we are to believe Lucianus, Macedonia produces serpents, so social to the human race, that they will play with children, and gently apply their mouths to the nipples of women's breasts to suck their milk; and as Louis de Marmol informs us, in Gruregra a mountainous country in the kingdom of Fez, the lions which are very numerous, are so tame and so timid, that the women with sticks in their hands, beat and drive them about like so many domestic dogs.

II. If the difference in our own species, which is produced by a diversity of climate, is not so great, it is sufficiently evident and apparent. It is manifest, that there are countries, where the inhabitants are remarkable for being of larger stature than they are in others, or for being more active, more strong, more healthy, or more beautiful; and that this difference

subsists with respect to all other things, that depend upon the two faculties that are common to both men and brutes, that is, the sensitive and the vegetative. Even in nations which border upon each other, this difference is sometimes observable.

III. Distinct dispositions of body, are attended with distinct qualities of mind; and from distinct temperaments, result distinct inclinations; and from distinct inclinations, distinct customs and manners. The first of these consequences is a necessary one; but the second is not so certain, because a man's free will may restrain the impetus of his inclination; but as is most commonly the case, men let their free will be led by the emotions, which are produced by the interior disposition of the machine; so that we may safely pronounce, that men in one country are more addicted to passion and anger, in another to gluttony, in another to be lascivious, and in another to be lazy or slothful.

IV. The inequality between men of different regions in the rational part, is generally supposed not to be less, but rather greater, than it is in the sensitive and vegetative; and it is not only in the conversation of the vulgar that we see this opinion upheld, but we see it supported in the writings of the learned; where we find one nation described as rude and savage, another as stupid, and another as barbarous; so that when according to this description of them, we come to compare one of these nations, with one of those who are esteemed cultivated and polished, we conceive that there is nearly as great a difference between the inhabitants of the one and the other, as there is between men and wild beasts.

V. I differ so widely from the common opinion in this particular, that with respect to the essential part of the use of the understanding, in the inhabitants of one nation and another, the difference is imperceptible. This sentiment cannot be better justified, than by shewing, that those nations which are commonly reputed rude and barbarous, are not inferior in ingenuity to the most cultivated ones, and perhaps in some instances excel them.

SECT. II.

VI. We will begin our comparisons with the nations of Europe. The Germans, who have been so stigmatized for heavy dull people, that Father Domingo Bohursius a French Jesuit, in his Conversations of Aristius and Eugenius, made no scruple of pronouncing, that he doubted whether it was possible to find a man of bright talents in the whole country: however, in answer to this charge, it may be alledged, that Germany has produced so many excellent authors in all kinds of literature, that it is hardly possible to number them. I doubt whether the before-named Frenchman, by rummaging for them through the series of all past ages, can point out two of his countrymen, who were equally eminent with Rabanus Maurus, and Albertus the Great. The epithets which Cardinal Baronius bestows upon Rabanus Maurus, are, that he was the resplendent star of his age, and the supreme Theologian of his time; and Sextus Senense, recognizes him as a man most perfectly accomplished in all kinds of letters; and the Abbot Trithemius, after celebrating him as a most excellent Theologian, Philosopher, Orator, and Poet, adds, that Italy never produced a man equal to this; and Trithemius was not ignorant, that Saint Thomas Aquinas was a native of Italy. Whom can France boast that exceeded Trithemius himself, who was so greatly venerated by Cornelius Agrippa; or who excelled the Abbot Rupertus, or Father Athanasius Kircher, who according to Caramuel was divinely learned; or Father Gaspar Schotti, and many others whom I shall omit? Neither should we forbear to mention that ray or flash of criticism, Gaspar Sciopius, who was the terror of all the learned men of his time, who at the age of sixteen begun to write books, which were the admiration of men of years and experience. I have in this literary map of Germany, only pointed out the mountains of greatest eminence, as I had not room to insert the lesser ones in it.

VII. The Hollanders, who in the days of antiquity were reputed for such stupid people, that it was common among the Romans, when they had a mind to describe a heavy dull man, or one that was slow of apprehension, to make use of the following proverbial expression, *auris Batava*; that he had the ears of a Hollander; however, the Dutch since then, have clearly proved, that imputation was a false one, for the opinion of their being people of ability, is at this day very fully established. Their civil government, and their industry in commerce, are the admiration of all the world; and there is scarce an art, that is not cultivated and brought to great perfection among

them. The two Williams of Nassau, Erasmus, and Hugo Grotius, are striking examples of their talents for policy and literature. Thus we see in this as well as other nations, the want of application has been construed into rudeness; but as soon as the neglect of study was remedied, the injustice of the imputation became manifest.

VIII. The abilities of the Muscovites also, have till of late days, been held in as much contempt as those of the antient Hollanders were; for it was but in the last century, that Urban Chevreau a celebrated wit of France, laid that a Muscovite was *the man of Plato*; by which expression, the French Author alluded to the poor definition that philosopher gave of a man, when he described him to be an animal without feathers, who walked upon two legs, *animal bipes implume*; this gave occasion to the joke of Diogenes, who having plucked the feathers off a cock, threw him into the school of Plato among his disciples, calling out with a loud voice, *see the Man of Plato*. Chevreau attempted to say, that the Muscovites were nothing like men but in their figure. But after the Czar Peter Alexiowitz had introduced arts and sciences among them, it soon became evident, that the Muscovites were as much men in all respects, as any of the other people in Europe. But if this never had been done, how could any one suppose that a people who were stupid and senseless, should have been able to form a most extensive empire; and to preserve it for so long a space of time as the Russians have preserved theirs? It required great ability to conquer such an empire; and to preserve it in the face of two such powerful enemies as the Turks and the Persians, much more. I am not ignorant, that Muscovy is a part of antient Scythia, whose moroders were reputed, and with reason, the most savage and barbarous of men; this barbarism, did not proceed from their want of natural talents, but from the want of those talents being cultivated; of which truth, the famous philosopher Anacharsis is a striking instance, who was the only man of that nation, that went to study in Greece. If numbers of other Scythians had done the same, Scythia might possibly have exhibited many Anacharsises.

SECT. III.

IX. If we attempt going out of Europe, every thing beyond those limits seems barbarous to us. When the imagination of the vulgar contemplates Asia, it represents to them, the Turks, the Persians, the Indians, the Chinese and the Japonese, as little better than so many congregations of satyrs, or demi-brutes. But notwithstanding this, there are none of these nations, who do not make as great advances in all the things they apply themselves to, as we do in whatever we study.

X. Their abhorrence for, nor their ignorance of the sciences in Turkey, is not so great as is generally imagined; for they have professors both in Constantinople and Grand Cairo, who teach astronomy, geometry, arithmetic, poetry, and the Arabic and Persian languages. But they have not so great an esteem for those faculties, as they have for politics; in their knowledge of which, there is scarce any nation who is equal to them, nor are there any subtilties appertaining to political arts, which they are ignorant of. The great English traveller, Mr. Chardin, in the relation of his voyage to India by land, tells us, that when he passed through Constantinople, he had much conversation with Senor Quirini the Venetian ambassador at the Porte; who assured him, that he had never treated with a man of equal penetration to the vizir of that time; and declared, that if he had a son whom he wished to breed a negociator, he would send him to the school of the Ottoman court preferable to any other. The Turks excel in all things that require dexterity of hand, and in all bodily exercises they take a fancy to. They are the best pen men in the world, and this is the reason, why they never would permit the art of printing to be introduced among them; and they are also the most active and dextrous rope-dancers in Europe. Cardanus relates wonders of two of them he saw in Italy, one of whom became a Christian, and lived an exemplary life, but always continued his old exercise; although his becoming a convert to Christianity, removed the suspicion that the vulgar entertained of his dealing with the devil. The dexterity of the Turks as archers, is so eminent, that the force and exactness with which they can shoot arrows is almost incredible. John Barclay, in the fourth book of his *Satyricon*, testifies to have seen a Turk penetrate with an arrow, a plate of steel, that was above an inch thick; and that he had seen another, who with the shaft of an arrow without any iron to it, split the body of a small tree. In the art of confectioning poisons, they are also very expert; for they will make them not only very active, but contrive them so,

as that they may be administered without creating suspicion; a thin vapour with which a piece of fine linen is impregnated, if it is afterwards made up in any form, and sent as a present to the person to be dispatched, is made the instrument to take away his life. This is a most horrid and execrable art; but although it proves the perverse and wicked disposition of those people, is an argument of their great talents, for acquiring a profound knowledge of all they apply themselves to understand.

XI. The Persians, are still more polished than the Turks. They have their colleges and universities, where they study arithmetic, geometry, astronomy, natural and moral philosophy, medicine, jurisprudence, rhetoric, and poetry; which last, they are passionately fond of, and write elegant verses, although they are generally too redundantly abounding in pompous metaphors; and they are so far from being guilty of that ferocious neglect of urbanity which we are apt to impute to all the Mahometans, that there are no people who excel them in expressions of civility, tenderness, and affection. Whenever a Persian invites any one to his house, or is desirous of manifesting his esteem for him, he generally makes use of these, or some such like expressions; *I intreat you that you will honour my habitation with your presence; and I shall be ready to devote myself to making it agreeable to you in every shape, and to indulging all your wishes; and should even be willing that the favourite females of my eyes, should be converted to carpets for you to set your feet on.*

XII. It is true that in the East Indies, we do not find the cultivation of letters, but we find there a more than ordinary capacity for attaining them. John Baptist Tavernier, speaking of the negroes or mulattoes of that country called Canarines, many of whom are employed in various occupations in Goa, the Philippine islands, and other places in the possession of the Portugueze and Spaniards, says, that the children of those who apply themselves to study, acquire more in six months, than the children of Europeans learn in a year; and says further, that he was told this by some of the religious of Goa who had the tutelage of them. I am persuaded that the Portugueze, when they first saw this sooty-faced race, believed their reason, was as obscure and dark as their complexions; and that they supposed themselves nearly as superior in natural talents to them, as men are to

brutes; and in how many parts of the globe, where we judge the inhabitants to be stupid, might we possibly see the same thing! for the metal of their understandings, for want of examining it by the touch-stone of study, has remained occult and unknown.

SECT. IV.

XIII. But the greatest injustice that the vulgar have been guilty of in this particular, is in the conception they have formed of the abilities of the Chinese. Why do I say the vulgar? When you hear men dignified with university honours, whenever they have a mind to heighten an extravagant action, or mode of proceeding that is contrary to all reason, say at every turn, *this could hardly have happened among the Chinese*; which amounts to the same thing, as making the conduct of the Chinese, the standard for measuring extreme barbarism. This certainly cannot tally very well with the idea those people entertain of themselves, which is, that they are the quintessence of policy, ingenuity, and penetration; for it is a proverb among them, that the Chinese have two eyes, the Europeans but one, and that all the rest of the world are totally blind.

XIV. The truth is, that they have great reason to believe this; for they excel in their civil government and policy all other nations. Their precautions for preventing civil, and avoiding foreign wars are admirable; and learned and wise men, are not held in such estimation by any other people in the world, as by them, and it is to such only, that they confide the reins of government. This alone, is sufficient to give them credit for being the most rational of mankind. The excellence of their inventive faculty is evident, from their being the first people, who hit upon the three famous inventions, of printing, gun-powder, and the nautical needle; for the knowledge of these in China, is supposed to be much anterior to our acquaintance with them in Europe; and there are also some well-grounded suspicions, that they were communicated to us from the Chinese. They excel exceedingly, in whatever arts they apply themselves to attain; and notwithstanding our utmost exertions, we in Europe, have not been able to equal, nor even to imitate them in many.

XV. There is great reason to believe they excel us in their knowledge of medicine, and in their mode of applying it. Their physicians, are both physicians and apothecaries; and it was formerly the custom in all nations to unite the two professions in one; would to God it was the same now! They keep in their houses all the medicines they make use of, which consist of various simples, whose virtues they have well examined and understand; which they collect, prepare, and apply. They are very attentive to, and take a long time in examining the pulse; and it is very common for them to be near an hour in exploring its movements. But the information they get from this circumstance, and the appearance of the tongue is such, that by revolving these particulars in their mind; they without asking any questions, either of the sick person or those who attend him, pronounce on what sort the distemper is, the symptoms that attend it, the time the patient was taken ill, together with all the antecedent and subsequent circumstances, that have accompanied, and will accompany it.

XVI. I am well aware, that this will appear incredible to our physicians; but the various accounts we have of China, some of which, have been written by men of most exemplary and unexceptionable characters, all agree in vouching the truth of these particulars, so that our refusing to give credit to them, would seem rash and unreasonable. But if I could possibly have entertained any doubt of this matter, our illustrious Don Joseph Manuel de Andaya y Hara, the worthy Prelate of Oviedo, would have removed them, who confirmed these relations to me, upon his own experience of a Chinese physician that practised in Manila, the capital of the Philippine Islands, of whom he told me wonderful things, both with respect to his prognostics and methods of cure. I am persuaded that some of our physicians about the court, have got the Book of Andrew Cleyer, the first physician of Batavia, intituled *Medicina Chinensium*, which was printed at Augsburgh. The diary of the learned of Paris in the year 1682, makes mention of it; and in that may be seen more at large, many of these accounts.

XVII. But, skilful as the Chinese physicians are in the practice of their art, the Chinese people are not behind them, in their skilful and wise regulations for the government of the physicians. If the physician, after having examined the pulse and the tongue, does not hit upon the distemper

and the symptoms of it, which rarely happens, he is dismissed as unskilful, and another called in. If he does hit upon the foregoing particulars, which is most commonly the case, they confide the cure to him, and he immediately goes home, and fetches from his house a bag of simples, which he gives directions for the using of, pointing out the manner in which, and the quantity of each that should be applied or taken. When he has compleated the cure, he is paid amply for his time and attendance, and also for the medicines he has expended; but if the patient does not recover, the physician is not paid, either for the one or the other; so that the money of the sick person, is saved if he does not get well; and the physician, loses both his time and his medicines if he does nor cure him. It is much to be wished, that such a regulation subsisted among us; for although Quevedo some time ago complained of the want of it, he did not know it was established in China. It is true, that he made the complaint in a bantering way, but I believe he felt the want of the regulation very sensibly.

XVIII. We may say in general of Asia, that this was the country in which arts and sciences originated; letters owed their birth to Phenicia; and were from thence transported into Egypt and Greece; and the knowledge of astronomy came from Chaldea, and from thence was circulated into various other countries.

SECT. V.

XIX. As to what regards Africa, we should consider, that it gave birth to a Cyprianus, and a Tertullian, and what is still more, to an Augustin; and that the Africans were at one time, as much superior to the Spaniards in military skill, as the Spaniards at this day are to the Africans; and that there was a time, when the conquest of all Spain cost the Africans less blood, than it has since cost Spain to conquer a few spots in Mauritania. The soil and the climate of Africa, are the same now they were formerly, and consequently capable of producing equal geniuses; and the fault of not cultivating them, should not be imputed to the soil or the climate, but to the want of opportunities of instruction, or to the neglect of application; but with all this, they are perhaps not so uncultivated as is commonly thought. Father Buttier, in his little treatise, intituled *An Examination of Vulgar*

Prejudices, gives us the copy of a speech, which the ambassador from Morocco made to Louis the Fourteenth, which was as eloquent and as much to the purpose, as if it had been composed by a learned European.

SECT. VI.

XX. The conceit, which upon the first discovery of America was entertained of its inhabitants, and which still subsists among the generality of people, is, that they are not so much directed by reason as by instinct; as if some Circe, in her peregrinations through that vast country, had transformed all their men into beasts. But with all this, there are abundant testimonies, that their capacities are in no wise inferior to ours. The illustrious Palafox, is not contented with allowing them to be equal to us in natural talents, but in the memorial he presented to the King in their favour, entitled *A Natural Display of the Indies*, declares they excel us. He there gives a relation of an Indian, whose person he knew, and who went by the name of *The Man of Six Trades*, from his understanding, and being able to work well at that number; and of another, who learned to build organs in an amazing short time; and of another, who in an amazing short time also, learned to play the organ. He there likewise gives an account of the exquisite address, with which an Indian recovered a horse that had been purloined from him by a Spaniard. The Indian, commenced a prosecution against the Spaniard for his horse, and when the trial came on, the beast was brought into court, where the Spaniard alledged that he had bought him, and had had him in his possession for several years, which he brought witnesses to confirm. The Indian, who had no evidence to prove his property in the horse, finding himself hard pressed, instantly threw his cloak over the horse's head, and requested that the Spaniard who insisted that he had been owner of the horse for so long a time, might tell of which eye he was blind; the Spaniard, who was taken by surprise, and much confounded with the question, answered at random, the right; upon which the Indian pulled the cloak off the horse's head, and manifested to the whole court, that he was not blind of either eye; this evinced the roguery of the Spaniard, and the Indian recovered his horse.

XXI. The Europeans under the command of Cortes, had scarce penetrated into the kingdom of Mexico, before they experienced many particulars, which convinced them, that the natives of that country were of the same species with themselves, and the children of the same common Father. We read in the History of the Conquest of Mexico, many military stratagems of the Indians, that were not inferior to those of the Carthaginians, the Greeks and the Romans; and many people have remarked, that the Crioles or children of Spaniards who are born in America, are more sprightly, and have more intellectual quickness, than those who are born in Spain; but whether what others add, that although their ingenuity manifests itself sooner, it does not last to so late a period of life, be just or not, I cannot pretend to determine.

XXII. It would be reasoning erroneously and grossly, to entertain a mean opinion of the capacities of the Indians, because upon their first intercourse with us, they gave pieces of gold for glass beads; for he would be more rude than they, who should conclude they were savage, upon this account. If we were to view glass free from our prepossession in favour of gold, glass would appear the most beautiful of the two; and with respect to what is sought after, for the purpose of ornament and ostentation, out of two things that are equally beautiful, that which is most scarce is always preferred. The Americans then in this instance, did no more than what is done by all the world. They had plenty of gold but no glass; and it was on this account that they concluded, and not without reason, that a string of beads, was a fitter ornament to adorn the neck of a princess, than a gold chain. A diamond, if we only regard the necessary utility of it, is not of superior value to a glass bead; but if we regard its lustre, it certainly excels it; and although, notwithstanding the principal difference between the two things, consists only, in the lustre and beauty of the diamond, the Asiatics sell the Europeans a diamond that weighs two ounces, for an amazing number of pounds sterling; and why is this? It can only be, because such diamonds are exceedingly scarce. The inhabitants of the island of Formosa, esteemed fine brass preferable to gold, because they had greater plenty of gold than fine brass, and continued to do so, till the Hollanders gave them to understand, the great estimation in which gold was held in other regions. If there was a greater plenty of gold all over the world, than there was of fine brass, the last of these metals would be preferred to the first. Upon the

arrival of the Dutch Admiral Matelief at the Cape of Good Hope in the year 1605, the African inhabitants of that country, gave him eight and thirty sheep and two bullocks, for a small quantity of iron, which did not in its value exceed twenty-pence; and the best of it was, that they were equally satisfied they had imposed upon the Hollanders, as the Hollanders were that they had imposed upon them. They had a super-abundance of cattle, and were in great want of iron. And in whatever country the same super-abundance of cattle, and the same want of iron prevails, they must purchase the iron with the same number of cattle.

XXIII. Father Lafitau a Missionary Jesuit, who was a long time among thole North American Indians, who, on account of their being esteemed the most barbarous of all, are called savages, gives great applauses of their government and civil policy, and compares them in these respects, to the antient Lacedemonians: and what is more extraordinary, he also bestows great panegyrics on their eloquence; and goes so far as to say, that he has known here and there one of them, whose orations were equal to those of Cicero and Demosthenes, and expresses some doubt, whether they may not be said to excel them. This relation of Father Lafitau, may be seen in the Memoirs of Trevoux of 1724, art. 106. It is possible, that this account is somewhat hyperbolical; but it should be considered, that Father Lafitau had a long and an intimate intercourse with these people, and there is no doubt, but a man who sees things in a near point of view, can judge better of them, than those who see them at a distance.

XXIV. Our intellectual sight, is exposed to the same defect that our corporeal one is, and is apt to represent distinct things less than they are. There is no man, let his stature be as gigantic as it will, who does not appear like a pigmy at a great distance. The same that happens with respect to the size of bodies, happens with regard to the stature of souls. Those nations which are very remote from us, appear so small in our eye as rational creatures, that we scarce allow them to be endued with the faculty of reason; but if we were to view them near, we should probably form a different judgment of them.

SECT. VII.

XXV. It may perhaps be objected to what I have been advancing, that the very absurd opinions entertained in matters of religion, by the bulk of the people of Asia, Africa, and America, without insisting, upon the total want of any religion among some of them, should induce us to form a very mean judgment of their talents.

XXVI. To this I answer in the first place, that although errors in matters of religion are the worst of all errors, they are no absolute proofs of the rudeness of those who assent to them. Nobody is ignorant, that the antient Greeks and Romans, who were exceedingly well skilled in arts and sciences, were extremely absurd with respect to the objects of their adoration. They worshiped as deities, men who had been adulterers, perfidious, and guilty of all sorts of wickedness. Rome, which as Saint Leon observes, domineered over all the other nations, was herself under the dominion of the errors of them all. When a man sets himself to search for a divinity from among his own species, it is a mark of a depraved imagination, and the question respecting his capacity, is not worth enquiring into, as we may naturally conclude, he has lost his reason before he makes the attempt. And he who walks blindfold, is not more terrified by a high precipice than a low one, as he is unable to discern the difference. I do not even know, whether when a man first begins to err in these particulars, he does not go the most extravagant lengths, who has been the best-informed; because in matters of religion, when the first error has taken root, it is easy for the person who is possessed with it, to confound the mysterious with the ridiculous, and by an affected subtilty, pretend to discover some hidden signs of divinity in those things, which in the eye of common sense, are the most remote from, or have the least to do with it.

XXVII. I answer secondly, that we have no certainty, that the idolatry of these various nations was so gross as it has been represented. With respect to the antient idolaters, some learned men have enforced this doubt very strongly, and have insisted, that there were solid reasons for supposing, that in the image they did not worship the wood, the metal, or the marble of which it was made, but some good genius or demon, whom they believed to have resided in it. Truly it seems incredible, that a statuary, such a one as Horace humourously describes in one of his Satyrs, should stand with his

hatchet in one hand, and having the other on the wood he was about to work, suspended and perplexed, whether he should carve the God Priapus, or Escanus; I say it is incredible, that such a man should suppose himself vested with sufficient authority to fabricate a Deity.

XXVIII. I say the same of the animated idols. How is it credible that the Egyptians, who were for some ages the repositories of the sciences, should chuse for the ultimate object of their adoration, a most vile snake, and even a dog, or an onion, which Juvenal ironically and with derision tells them, was raised up to them in their own gardens? *O sanctas gentes, quibus hæc nascuntur in hortis numina*. It is much more reasonable to suppose, that that nation, who were much addicted to represent every thing enigmatically, and by symbols, should adore these vile creatures in some mystical sense, which these served as a sort of hieroglyphics to explain the signification of, and that their adoration of them, was not absolute but respective. The same reasoning might be brought to apply to other nations, both of former and modern times, as well as to them.

XXIX. I am confirmed in this opinion, by what I have read concerning the superstition which prevails in the island of Madagascar. The inhabitants of that island, worship a cricket, and every one rears his own with great care and veneration. Some French ships in their voyage to India in the year 1665, touched at Madagascar, and being apprized of the superstition of the natives, a curious Frenchman, asked one of them whom they respected as a wise man, what could induce them to worship so vile an animal, who answered, *that they worshiped the principal and head of all, that is the Creator in the creature, and that it was necessary to direct their adoration to a sensible object, in order to fix the attention*. Who could have expected to meet with so delicate a sentiment, in such a country? I do not insist, that the reply exempts them from the note of superstition, but it proves, that they are not stupid and insensible. If the same observation the Frenchman made on the absurdity of the worship of the people of Madagascar, had been made to an antient Egyptian, he in all probability, would have returned the same answer to it in substance, which the Madagascar man gave.

XXX. With regard to people who are supposed to have no religion at all, I must declare, that I doubt whether there are any such people in the world. The voyagers who assure us there are such, might possibly from their want of sufficient intercourse with them, or from the want of understanding their language, not be able to penetrate their sentiments on that head; for all nature proclaims the existence of a Creator with so loud a voice, that the most sleepy reason cannot fail to be awakened with her cries.

SECT. VIII.

XXXI. There is then, scarce any people whatever, if you examine deeply into things, who can with justice be deemed barbarous. I will not however deny, that there is not between particular nations, some inequality in the use or application of their reason. Yes, but this depends in some measure, on the disposition of their organs, and the climate in which they are born, and these possibly may have some influence to promote this disposition. But if I was to be asked which are the most penetrating and acute nations, I should answer ingenuously that I could give no judgment in the matter that might safely be relied on. I see that the sciences, at one time flourished among the Phenicians, at another among the Chaldeans, at another among the Egyptians, at another among the Greeks, at another among the Romans, and at another among the Arabs, and that at last, they extended themselves to almost all the European nations. I observe also, that the inhabitants of every country into which they were not introduced, were looked upon as rude and uncivilized; but it was generally remarked, that after they came to be introduced into one of these last, the natives of it did not make less advances in them, than those did, who had the happiness to be the first visited by them. Perhaps if the world lasts much longer, and there should happen great revolutions of empire in it, as Minerva goes wandering about the earth, and continues to shift her station according to the violent agitations she receives from the impulse of Mars; I say if such revolutions should happen, the Iroquois, the Laplanders, the Troglodytes, the Garamantes, and other people, whom we now look upon with disdain, and whom we with repugnance admit to be members of our species, may one day possess the sciences in an eminent degree; so that experience will hardly assist us to determine, the inequality of ingenuity, that prevails in different nations.

XXXII. Much less then can we determine the point by physical reasonings. Many have endeavoured to establish, that this inequality bears proportion, to the predominance of the elemental qualities which prevail in different countries; and it is commonly said, that humid and cloudy climates, produce heavy dull spirits, and serene and dry ones, sprightly and penetrating ones. Aristotle gives the preference in this matter, to those who inhabit the hot countries. Agreeable to the first of these opinions, the Venetians and the Hollanders, should be very stupid dull people, for the first live in marshes surrounded by standing waters, and the last in a muddy low country, that might be said to have been stolen from the sea, and which is surrounded by standing waters also; and agreeable to the second, the negroes of Angola, should be more acute than the English. It does not appear to me, that any reasonable man should admit of either of these consequences. But it is not necessary for us to dwell longer on this subject at present; as we have shewn at large in our defence of the women, that we cannot infer there is any inequality of understanding, produced by the predominance, the sensible qualities have on the temperament; and that therefore, we are under a necessity of acknowledging, that the influence our native country may have over it, is the effect of some occult cause, which is impenetrable to our reason, or at least, which has not been penetrated by it as yet.

XXXIII. When I say that by experience, we can hardly distinguish the inequality of ingenuity of nations, I would be understood to mean, in point of the essential qualities, of penetration, solidity, and clearness; and not in point of the accidental ones, of superior quickness and readiness, or those of being more slow and tardy of comprehension; for with respect to these, it is evident that some nations exceed others. Thus it is manifest, that the Italians and French are more quick and ready than the Spaniards; and even in Spain itself, there is a great difference between the inhabitants of one province and those of another in these respects; for it is remarked, that in Asturias, the people are quicker of apprehension, and are more ready at explaining themselves, than they are in any of the other provinces; and the experience of this, should be sufficient to dissuade us from falling-in with that general notion, that rainy countries produce heavy dull people; as it is well known, that the heavens may rather be said to inundate, than refresh this country, so that it may be truly called,

Nimborum patriam, loca fœta furentibus austris.

XXXIV. But if I was to give my opinion which of the nations of Europe should be preferred to the rest, with respect to their penetration, I should incline to the sentiment of Heidegger a German author, who concedes this advantage to the English. It is certain that Great Britain, since literature, arts, and sciences, were first cultivated in it, has produced a copious harvest of authors of the first class. The recital of those only, which have arisen in that country, out of the Benedictine and Seraphic orders of religious, would be tedious and tiresome. But I cannot forbear to mention, that we are indebted to each of these two orders, for three stars of the first magnitude. To the Benedictine, for the venerable Bede, the famous Alcuinus, and the celebrated Calculator Suiset. To the Seraphic, for Alexander de Ales, the subtil Scotus, and his disciple William Ocman. Cardanus in his Treatise (de Subtilit. lib. 16. de Scient.) graduates among the twelve most acute geniuses of the world, the subtil Scotus and the Calculator Suiset, in the fourth and fifth rank, of whom he says: *Barbaros ingenio nobis haud esse inferiores, quandoquidem sub Brumæ cælo, divisa toto orbe Britannia duos tam clari ingenii que viros emisserit.*

XXXV. Neither should we conceal, that at the time when the other nations of Europe scarce knew what the mathematics meant, the two beforementioned orders of Religious exhibited two illustrious English mathematicians; Roger Bacon, and Oliver of Malmsbury; of the first of whom, the vulgar feigned the same tale, they did of Albertus Magnus, that is, that he had made a brazen head, which answered all questions that were put to it. The other was not less famous than he, of whom John Pitseus relates, that he had invented some machinery, by the help of which he was enabled to fly, although he never could attain at doing it, for above the distance of about a hundred and twenty yards at a time; but that was doing more, than any other man ever did before him.

XXXVI. On physical subjects, England has furnished a greater number of original authors, than all the other nations of Europe put together; and the French, who are very tenacious of the reputation of their country for producing men of ingenuity, are obliged to confess, that the English are

superior to them in philosophical abilities. It may without rashness or exaggeration be said, that all the advances which have been made in physical knowledge for a century past, are owing to my Lord Chancellor Bacon. It was he that broke through the strait limits, within which, Physics till his time had been imprisoned; and he was the man who threw down the columns, on which to mark the boundaries of human knowledge in natural things, *ne plus ultra* had been inscribed. The most learned Peter Gassendo, was nothing else but a faithful disciple of Bacon's, and what Bacon had said in a summary way, he repeated more at large, in his extensive philosophical writings. All that Descartes said which was of any real value, he took from Bacon. After Bacon, we may reckon as great originals, Mr. Boyle, and the most subtil Sir Isaac Newton; as also John Locke^[2], Sir Kenelm Digby, and many others; but the misfortune is, that the lustre of their ingenuity, was tarnished with the same religious blemish my Lord Bacon's was; and when they had once strayed out of the right path, they flew with such velocity, that the extent of their wanderings, was great as the liveliness of their imaginations. But with all this, there has not been wanting in England, since it was blemished with heresy, a Sir Thomas More, who was as celebrated for his Catholic constancy, as he was for his eminence in the sciences.

XXXVII. I must also say, that I have always observed in the English philosophers, great frankness, and that they gave a simple plain relation of the result of their experience and labours, free from all artifice and deceit; which is a thing not very common with those of other nations; and I have remarked this particularly, in Bacon, Boyle, Sir Isaac Newton, and Sydenham the physician; and it has afforded me great pleasure, to see, how without boasting they have declared what they know, and without blushing, have confessed what they are ignorant of. This is the true mark and characteristic of sublime geniuses. Oh! how much it is to be lamented, that such great lights, should be obscured by heretical prejudices!

END OF THE FOURTH VOLUME.

FOOTNOTES

[1] The translator thinks, as he has not translated that Discourse, it will not be amiss to insert here the Author's sentiment on this subject. In the Essay or Discourse referred to, after reciting the arguments that have been used to prove the invention came from China, and the claims that have been made on the behalf of a variety of people, to their being the inventors; he gives it as his opinion, that Bertoldus Schuvart, a German Franciscan friar, and an eminent chemist, was the man who invented it, or at least was the person who, brought the invention to perfection.

[2] This is the same John Locke, of whose writings, as also of those of Rapin, Sidney, and Bishop Hoadly, the late David Hume, in his History of Great Britain, gives the following description: "Compositions of the most despicable kind both for style and matter, which have been extolled, and propagated, and read; as if they had equalled the most celebrated remains of antiquity."

Vid. vol. viii. pag. 323, of the last edition of Hume's History of Great Britain, published in 1778.

ERRATA.

[Page 68, line 27](#), *read* such for example as spiritual entities.

[Page 101, line 8](#), *for* that, *read* and that.

[Page 104, line 16](#), *read* and after turning, &c.

[Page 129, line 25](#), *for* executed, *read* executes.

[Page 157, line 12](#), *for* ascends, *read* ascend; and line 21, *for* lerned, *read* learned.

[Page 158, line 20](#), *for* but, *read* yet.

[Page 169, line 13](#), *read* many of those instruments.

[Page 187, line 20](#), *read* proper resolution.

[Page 212, line 9](#), *for* had, *read* has.

[Page 213, line 14](#), *for* raise, *read* arise.

[Page 271, line 3](#), *for* reasonable, *read* reasoning.

[Page 272, line 6](#), *for* qulities, *read* qualities.

[Page 273, line 10](#), *for* is, *read* are.

[Page 297, line 14](#), *read* as it is called.

[Page 352, line 6](#), *for* eben, *read* been.

Transcriber's Note: The errata have been corrected, along with a few other minor printing errors.

*** END OF THE PROJECT GUTENBERG EBOOK ESSAYS, OR
DISCOURSES, VOL. 4 (OF 4) ***

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